Master's Thesis
in the Master's Programme Design & Computation (M.A.)

Generative Artificial Intelligence for Early-Stage Urban Massing

Development of a Context-Aware Generative Modelling Approach for Regulation-Grounded Urban Massing

submitted by

Timo Nikisch
Student ID: 4072339 (UdK Berlin) / 169005089560 (TU Berlin)

Berlin University of the Arts, Faculty of Design
Technische Universität Berlin, Faculty I – Humanities and Educational Sciences

First examiner: Prof. Albert Lang
Second examiner: Johannes Pointer

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Code Repository

The accompanying code repository documents the computational workflow of this thesis, including data processing, model training, evaluation scripts, and experiment configurations. References to the raw input data and the derived training data are documented there as well.

github.com/Timonksch/Generative-Artificial-Intelligence-for-Early-Stage-Urban-Massing

Abstract

This thesis investigates how generative artificial intelligence can support early stages of urban planning and design. To this end, it develops and evaluates a series of machine-learning models for generating parcel-scale building volumes in context-sensitive infill situations. In such settings, building volumes cannot be assessed as isolated forms but must be understood in relation to the surrounding urban fabric and regulatory target values for development intensity. Parametric and optimisation-based tools already support rapid massing modelling and feasibility studies. However, their reliance on predefined rules limits their suitability for situations in which contextual spatial relationships must be inferred from the existing built environment and cannot be fully specified in advance. Against this background, the thesis examines whether machine-learning models can generate parcel-scale building volumes that respond to the surrounding urban fabric, can be steered through target values for GRZ, GFZ, and building height, and can produce different variants without losing contextual fit or alignment with the prescribed targets.

Using Berlin as a case study, the thesis constructs a voxel-based dataset from CityGML LoD1 building volumes, cadastral parcels, and street-network data, comprising 13,164 parcel-centred samples. Urban form is represented through a shared voxel grid with spatial context channels. The data are prepared at a native resolution of 256^3 and downscaled to 64^3 for training. Selected regulatory metrics are derived deterministically from parcel geometry and voxel occupancy. On this basis, the thesis proceeds through three successive phases, each investigating a different machine-learning architecture for massing modelling and building on the results and limitations of the preceding phase. Phase I develops a 3D U-Net without explicit regulatory conditioning to examine how well parcel-scale building volumes can be inferred from the surrounding urban fabric alone. Phase II extends this architecture through FiLM-based conditioning on GRZ, GFZ, and building height and investigates whether these target values can steer the predicted form. Phase III introduces a two-stage latent diffusion pipeline. It consists of a variational autoencoder for learning a latent representation and a conditional diffusion model for stochastic sampling. This phase examines whether controlled variants can be generated under fixed spatial and regulatory conditions.

The results show that deterministic, context- and target-conditioned voxel prediction can produce plausible parcel-scale building volumes within the developed setup. Across the Berlin samples, the surrounding urban fabric provides a robust contextual basis for predicting likely parcel-scale building volumes. Conditioning on GRZ, GFZ, and building height improves alignment with the prescribed target values and enables controllable responses to changes in these targets. The thesis develops a generative modelling approach that accounts for regulatory target values and connects urban form, selected regulatory metrics, and generative methods within an exploratory setup for early-stage infill studies. However, the reliable generation of controlled stochastic variants investigated in Phase III remains unresolved in the present latent diffusion setup.

List of Abbreviations

Abbreviation	Meaning
2D	Two-dimensional
3D	Three-dimensional
AdamW	Adam optimiser with decoupled weight decay
AI	Artificial intelligence
ALKIS	Amtliches Liegenschaftskatasterinformationssystem
BauGB	Baugesetzbuch (Federal Building Code)
BauNVO	Baunutzungsverordnung (Land Use Ordinance)
BCE	Binary cross-entropy
BIM	Building Information Modelling
BVerwG	Bundesverwaltungsgericht (Federal Administrative Court)
CAD	Computer-aided design
CityGML	City Geography Markup Language
CNN	Convolutional neural network
DDIM	Denoising Diffusion Implicit Models
$\Delta\mathrm{Vol}$	Relative volumetric error
EMA	Exponential moving average
EPSG	European Petroleum Survey Group code for a coordinate reference system
FiLM	Feature-wise Linear Modulation
FIS	Fachübergreifendes Informationssystem (as in FIS-Broker)
FN	False negative
FP	False positive
GAN	Generative adversarial network
GFZ	Geschossflächenzahl (floor area ratio)
GIS	Geographic Information System
GN	Group Normalisation
GNN	Graph neural network
GPT	Generative Pre-trained Transformer
GRZ	Grundflächenzahl (ground coverage ratio)
GT	Ground truth
ID	Identification number
IoU	Intersection over Union
JSON	JavaScript Object Notation
KL	Kullback–Leibler divergence
LoD	Level of Detail (e.g. LoD1 and LoD2)
M.A.	Master of Arts
MBO	Musterbauordnung (Model Building Code)
MLP	Multilayer perceptron

Abbreviation	Meaning
n/a	Not available or not applicable
NeRF	Neural Radiance Field
NPZ	Compressed NumPy archive
OECD	Organisation for Economic Co-operation and Development
P1	Run-label prefix for Phase I
P2	Run-label prefix for Phase II
P3A	Run-label prefix for Phase III Stage A
P3B	Run-label prefix for Phase III Stage B
ReLU	Rectified Linear Unit
SiLU	Sigmoid Linear Unit
TP	True positive
TTA	Test-time augmentation
TU	Technische Universität (Technical University)
UdK	Universität der Künste (University of the Arts)
U-Net	Encoder–decoder neural network with skip connections
UTM	Universal Transverse Mercator
VAE	Variational autoencoder
VE	Absolute volume error
ZPE	Z-profile error

Glossary

Term	Definition
§ 34 BauGB	Provision of the German Federal Building Code governing the admissibility of development in built-up areas where admissibility is not conclusively determined by a binding development plan. Among other requirements, a proposal must fit the character of its immediate surroundings with regard to type and intensity of land use (*Art und Maß der baulichen Nutzung*), building arrangement (*Bauweise*), and the plot area to be built over (*überbaubare Grundstücksfläche*). The thesis uses the provision as a conceptual reference for context-dependent comparison and does not perform a legal admissibility test.
Activation functions and logits	Activation functions introduce non-linearity after convolutional or linear layers. The architectures use SiLU, defined as $x \cdot \mathrm{sigmoid}(x)$, in their feature-processing blocks. Logits are raw and unbounded network outputs. In Phases I and II and in the VAE decoder, a sigmoid maps occupancy logits to probabilities before thresholding. The Stage B diffusion U-Net instead outputs a raw noise prediction.
Adjusted relative error	Tolerance-aware relative error for GRZ, GFZ, and height. A resolution-derived tolerance is subtracted from the absolute deviation before the remaining difference is expressed relative to the reference value. Deviations within the tolerance contribute no adjusted error. For GFZ, part of the measured resolution shift remains beyond this tolerance.
Administrative legibility	Concept after Scott describing how states render territory readable through standardised categories, measurements, and representations. It supports the critique of metric-based spatial abstraction developed in Chapter 2 and revisited in the discussion.
Agency, transparency, and accountability	Agency concerns who or what shapes the conditions and outcomes of a design process. Transparency concerns whether data, assumptions, transformations, and model limitations can be traced and examined. Accountability concerns who remains responsible for interpreting, selecting, and acting on model outputs. The models developed in this thesis support human judgement and do not assume planning responsibility.
ALKIS	Amtliches Liegenschaftskatasterinformationssystem, the official German real-estate cadastre information system. It supplies the cadastral parcel geometries used in the dataset.

Term	Definition
Artificial intelligence, machine learning, and deep learning	Artificial intelligence is the broader field concerned with computational systems performing tasks associated with human intelligence. Machine learning refers to methods that learn statistical relationships from data, reducing dependence on explicitly programmed rules. Deep learning denotes machine learning based on multilayer neural networks and includes the CNN, U-Net, VAE, and diffusion architectures used in this thesis.
BauGB, BauNVO, and MBO	German planning and building-law framework comprising the Baugesetzbuch, the Baunutzungsverordnung, and the Musterbauordnung. The BauGB provides the statutory basis for spatial planning and development control. The BauNVO specifies land-use categories and quantitative measures such as GRZ and GFZ. The MBO informs the state building codes governing technical construction requirements.
Bebauungsplan	Legally binding local development plan containing statutory determinations on land use and built form. Where such a plan does not conclusively determine the admissibility of a proposal, § 34 BauGB may apply within a built-up area.
Binarisation threshold	Probability threshold $\hat{t}$ that converts predicted occupancy probabilities into a binary volume. It is calibrated on the validation split and subsequently applied unchanged to the test split.
Binary cross-entropy (BCE)	Per-voxel classification loss between predicted occupancy probabilities and binary ground truth. The weighted form applies a positive class weight w^+ derived from the ratio of empty to occupied voxels. This addresses the strong class imbalance of sparse occupancy grids.
Building height	Regulatory metric defined as the vertical occupied span of a building volume. In the voxel representation it is computed from the difference between the exclusive upper and inclusive lower occupied indices, multiplied by the voxel size. It is evaluated with a tolerance of one vertical voxel step and forms one component of the conditioning vector.
Building typology	Recurring building or block configuration such as a perimeter block, parallel slab development, or detached housing. Qualitative evaluation considers whether predictions preserve relevant typological characteristics of their surroundings.
BVerwG	Bundesverwaltungsgericht, the German Federal Administrative Court. Its case law informs the interpretation of § 34 BauGB, including the delimitation of the immediate surroundings and the assessment of development intensity.

Term	Definition
CityGML and Level of Detail (LoD)	CityGML is an open standard for the semantic and geometric representation of three-dimensional city models. The Berlin building dataset is provided as CityGML LoD1. Level of Detail describes the degree of geometric and semantic detail. LoD1 represents buildings as simplified extruded volumes with aggregated heights, while LoD2 additionally represents differentiated roof geometry.
Classifier-free guidance	Sampling method for conditional diffusion models that combines a condition-dependent noise prediction with a reference prediction. A guidance scale s controls the influence of the conditioning signal. Stage B uses a dataset-mean regulatory vector for the reference branch, with no null condition. This variant is referred to as mean-conditional guidance.
Conditioning	Provision of additional information that steers a model's output. Phases I and II receive the spatial context tensor, while Phase II additionally receives parcel-specific targets for GRZ, GFZ, and building height. In Phase III, Stage A learns a latent representation from target occupancy volumes, while Stage B receives spatial context and regulatory targets for conditional sampling. Phase I is described as unconditional only with respect to explicit regulatory conditioning and still predicts massing from spatial context.
Conditioning dropout	Training procedure that randomly replaces the regulatory conditioning vector for selected samples. In Stage B the vector is replaced by the dataset-mean condition. This trains the reference branch later used for mean-conditional guidance.
Conditioning vector	Global regulatory vector $c = (\mathrm{GRZ}, \mathrm{GFZ}, H) \in \mathbb{R}^3$. It is z-score normalised using training-split statistics and supplied separately from the spatial context tensor.
Context-sensitive infill	Densification within an existing built environment whose massing is assessed or developed in relation to surrounding buildings, parcels, streets, and density conditions. The thesis focuses on situations in which these contextual relations are not fully captured by predefined quantitative rules.
Context tensor	Multi-channel spatial input $X \in \mathbb{R}^{4\times D\times H\times W}$ consisting of the target parcel mask C_0, the neighbouring building height field C_1, the parcel-boundary structure C_2, and the street mask C_3.
Contextual admissibility	Aspect of admissibility under § 34 BauGB concerned with whether a proposal fits the characteristics of its immediate surroundings where binding plan determinations do not conclusively govern the proposal. The thesis uses this contextual principle as a conceptual reference and does not assess the other statutory requirements of legal admissibility.

Term	Definition
Contextual fit	Spatial agreement of a predicted building mass with observable characteristics of its surroundings, including placement, orientation, footprint scale, and vertical regime. It is assessed qualitatively across all phases. Phase I isolates the extent to which such agreement can be recovered from spatial context alone.
Controllability	Capacity of a conditioned model to respond measurably and directionally to deliberate changes in regulatory targets while spatial context remains fixed. It is evaluated by scaling the conditioning values jointly and, for the final Phase II model, one variable at a time.
Convolutional neural network (CNN)	Neural network whose layers apply learned local filters across a spatial grid. Each filter produces a feature map recording its response at every position. Successive layers combine these maps into increasingly abstract spatial representations.
Coordinate channels	Additional input channels containing normalised x, y, and z coordinates. They provide explicit positional information within the voxel grid. The experiments did not show a consistent advantage over the coordinate-free context representation.
Coordinate reference system	Defined spatial reference frame assigning coordinates to geographic positions. All geospatial data are processed in EPSG 25833, corresponding to UTM Zone 33N and representing the Berlin study area in metric coordinates.
Data augmentation and test-time augmentation (TTA)	Training-time augmentation applies random horizontal $90°$ rotations and flips to input and target tensors. It exposes the model to equivalent orientations without adding new samples and is disabled for validation and test data. Test-time augmentation averages predictions over rotations of $0°$, $90°$, $180°$, and $270°$ before binarisation and is treated as an inference setting.
DDIM sampling	Denoising Diffusion Implicit Models (DDIM) sampling applies a reduced sequence of reverse denoising steps to generate a latent output. The thesis uses a deterministic DDIM trajectory with $\eta = 0$. Variation between samples originates from different initial noise tensors.
Design space	Set of forms reachable by a generative system. Parametric systems delimit this space through explicitly authored parameters and rules. In learned systems, the reachable space is shaped by the training distribution, representation, model architecture, conditioning mechanism, and training objective.
Deterministic prediction	Mapping that returns the same output for the same input without random variation. Phases I and II perform deterministic prediction. Phase III introduces stochastic sampling.
Dice score	Evaluation metric for overlap between binarised predicted and reference volumes. It combines precision and recall into a single value. It is distinct from the soft Dice loss used during training.

Term	Definition
Diffusion model	Generative model that progressively adds noise in a forward process and learns a reverse process that predicts and removes this noise. The Stage B model operates in latent space and receives the noisy latent, diffusion timestep, spatial context, and regulatory condition. Its training objective minimises the mean squared error between added and predicted noise.
Dropout and Dropout3d	Regularisation method that randomly deactivates model activations during training. Dropout3d deactivates entire feature channels and is used at the bottleneck of the Phase I and Phase II networks. It is distinct from conditioning dropout.
Early-stage massing study	Exploratory investigation of building volumes, density, footprint, and height before detailed architectural design. Such studies support early feasibility assessment and comparison of possible development envelopes.
Edge-adjacent neighbour filter	Dataset-construction rule comparing each target parcel with the two edge-adjacent parcels sharing the longest boundaries. Samples are retained when target and neighbours remain within defined GRZ and GFZ tolerances. The rule selects a contextually consistent dataset subset and does not reproduce legal planning judgement.
Encoder–decoder architecture	Network structure that compresses an input into a lower-resolution feature representation and subsequently restores it to the output resolution. Encoder stages perform downsampling, decoder stages perform upsampling, and the lowest-resolution stage forms the bottleneck.
Exponential moving average (EMA)	Running average of model weights maintained during diffusion training. EMA-smoothed weights are used for Stage B sampling and evaluation to reduce short-term variation in the trained parameters.
Failure mode	Recurring and characteristic error pattern identified through quantitative or qualitative evaluation. Phase III failure modes include diffuse background activation, excessive occupied volume, and fragmented geometry.
Feature map	Intermediate tensor produced by a convolutional layer. Each feature channel represents the spatial response of a learned filter. FiLM modifies these channels using parcel-level regulatory conditions.
Feature-wise Linear Modulation (FiLM)	Conditioning method that applies channel-wise scale and shift parameters γ and β to intermediate feature maps. A small multilayer perceptron derives these parameters from the normalised regulatory vector. Phase II uses FiLM throughout the encoder, bottleneck, and decoder.

Term	Definition
Generative adversarial networks and graph neural networks	Generative adversarial networks train a generator and discriminator in competition. They can produce sharp outputs but are sensitive to unstable training and do not directly guarantee geometric accuracy. Graph neural networks operate on nodes and edges and are suited to relational structures such as layouts and adjacency graphs.
Generalisation	Ability of a model to perform on samples not used during training. Evaluation uses a held-out test split. An additional sensitivity analysis examines test parcels whose context windows do not overlap any training sample.
Generative model	Model that learns properties of a data distribution and can produce outputs consistent with that distribution. Relevant model families include GANs, VAEs, and diffusion models. In this thesis, the term generation also describes the broader design-oriented production and exploration of model-based massing outputs.
GFZ	*Geschossflächenzahl*, translated as floor area ratio. It denotes total floor area relative to parcel area. Legal floor-area calculations depend on the classification of full storeys, which cannot be reconstructed from the available LoD1 data. The thesis instead estimates a floor-area equivalent from occupied voxel volume, parcel area, and an assumed storey height of 3.0 m. It does not perform a legal full-storey calculation.
GIS	Geographic information system for storing, processing, and analysing spatial data. Dataset construction combines GIS operations on buildings, cadastral parcels, and streets with subsequent rasterisation and voxelisation.
Ground truth	Reference data used to train or evaluate a model. In this thesis it is the binary voxel volume Y derived from the existing LoD1 building geometry located on the target parcel.
Group Normalisation	Normalisation method operating within groups of feature channels and independently of the batch dimension. It is suitable for three-dimensional networks whose memory requirements permit only small physical batch sizes.
Gründerzeit	Period of rapid German urbanisation associated in Berlin with dense nineteenth-century and early twentieth-century perimeter-block development. Buildings from this period often have storey heights above the fixed 3.0 m assumption used for the floor-area equivalent.
GRZ	*Grundflächenzahl*, translated as ground coverage ratio. It denotes the proportion of a parcel covered by building footprints. In the voxel representation, the occupied volume is projected onto the horizontal plane to obtain the footprint mask used in the calculation.

Term	Definition
Hyperparameter	Configuration value selected before or outside model optimisation. Examples include learning rate, network width, network depth, loss coefficients, conditioning-dropout probability, and the diffusion noise schedule.
Inference	Application of a trained model to input data without updating its weights. In Phases I and II, inference produces occupancy logits that are converted into binary predictions. In Phase III, Stage A inference reconstructs an occupancy volume through the VAE, while Stage B inference consists of latent diffusion sampling followed by decoding with the frozen VAE.
Intersection over Union (IoU)	Primary voxel-level overlap metric defined as the number of occupied voxels shared by prediction and ground truth divided by the number occupied in either volume after binarisation.
Kullback–Leibler divergence	Measure of difference between probability distributions. In the VAE it regularises the encoder posterior toward a standard Gaussian prior. Its coefficient can be increased gradually during the first training epochs, a procedure referred to as KL annealing.
Latent diffusion	Diffusion approach that performs noising and denoising in a compressed latent representation, avoiding direct voxel-space diffusion. Phase III consists of Stage A, which trains the VAE representation, and Stage B, which trains a conditional diffusion model in the fixed latent space.
Learned representation	Internal encoding of patterns acquired from training data through model optimisation. In contrast to the explicit relations of a parametric model, a learned representation is distributed across network weights, activations, and intermediate features.
Learned spatial prior	Context-derived expectation of likely urban form acquired from the training data. It guides predictions toward recurring configurations represented in the dataset and may limit responses that depart substantially from those configurations.
Mean-conditional guidance	Variant of classifier-free guidance in which the reference prediction is computed from the dataset-mean regulatory condition. It corresponds to the mean-replacement conditioning dropout used during Stage B training.
Model capacity	Amount of representational capacity available to a neural network. The experiments vary it through the number of base feature channels and the depth of the encoder–decoder hierarchy.
Multilayer perceptron (MLP)	Fully connected neural network composed of linear layers and non-linear activation functions. FiLM modules use MLPs to transform regulatory vectors into modulation parameters. Stage B also uses an MLP to process the diffusion timestep embedding.

Term	Definition
Nähere Umgebung	Immediate surroundings within the meaning of § 34 BauGB. Their spatial extent is determined case by case through the reciprocal influence between the proposal and its surroundings and may differ between assessment criteria.
Native and training resolution	Dual-resolution representation used by the dataset. Samples are constructed at a native voxel size of 0.5 m on a 256^3 grid and downsampled to a voxel size of 2.0 m on a 64^3 grid for training. Average pooling reduces each group of neighbouring cells to a mean value before the target volume is re-binarised.
Negative result	Finding that a tested approach does not achieve its stated objective within defined experimental conditions. The Phase III result is localised to the tested latent-diffusion pipeline. Within this pipeline, the loss of stable conditional geometry emerges from the interaction of latent representation, denoising, conditioning, dataset scale, and sparse voxel occupancy. It cannot be attributed to one component alone.
Noise schedule	Sequence controlling how much Gaussian noise is added at each diffusion timestep. Stage B compares linear and cosine schedules and different timestep horizons. The final model uses the selected linear schedule.
Normativity	Presence of value-laden assumptions concerning what is considered relevant, desirable, typical, or plausible. In this thesis, normativity enters through data selection, regulatory metrics, filtering rules, spatial representations, model objectives, and evaluation criteria.
Out-of-distribution input	Input that lies outside or at the margins of the spatial configurations and regulatory values represented in the training distribution. Model behaviour is less supported by empirical training examples in such regions. Conditioning sweeps remain close to the native values to limit this effect.
Parametric design	Form generation through explicitly authored parameters, dependencies, rules, and constraints. Variants are produced by changing these parameters, while the relationships between them remain defined by the designer.
Parcel	Cadastral land unit used as the unit of analysis. Each dataset sample is centred on one target parcel whose existing building volume supplies the ground truth for massing prediction.
Pattern language	Concept developed by Christopher Alexander describing recurring spatial solutions as an interconnected system of reusable patterns. Chapter 2 treats it as an early formalisation of relational and rule-based design knowledge.
Perimeter block	Urban typology in which buildings form a largely continuous edge along the surrounding streets and enclose an internal courtyard. It is characteristic of substantial parts of Berlin's dense urban fabric.

Term	Definition
Precision and recall	Complementary metrics derived from true-positive, false-positive, and false-negative voxel counts. Precision measures the proportion of predicted occupied voxels that are also occupied in the ground truth and penalises excessive predicted mass. Recall measures the proportion of ground-truth occupied voxels recovered by the prediction and penalises missing mass.
Prediction, reconstruction, and sampling	Terms distinguishing the technical operations of the three phases. Phases I and II predict an occupancy volume from their inputs. Stage A reconstructs an input volume through the VAE. Stage B samples a latent output from noise under spatial and regulatory conditioning.
Proof of concept	Demonstration that an approach is feasible in principle within a delimited experimental setup. It does not establish operational maturity, legal validity, or transferability to other cities and planning contexts.
Regulatory metric	Quantitative measure describing a planning-relevant property of built form. The selected metrics are GRZ, GFZ, and building height. They are computed deterministically from parcel geometry and voxel occupancy.
Regulatory target	Prescribed value of GRZ, GFZ, or building height supplied to a conditioned model. The resulting massing is expected to respond to these values while retaining its spatial relationship with the surrounding context.
Regulatory target alignment	Agreement between a prescribed regulatory target and the corresponding metric computed from a model output. It is evaluated through direct value comparison and tolerance-adjusted errors.
Representational abstraction	Deliberate simplification of the modelling problem. Buildings are represented as LoD1 envelopes, continuous space as discrete voxel occupancy, and selected aspects of regulation through GRZ, GFZ, and height. These choices enable learning and evaluation while excluding architectural detail and many social, functional, and legal dimensions of urban form.
Reproducibility	Capacity to repeat an experiment under documented conditions. The thesis supports reproducibility through fixed random seeds, a fixed split manifest, shared configuration structures, documented inference settings, and stored model checkpoints.
Research-through-design	Methodological framing in which knowledge is produced through iterative making, testing, comparison, and reflective interpretation of artefacts. The relevant artefacts include the dataset, model architectures, training experiments, and resulting massing outputs.

Term	Definition
Resolution shift	Systematic change in derived GRZ, GFZ, or height values caused by reducing the native voxel resolution to the coarser training grid. It is measured separately so differences introduced by discretisation can be distinguished from model behaviour.
Resolution tolerance	Metric-specific tolerance derived from the discrete evaluation grid. It corresponds to one resolvable footprint cell, one occupied voxel converted into a floor-area equivalent, or one vertical voxel step. Deviations within this tolerance are excluded from the adjusted model error.
Sampling window	Fixed cubic spatial extent of 128 m per axis centred on the target parcel. It defines the parcel and neighbourhood context available to the model and is represented at native and training resolution.
Setback	Horizontal distance between a building or part of a building and a parcel edge, street line, or lower building section. Fine setbacks may be reduced or removed by the 2.0 m training resolution.
Soft Dice loss	Differentiable overlap loss computed from continuous occupancy probabilities. It complements BCE through global volumetric overlap optimisation alongside BCE's voxel-wise evaluation.
Spatial overlap between dataset splits	Overlap between the fixed context windows of samples assigned to different dataset subsets. The parcel-level random split permits neighbouring training and test parcels to share parts of their urban context. A separate non-overlapping test subset is examined as a sensitivity analysis.
Statistical inference	Derivation of outputs from statistical relationships learned from data. In the conceptual discussion, the term describes how urban form becomes mediated through learned patterns. It is broader than inference in the technical machine-learning sense, which denotes applying a trained model without updating its weights.
Stochastic sampling	Production of different outputs for identical spatial and regulatory inputs through random initial noise. Phase III tests whether this variation can remain stable, context-sensitive, and aligned with regulatory targets.
Surface weighting	Loss weighting that assigns greater importance to building boundaries. A boundary mask is derived from morphological dilation and erosion of the ground-truth volume. Boundary voxels receive an increased loss weight.
Systems thinking and cybernetics	Perspectives that understand cities and design processes as systems of interacting components, information, control, and feedback. Chapter 2 uses these traditions to connect process-oriented design with later computational and generative approaches.

Term	Definition
Tensor	Multi-dimensional numerical array used to represent model inputs, feature maps, latent codes, and outputs. Relevant dimensions include batch, channels, depth, height, and width.
Timestep embedding	Vector representation of the current diffusion timestep. Stage B obtains it from a sinusoidal encoding processed by an MLP and supplies it to the conditional convolution blocks so feature processing can respond to the current noise level.
Training objective	Loss function minimised during model optimisation. Phases I and II use weighted BCE and soft Dice with volume regularisation. Stage A uses a VAE objective combining reconstruction loss and KL divergence. Stage B uses a mean-squared-error noise-prediction objective.
Training run	Complete optimisation procedure for one model configuration. Training proceeds in epochs and mini-batches. Gradient accumulation combines several physical batches before an optimiser update. AdamW updates the weights using a scheduled learning rate, weight decay, and gradient clipping. Checkpoints store selected training states for later evaluation.
Training, validation, and test split	Fixed partition of the 13,164 retained samples into 70 % training, 15 % validation, and 15 % test data. The validation split guides checkpoint selection, threshold calibration, and phase-specific model choices. The held-out test split is reserved for final evaluation.
U-Net (3D)	Convolutional encoder–decoder network with skip connections operating on three-dimensional voxel grids. Skip connections transfer higher-resolution encoder features to corresponding decoder stages and help preserve spatial detail. U-Nets form the backbone of Phases I and II and of the Stage B denoising model.
Urban fabric	Spatial configuration and relationships of buildings, parcels, streets, and open spaces. These elements provide the contextual information from which the models infer likely parcel-scale massing.
Urban massing	Simplified representation of building volumes and their spatial arrangement without detailed architectural articulation. It is used in early design stages to study footprint, density, height, and volumetric relationships.
Urban structure types	Berlin-wide classification of block-scale development patterns documented in the *Umweltatlas Berlin*. The thesis uses these types to characterise the range of density, height, and morphology represented across the study area.

Term	Definition
Variational autoencoder (VAE)	Probabilistic encoder–decoder model. The encoder maps each binary target volume to a Gaussian posterior over latent codes. A latent sample is drawn through the reparameterisation rule $z = \mu + \sigma \odot \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, I)$, and the decoder reconstructs the voxel volume. KL regularisation moves the posterior toward a standard Gaussian prior. Stage A compresses the 64^3 target volume to a 32^3 latent grid with four channels. Its weights remain frozen during Stage B training and sampling.
Volume regularisation	Additional training-loss term weighted by λ_{vol}. It penalises the relative difference between predicted and ground-truth occupied voxel counts and supports consistency of total building volume.
Volumetric error	Measures describing disagreement in total occupied volume. Relative volumetric error $\Delta\mathrm{Vol}$ divides the absolute voxel-count difference by the ground-truth volume. Absolute volume error VE reports the mean absolute discrepancy in occupied voxels.
Voxel and voxelisation	A voxel is a discrete volumetric cell in a three-dimensional grid. Voxelisation converts polygonal building geometry into occupied and empty cells. Target buildings and predictions are represented as binary occupancy volumes. Parcel, boundary, height, and street information are aligned to the same regular grid as spatial input channels.
Z-profile error (ZPE)	Mean absolute difference between predicted and ground-truth occupied voxel counts at each vertical slice. It measures whether building mass is distributed across the correct height levels even when total occupied volume is similar.
Z-score normalisation	Standardisation of each regulatory conditioning variable using the mean μ and standard deviation σ of the training split, $\tilde{c} = (c - \mu)/(\sigma + 10^{-8})$. It places GRZ, GFZ, and height on comparable numerical scales and maps the training-set mean condition close to zero.

1 Introduction

This thesis brings urban planning and machine-learning-based generative modelling together around the question of how built form can be assessed, represented, and produced under contextual and regulatory constraints. The introduction defines this problem space, formulates the guiding research questions, and outlines the scope, contributions, and structure of the thesis.

1.1 Motivation and Problem Space

Urban planning translates collectively articulated goals into built form, where questions of housing supply, density, accessibility, and overall neighbourhood character are negotiated and take physical shape in buildings, blocks, and streets. In doing so, it operates between legal constraint and design reasoning, with outcomes shaped by site-specific intent and the statutes that determine what counts as admissible in a given setting (Fainstein, 2010; Bentlin, 2019; OECD, 2017). Central to this process are quantitative regulatory metrics such as ground coverage ratio, floor area ratio, and maximum building height, which relate building mass to plot area and help standardise assessments of urban density and built form (BauNVO §§ 16–21a; BauGB).

Over the past decades, digital tools have become central to assessing the development potential of building sites and architectural proposals through regulatory metrics and compliance checks (Aish & Woodbury, 2005; Gerber & Lin, 2013). In particular, contemporary parametric planning platforms for early-stage feasibility analysis position themselves as tools for rapid site evaluation and automated comparison of design variants, framing planning as a process that can be accelerated through automation (Forsberg, 2025; Turrin et al., 2011). This automation rests on computational procedures in which variants are generated by adjusting parameters within predefined constraint systems, with site context entering the process mainly through variables specified in advance and not through patterns derived from existing urban form (Davis, 2013). Such approaches are suited to larger development sites with explicit constraints and clearly defined optimisation targets, but become limited in planning settings where admissibility is assessed through comparison with the existing urban fabric and depends on contextual fit that cannot be fully represented through predefined variables and quantitative thresholds alone (BauGB § 34; Pasquale, 2015; Turrin et al., 2011).

Against these limitations, machine-learning-based generative approaches have attracted growing attention for their potential to infer recurring spatial patterns directly from existing built environments. Such models promise to represent aspects of urban context that are difficult to formalise and to generate massing proposals conditioned on GRZ, GFZ, and height targets, allowing design alternatives to be explored without defining all spatial relationships in advance (He et al., 2025; Nauata et al., 2020; Zhang et al., 2024). Building on this potential, the thesis investigates how machine-learning models can relate targets for GRZ, GFZ, and height to the surrounding urban fabric and whether such models can support the exploration and comparison of early-stage massing options without directing the process towards a single optimised solution.

1.2 Regulation, Abstraction, and Learned Spatial Relations

Historically, legal and administrative frameworks such as planning laws, land-use plans, and building codes have governed urban planning by translating contested planning objectives into applicable rules and predictable conditions for implementation (Bentlin, 2019; Scott, 1998). While this mode of governance ensures legal consistency, its reliance on standardisation and metrics has been criticised for reducing the complexity of urban space to what fits within an administrative frame (Kitchin, 2014; Scott, 1998).

Against this background, machine-learning models offer a different mode of abstraction by learning from existing data how density, height, and massing vary across parcel and neighbourhood contexts, instead of relying solely on predefined rule structures and manually specified contextual assumptions (He et al., 2025; Nauata et al., 2020). At the same time, what such models can and will learn depends on the training data and may reinforce prevailing spatial patterns and historical precedents, limit the emergence of novel configurations, and shift normative assumptions from codified rules into training data and model design, where they become embedded in learned representations (Kitchin, 2014; Pasquale, 2015).

This changes how regulatory metrics can be interpreted and applied, with parametric models treating such values as fixed constraints that bound the design space and generative models using them as conditioning signals that steer variation without fully prescribing the resulting form (Davis, 2013; He et al., 2025; Rombach et al., 2022).

1.3 Research Problem and Gap

While contemporary tools for early-stage urban planning and feasibility studies enable the rapid generation and assessment of massing variants through predefined rules and parameters, they remain limited where contextual massing relations must be inferred from existing urban form. Recent academic research on generative urban design, reviewed in Chapter 2, has shown that urban form can be synthesised from learned representations. However, many of these approaches prioritise formal or visual coherence under abstracted or weakly constrained conditions, often encoding surrounding context only partially and introducing regulatory metrics as external constraints or evaluation criteria instead of integrating them as conditioning signals, limiting the contextual and regulatory alignment of generated massing (He et al., 2025; Rasoulzadeh et al., 2025; Zhang et al., 2024).

This points to a representational problem at the intersection of parametric planning tools and generative models, where regulatory metrics, the surrounding urban fabric, parcel-scale geometry, and deterministic evaluation are not yet brought together within a shared computational representation. Parametric platforms embed regulation within predefined design spaces, while research models learn form without fully integrating regulatory metrics and parcel relations into directly evaluable outputs, with both strands connecting regulation, context, and generative modelling only partially (Davis, 2013; He et al., 2025). Against this gap, the thesis develops a parcel-scale generative approach in which urban context can be learned from data, selected regulatory metrics can condition model outputs, and resulting massing can be assessed within a shared representation.

1.4 Research Questions

The research questions follow from the need to connect contextual representation, regulatory conditioning, generative variation, and deterministic evaluation within a shared modelling setup. They ask how parcel-scale urban massing can be represented computationally, how selected regulatory targets can guide model outputs, how variation can be introduced and assessed, and how the resulting outputs can serve as exploratory artefacts. The research is structured around one main question and four related sub-questions.

How can a machine-learning-based generative modelling approach be designed, trained, and evaluated to produce parcel-scale urban massing that responds to the surrounding urban fabric and adjusts to regulatory targets for GRZ, GFZ, and height, and how can the approach be tested to determine whether distinct massing options can be produced under defined constraints?

This question frames the thesis as an exploration of generative modelling for planning reasoning, without assuming that such models should replace existing tools or automate planning decisions.

The main question is pursued through four sub-questions:

1. *How can urban form, parcel geometry, and the surrounding urban fabric be translated into a computational representation that is learnable, interpretable, and evaluable?*

   The focus here is on representation as a design decision that determines what the model can learn, how parcel relations are encoded, and how outputs can be assessed.

2. *How can metrics such as ground coverage ratio, floor area ratio, and building height be incorporated into generative models to guide massing outputs?*

   Control is addressed through the model's sensitivity to regulatory target values and their effect on predicted geometry under fixed spatial context.

3. *To what extent can output variation be introduced and assessed while maintaining massing quality and alignment with regulatory targets?*

   Variation is examined by comparing generated massing alternatives in terms of formal differences, contextual fit, regulatory target alignment, and sampling stability.

4. *How can model outputs be positioned as exploratory artefacts for structured comparison within a conceptual early-stage planning workflow?*

   The final question addresses the conceptual role of model outputs in early-stage planning, examining how they can support comparison and reflection on form, density, contextual fit, and regulatory target alignment, while leaving design judgement and planning decisions to human actors.

These sub-questions structure the thesis from dataset construction and spatial representation to the iterative development and examination of different model architectures across the three experimental phases, with a focus on the trade-offs between control, contextual fit, and variation.

1.5 Scope and Limits

Focusing on parcel-scale building massing in Berlin, the thesis uses voxel-based representations derived from LoD1 building masses to encode urban form and quantitative metrics such as ground coverage ratio, floor area ratio, and maximum building height. It examines whether trained models can reproduce observed massing patterns, steer model outputs through selected regulatory targets, and test controlled variation, while reduced resolution limits geometric precision and architectural detail. Accordingly, the thesis does not evaluate model outputs as architecturally desirable, legally admissible, economically feasible, or socially and ecologically appropriate proposals, nor do the findings establish whether the behaviour or performance of the models developed here transfers to other urban contexts.

The models are developed as analytical and exploratory instruments for structured comparison in early-stage planning and design reasoning, not as production-ready planning tools or substitutes for professional judgement. This framing assumes that existing urban form contains learnable regularities, that regulatory metrics capture aspects of density and volume that can be translated into computational inputs, and that learned models can support design inquiry as long as their representational limits remain explicit. Within this scope, the thesis investigates how urban form, regulatory metrics, and generative modelling can be brought into a shared computational representation.

1.6 Contributions

The thesis contributes to urban planning, computational design, and generative modelling by linking regulatory metrics with context-aware massing modelling. Conceptually, it treats such metrics not only as evaluation criteria but also as model inputs and training targets that steer massing outputs in relation to surrounding urban form. Methodologically, it combines a research-through-design approach with comparative experimentation to make trade-offs between contextual fit, regulatory target control, and output variation explicit (Frayling, 1993; Zimmerman et al., 2007).

On a technical level, the thesis develops and compares three parcel-scale urban massing models, comprising an unconditional model receiving only the surrounding urban context of a target parcel, a deterministic model additionally guided by regulatory targets, and a stochastic variant designed to test controlled sampling under fixed spatial and regulatory conditions. All three models are trained on the same dataset and compared through shared quantitative metrics, documented inference settings, and qualitative analysis of representative outputs, enabling comparison under stable conditions. The central technical result is that deterministic conditioning can produce consistent, steerable, and target-aligned massing within the dataset constructed for this thesis, whereas controlled stochastic variation remains unresolved in the present setup.

From a design perspective, the contribution lies less in producing open-ended design solutions than in making regulation-aware model behaviour inspectable under defined inputs. This includes examining where conditioning performs reliably, where alternative outputs under fixed conditions remain limited, and how learned massing models may complement predefined parametric design spaces in context-sensitive infill situations, while remaining bounded by the representational and empirical scope of the thesis.

1.7 Structure of the Thesis

The thesis moves from conceptual and historical framing to the research design, then to dataset construction and the development and evaluation of the phase-specific machine-learning models, before turning to discussion and concluding synthesis.

Chapter 2 establishes the foundations and related work of the thesis. It reconstructs the urban planning and legal context in which regulation and spatial judgement are linked, reviews relevant machine-learning-supported and generative approaches in urban design, and distils the persistent gaps to which the thesis responds.

Chapter 3 explains the research-through-design approach and the phased logic of the thesis. It shows how the research questions are translated into successive modelling phases and defines the conventions of comparability, reproducibility, and claim-making that structure the experiments.

Chapter 4 defines the urban context and data foundation of the subsequent experiments. It describes Berlin's urban diversity, the selected data sources and metric definitions, the voxel-based spatial representation and processing workflow, and the dataset characteristics and limitations that shape what the models can learn.

Chapter 5 documents the iterative research process, methods, model evolution, and results. After establishing the shared modelling setup and evaluation protocol, it presents the three modelling phases and the cross-phase comparative evaluation. It traces how the sequence moves from context-only prediction to explicit regulatory conditioning and finally to testing controlled stochastic sampling, and how these approaches perform in relation to overlap quality, target steering, and variation.

Chapter 6 provides the discussion and interpretation. It analyses the progression from Phase I to Phase III, examines questions of representation, normativity, responsibility, and design agency, and positions the developed approach within a conceptual early-stage planning workflow.

Chapter 7 concludes the thesis through key findings, limitations, outlook, and a final reflection. It clarifies the validity and limits of the approach and returns to the relation between computational abstraction and planning judgement.

Across these chapters, the thesis develops a comparable parcel-scale modelling sequence under defined conditions within a case-study setting. By holding representation, dataset, and evaluation conventions consistent across phases, it creates an experimental basis for comparing different modelling logics and conceptually positioning their potential role in early-stage planning.

2 Foundations and Related Work

This chapter reviews how urban planning has been shaped by regulatory abstraction, how digital and parametric tools have translated planning problems into computational procedures, and how machine-learning-based generative models shift attention toward learned relations between data, context, and form. It identifies the conceptual and methodological gap between rule-based planning instruments, parametric feasibility tools, and machine-learning-based massing modelling.

2.1 Urban Planning Context and Legal Framework

This section situates the thesis within planning traditions concerned with administrative legibility, modernist planning, and systems thinking, before turning to digital planning tools and the German legal framework governing contextual admissibility and development intensity.

2.1.1 Spatial Order and the Critique of Planning Rationality

Contemporary planning tools can be read against a broader historical lineage in which the ordering of space has been linked to administrative legibility and state intervention (Scott, 1998). Examples range from ancient grid cities and the Roman *castrum* to Renaissance ideal cities, baroque capitals, and Haussmannian Paris, where geometry and visibility served to make urban space readable, governable, and representationally coherent (Battini, 2018; Magli, 2007; Rykwert, 1976; Harvey, 2003; Scott, 1998).

Against the background of nineteenth-century industrialisation and rapid urban growth, modernist planning gave this administrative ordering new force by recasting urban development as a technical problem of hygiene, infrastructure, circulation, and density (Bentlin, 2019; Scott, 1998). It treated the city as a rational system, using tools such as the master plan to standardise, control, and direct top-down urban expansion. By the 1920s, this ambition was visible in Le Corbusier's unbuilt *Plan Voisin* for Paris and later reappeared in projects such as Brasília and Chandigarh, where related principles took built form through monumental axes, functional separation, and geometrically ordered urban space (Le Corbusier, 1925/1987; Scott, 1998; Kitchin, 2014).

Jane Jacobs' *The Death and Life of Great American Cities* (1961) remains a central critique of this rationalising planning logic. For Jacobs, urban vitality emerges from the complexity of everyday city life, where mixed uses, short blocks, and informal neighbourhood practices support social interaction. Her street-level perspective challenged the modernist bird's-eye view by arguing that public order depends less on institutional control than on everyday forms of voluntary surveillance, captured in her notion of "eyes on the street" (Jacobs, 1961/1993; Scott, 1998). Christopher Alexander advanced a related critique by arguing that rigid functional separation reduces the city to an artificial tree-like hierarchy, while real urban life is structured through overlapping relations. Together, Jacobs and Alexander point toward relational and adaptive design approaches and expose the limits of top-down order when local knowledge and informal urban practices are suppressed (Alexander, 1965; Jacobs, 1961/1993; Scott, 1998; Steenson, 2017).

2.1.2 Systems Thinking and Generative Design Logics

The need to engage urban complexity was taken up by systems thinking and cybernetic design theory in the 1960s, as architects and theorists reconceived the city as an adaptive system shaped by information exchange, local interaction, and feedback (Pask, 1969, 1976; Scott, 1998). Against growing scepticism toward modernist and technocratic forms of planning, design increasingly shifted from fixed outcomes to the processes through which spatial organisation is generated. Form was understood less as a predetermined object than as an emergent result of interacting constraints, establishing a conceptual basis for later computational approaches (Harvey, 2006; Pask, 1969; Steenson, 2017).

Christopher Alexander articulated this shift in *Notes on the Synthesis of Form* (1964), describing design problems as networks of interdependent requirements whose coherence emerges through constraint resolution. In *A Pattern Language* (1977) and *The Timeless Way of Building* (1979), this logic became a system of reusable spatial patterns operating across scales, through which coherent environments could emerge from local decisions (Alexander, 1964, 1977, 1979; Salingaros, 2000; Steenson, 2017).

A comparable process-oriented logic appears in the work of Cedric Price, whose Fun Palace (1961–1964) and Potteries Thinkbelt (1966) treated architecture as a temporal infrastructure shaped by user interaction, communication, and feedback. Drawing on cybernetics and game theory, these projects functioned less as fixed buildings than as adaptive spatial programmes whose organisation could respond to changing events and behaviours (Mathews, 2006; Pask, 1969; Steenson, 2017). Frei Otto approached related questions through physical form-finding, using soap films, hanging chains, and material forces to generate structural configurations that anticipated later ideas of material computation (Leach, 2014; Schumacher, 2009).

Together, these approaches reframed design as a mediating practice in which conditions are established, outcomes observed, and results progressively refined. The designer no longer determines form directly, but coordinates relationships between different actors and systems, corresponding to Pask's understanding of the architect as a system designer (Pask, 1969). Yet pattern languages, cybernetic approaches, and physical form-finding remained largely manual or analogue. Although they enabled designers to engage with complexity, they could not yet translate it into scalable data-driven computation. This limitation points toward later computational approaches in architecture and urban planning, in which generative processes could be implemented and assessed through digital models.

2.1.3 The Digital Turn

Computer-aided design (CAD) transformed architectural and urban planning by introducing computational environments for structuring and managing spatial relations. While early CAD systems of the 1980s primarily accelerated drafting through digital geometry, they retained the representational logic of manual drawing (Davis, 2013). Building Information Modelling (BIM) expanded this development by linking digital geometry to relational data structures containing material, cost, and performance information, enabling interdisciplinary coordination and automated consistency checking (Eastman, 1999). Parametric design extended this logic further by defining form through explicit dependencies and adjustable parameters, allowing regulatory metrics to guide iterative workflows and the rapid comparison of compliant massing variants (Davis, 2013; Schumacher, 2009; Gerber & Lin, 2013; Turrin et al., 2011).

Despite their increased flexibility and speed, these systems are deductive. Whether based on static CAD geometry, semantic BIM databases, or parametric graphs, spatial relationships must still be specified in advance by the designer and executed within authored rules and constraints. Such systems lack trainable representations and cannot infer spatial structure beyond what has been manually encoded from precedent or context (Davis, 2013; Goodfellow et al., 2016). When regulatory metrics function primarily as standardised limits, digital modelling reproduces the abstraction logic associated with earlier planning paradigms (Kitchin, 2014). This deductive closure forms the computational baseline against which contemporary AI-supported tools and generative models must be evaluated, particularly in planning contexts where admissibility depends on translating contextual judgement into operational parameters.

2.1.4 German Planning Law and Contextual Admissibility

In Germany, urban planning is formalised through legal instruments that structure spatial planning and development control. The Baugesetzbuch (BauGB; Federal Building Code) provides the statutory basis, the Baunutzungsverordnung (BauNVO; Land Use Ordinance) operationalises land-use regulation through use categories and quantitative measures, and the state building codes (*Landesbauordnungen*), often derived from the Musterbauordnung (Model Building Code), define technical construction requirements (BauGB, 2026; BauNVO, 2023; MBO, 2023).

Within this framework, § 34 BauGB governs admissibility in built-up areas where it is not conclusively determined by a binding development plan. Among other requirements, a proposal must fit the character of its immediate surroundings with regard to type and intensity of land use (*Art und Maß der baulichen Nutzung*), building arrangement (*Bauweise*), and the plot area to be built over (*überbaubare Grundstücksfläche*) (BauGB, 2026). The spatial extent of these surroundings is determined case by case, reaching as far as the proposal may affect its surroundings and as far as those surroundings in turn influence the site, and it is delimited separately for each assessment criterion (BVerwG, 2014). When assessing development intensity, externally perceptible characteristics such as absolute footprint, storey count, building height, and, in open development, the relationship between the building and surrounding open space are considered together (BVerwG, 2016).

For this thesis, § 34 BauGB serves as a conceptual reference for context-dependent comparison, while GRZ and GFZ serve as reproducible indicators of density and built extent. To create a contextually consistent training subset, the dataset filter described in Section 4.4.1 compares target parcels with two edge-adjacent neighbours using GRZ and GFZ tolerances. This adjacency-and-similarity rule is a data-selection procedure and does not reproduce planning judgement. It makes selected aspects of density, built extent, and volumetric intensity comparable but excludes other spatial relations, legal criteria, and interpretive considerations. Formalising such measures in digital planning systems may privilege measurable conformity over situated assessment (Kitchin, 2014; Scott, 1998; Cortiços et al., 2025; Gerber & Lin, 2013).

2.2 AI in Urban Design: State of the Art

Building on this translation of planning rules into measurable and computationally tractable criteria, the chapter turns to modern parametric planning platforms and then to the machine-learning architectures most relevant to this thesis.

2.2.1 Emerging AI-Supported Tools

In recent years, professional planning platforms such as Autodesk Forma, Spacio, and Parametric's Hektar have positioned themselves as tools for the automated generation of high-performing, regulation-compliant design variants in early-stage urban planning (Cortiços et al., 2025; Forsberg, 2025). Although often marketed as AI-driven or generative, these tools are largely parametric and use "generative" mainly to describe the production of multiple design variants, not data-driven generation based on learned representations, as contrasted in Figure 2.1 (Kingma & Welling, 2013; Goodfellow et al., 2014; Ho et al., 2020; Turrin et al., 2011; Davis, 2013; Batty, 2018).

Nevertheless, they mark a notable shift from traditional desktop-based CAD software toward web-based, interactive planning environments. Spatial elements are modelled as adjustable structures whose relationships update through dependency graphs when parameters change, while preconfigured models and slider-based interfaces allow users to generate variants and compare them against regulatory and environmental metrics such as solar exposure, daylight, and energy demand (Davis, 2013; Cortiços et al., 2025). As these adjustable variables define high-dimensional search spaces, computed metrics are often translated into objective functions and explored through heuristic methods to balance competing performance and compliance goals (Deb et al., 2002; Gerber & Lin, 2013; Turrin et al., 2011). More recently, machine learning has been integrated into such workflows for tasks such as performance prediction or variant selection, while the underlying design spaces are defined by human-authored parameters and constraints (Davis, 2013; Kitchin, 2014).

These platforms are strongest in large-scale development tasks where context can be simplified and regulatory values can be translated into algorithmic constraints (Gerber & Lin, 2013; Turrin et al., 2011; Cortiços et al., 2025; Forsberg, 2025). Their capabilities are more limited in context-sensitive infill situations, where admissibility depends on heterogeneous surroundings and interpretive judgement that cannot be fully scripted in advance (Davis, 2013; BauGB, 2026).

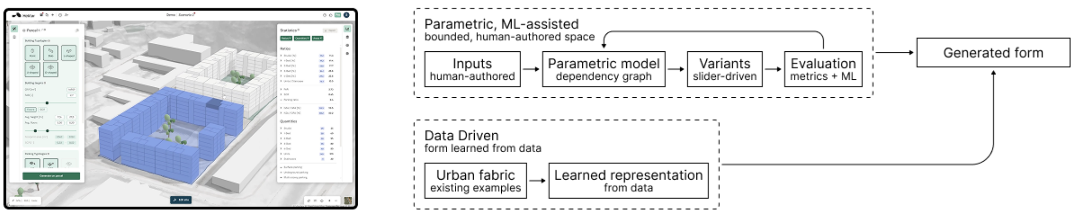

Figure 2.1: Schematic comparison of parametric and data-driven workflows

Screenshot source: Hektar (2026), *Generate housing volumes in 30 seconds*, YouTube, accessed 28 June 2026.

2.2.2 Foundational Architectures

The following sections introduce selected machine-learning architectures that underpin the models developed in this thesis and have also been used in related work on architecture, urban planning, and spatial generation.

Convolutional neural networks (CNNs) learn hierarchical representations from regularly gridded data, such as images, feature maps, or voxel grids, through convolutional operations, nonlinear activations, and, where used, pooling or downsampling (LeCun et al., 2015; Goodfellow et al., 2016). Shared filters detect local patterns across the input, while successive layers aggregate them into increasingly abstract representations (Figure 2.2). In generative systems, CNNs can serve as encoders, decoders, feature extractors, or discriminators, making them relevant for image-like, occupancy-field, and voxel-based representations in which spatial structures are captured through local neighbourhood patterns (Isola et al., 2017; Li et al., 2019; Nauata et al., 2020).

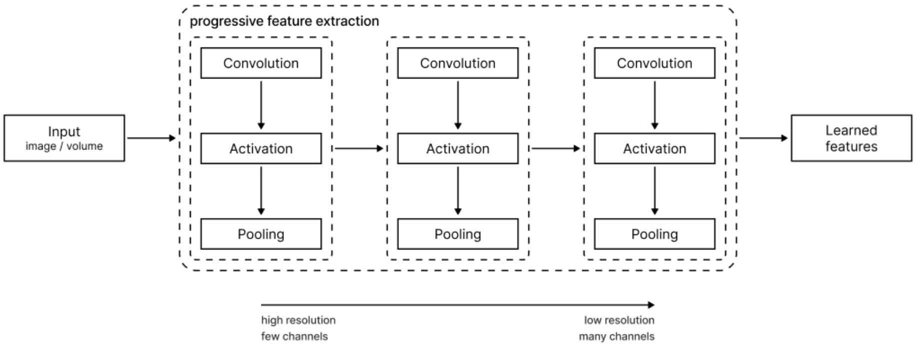

Figure 2.2: CNN for Spatial Feature Extraction

The **U-Net** is a CNN-based encoder–decoder architecture for dense prediction tasks in which the output is spatially aligned with the input (Ronneberger et al., 2015; Isola et al., 2017). A contracting path aggregates broader spatial context, while an expanding path reconstructs spatial detail. Skip connections transfer high-resolution features from encoder to decoder, preserving fine-grained boundaries that would otherwise be weakened during downsampling (Figure 2.3). This structure makes U-Nets suitable for tasks such as image-to-image translation, segmentation, and generative pipelines requiring structured outputs, while also commonly serving as denoising backbones in diffusion models (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022).

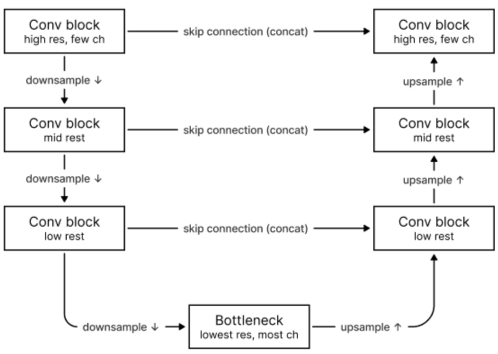

Figure 2.3: U-Net Encoder–Decoder with Skip Connections

Variational autoencoders (VAEs) encode input data into a structured latent space from which samples can be reconstructed or generated (Kingma & Welling, 2013). An encoder maps data to a latent distribution, while a decoder reconstructs samples from it through a latent bottleneck (Figure 2.4), with Kullback–Leibler regularisation supporting interpolation and sampling. VAEs often serve in generative pipelines as compression modules that map high-dimensional data into lower-dimensional representations where generation becomes more efficient, introducing trade-offs between efficiency and geometric precision, as fine details may be attenuated during encoding (Rombach et al., 2022).

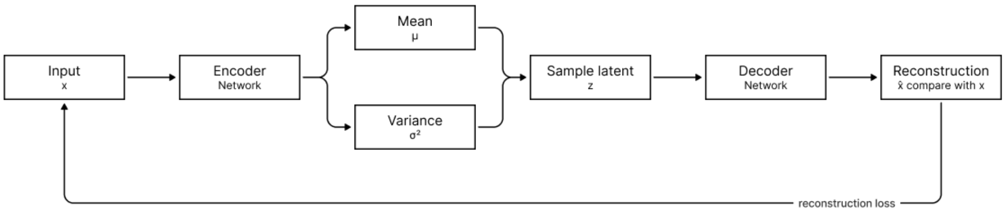

Figure 2.4: Variational Autoencoder with Latent Bottleneck

Diffusion models generate data through a sequence of forward noising and reverse denoising steps (Figure 2.5) that transform an initial random state into an output (Ho et al., 2020; Dhariwal & Nichol, 2021). Conditioning can be introduced by providing additional inputs during this denoising process, with classifier-free guidance offering one common way to modulate the influence of these inputs (Ho & Salimans, 2022). When efficiency is a concern, sampling can be accelerated through approaches such as progressive distillation, while latent diffusion performs denoising in a compressed latent space, typically learned by a VAE, reducing computational cost while maintaining output quality. Its performance, however, depends strongly on the stability of the learned latent representation and the effectiveness of the conditioning mechanism (Salimans & Ho, 2022; Rombach et al., 2022).

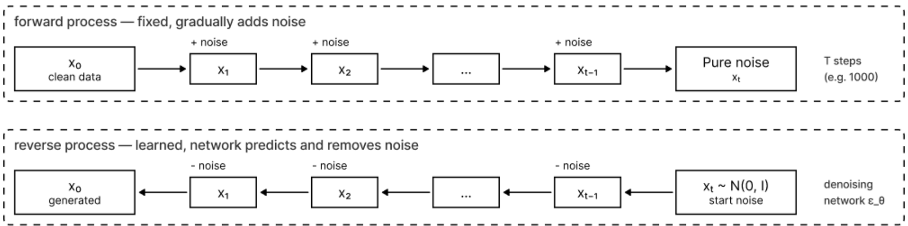

Figure 2.5: Diffusion Denoising Process

Although less central to this thesis, several other approaches exemplify the broader range of methods used in urban and architectural generation tasks. **Generative adversarial networks** (GANs) can produce sharp outputs, but adversarial training is inherently unstable, and the objective often prioritises visual plausibility without enforcing geometric correctness (Goodfellow et al., 2014; Salimans et al., 2016; Nauata et al., 2020). **Graph neural networks** (GNNs) operate on relational structures, making them useful for layout generation but less suited to parcel-scale massing (Nauata et al., 2020). **Diffusion-based semantic occupancy models**, such as UrbanDiffusion, generate urban scenes under bird's-eye-view semantic conditioning, but focus on scene-level occupancy and semantic structure instead of regulatory massing at parcel scale (Zhang et al., 2024).

2.2.3 Generative Approaches in Architecture and Urban Design

Generative machine-learning approaches for architecture, urban planning, and general 3D generation can be grouped by the spatial scale at which they operate.

Early object-level work in general 3D generation established 3D shape learning on isolated geometries using voxel-based models such as 3D-GAN (Wu et al., 2016) and point-based representations for sparse data (Qi et al., 2017). Later approaches extended toward latent-space, point–voxel, and diffusion-based methods (Achlioptas et al., 2018; Zhou et al., 2021; Qi et al., 2023), while recent systems such as Point-E, MeshAnything, and Magic3D continue to focus on generic 3D shape and asset generation without architecture-specific constraints (Nichol et al., 2022; Chen et al., 2024; Lin et al., 2023). Even architecture-oriented works such as ArchComplete remain limited to building-scale synthesis and do not account for parcel-level relations or surrounding urban form (Rasoulzadeh et al., 2025).

A second line of work focuses on architectural layouts, where generation is conditioned on relational structure alongside class labels. House-GAN generates floor plans from program graphs encoding adjacency and functional relationships (Nauata et al., 2020; Nauata et al., 2021), while related generative layout approaches use GAN-based, vector-based, raster-based, and diffusion-based representations (Li et al., 2019; Shabani et al., 2022). These systems align more closely with constraint-based design logics, yet focus on building-internal relations without integrating site geometry, urban context, and regulatory parameters (He et al., 2025).

Urban-scale approaches incorporate spatial context through semantic maps or geographic priors, including OpenStreetMap-derived inputs for scene synthesis (Shang et al., 2024), coarse-to-fine sparse voxel representations in XCube (Ren et al., 2024), and bird's-eye-view maps in UrbanDiffusion (Zhang et al., 2024). They show that large-scale spatial distributions can be learned under structured conditioning, but their inputs encode geometric or semantic context without supporting deterministic extraction of regulatory metrics such as GRZ, GFZ, or building height.

Across these approaches, generative modelling expands from isolated objects through relational layouts to urban-scale scene generation (Figure 2.6). Conditioning becomes increasingly important, with object-level systems emphasising shape representation, layout models encoding internal spatial relations, and urban-scale models relying on semantic or geographic context.

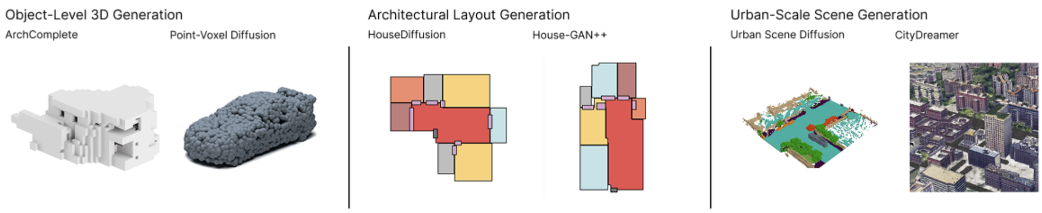

Figure 2.6: Generative Model Families by Spatial Scale

Source images: ArchComplete and Point-Voxel Diffusion; HouseDiffusion and House-GAN++; UrbanDiffusion and CityDreamer (Rasoulzadeh et al., 2025; Zhou et al., 2021; Shabani et al., 2022; Nauata et al., 2021; Zhang et al., 2024; Xie et al., 2024).

2.3 Synthesis and Positioning

From modernist planning to parametric workflows and contemporary machine-learning-based generative approaches, complex urban situations are repeatedly translated into typologies, variables, ratios, and optimisation targets, enabling comparability and administrative tractability while reducing the relational and contextual basis of admissibility (Scott, 1998; Kitchin, 2014). This creates a tension in provisions such as § 34 BauGB, where admissibility is defined not by isolated thresholds but by comparison with the surrounding built environment (BauGB, 2026).

This tension persists across computational approaches, as parametric systems enable systematic variation within authored rule sets, optimisation-based platforms treat regulatory parameters mainly as constraints, and contemporary machine-learning-based generative models introduce distributional learning while often remaining weakly connected to local urban context and deterministic verification. These limitations define three interrelated gaps that position this thesis.

1. *Regulation beyond thresholds.*

   German planning law, especially § 34 BauGB, assesses admissibility by whether a proposal fits the surrounding area in terms of the type and intensity of land use, building arrangement, and the plot area to be built over (BauGB, 2026). Metrics such as GRZ and GFZ derive from the BauNVO and serve here as indicators of density and built extent (BauNVO, 2023). Computational systems often treat these parameters as universal, decontextualised constraints, reproducing a simplification logic in which quantitative abstraction suppresses relational nuance.

2. *Typology without situatedness.*

   Object-level systems train on isolated geometries, layout models encode internal adjacency, and urban-scale pipelines condition on semantic maps, yet few approaches integrate parcel geometry, neighbouring massing, and regulatory data within a unified setup. Current models can thus reproduce statistical regularities of form, but rarely capture the relational dependencies through which context-sensitive urban coherence emerges.

3. *Plausibility without checkability.*

   Planning decisions require reliable and deterministic computation of coverage ratios, floor areas, heights, and geometric relations. Representations optimised for perceptual realism do not meet this requirement, as statistical resemblance does not ensure agreement with regulatory metrics. Without direct metric computation and fixed evaluation protocols, generative models cannot reliably interface with planning applications.

Positioned at the intersection of these gaps, the thesis represents urban form as discrete voxel geometry to enable deterministic measurement of density, height, and volumetric fit. GRZ, GFZ, and height are introduced as conditioning variables that serve as measurable proxies for selected aspects of parcel-scale massing and remain comparable across cases, without being treated as exhaustive legal criteria. The aim is not to automate planning judgement, but to test, as a proof of concept, how contextual form relations and selected regulatory indicators can be combined within an evaluable generative modelling workflow.

3 Research Design and Methodology

As questions about representation, control, and the sampling of variants cannot be resolved through analysis alone, the thesis adopts a research-through-design approach. Following Frayling's (1993) distinction between research into, for, and through design, it treats the phased development of successive machine-learning models as a mode of producing knowledge. In this context, experimental model development becomes the primary mode of inquiry, with design intent, changing model behaviour, and reflective interpretation informing one another throughout the research process. Each experiment addresses a specific methodological question, and its outputs show how representation, conditioning, and sampling affect modelled urban form (Gaver, 2012; Zimmerman et al., 2007).

This approach aligns with the context of the interdisciplinary *Design & Computation* programme at Technische Universität Berlin (TU Berlin) and Universität der Künste Berlin (UdK Berlin), in which technological development, design practice, and critical reflection on societal questions are examined together (Lang, n.d.). Technical performance is treated as one analytical dimension alongside interpretive comparison, representational critique, and reflection on the role of generative models in urban planning.

3.1 From Questions to Phases

The stated research questions address distinct but connected aspects of how urban form can be represented computationally, how regulatory parameters can guide modelled massing, and whether output variation can be introduced without undermining contextual fit and regulatory target alignment. As representation, control, and variation form a methodological sequence, treating them within a single model would obscure their respective assumptions and trade-offs. This motivates the phased structure adopted in the thesis, in which different modelling logics are examined separately.

Phase I establishes a baseline for context-based massing prediction using only the fixed spatial representation and surrounding urban context, asking how far parcel-scale form can be inferred when no explicit regulatory targets are provided. Phase II introduces target-based control by incorporating GRZ, GFZ, and building height as conditioning variables, allowing their influence on spatial outcomes and target alignment to be examined. Phase III tests stochastic variation by introducing sampling under identical contextual and regulatory inputs, shifting the focus from deterministic control to the stability of variation under fixed conditions. Terminologically, Phases I and II are deterministic prediction tasks, Phase III Stage A (VAE) is reconstruction, and Stage B (diffusion) is sampling. The term generation is reserved for the broader design-oriented framing of producing and exploring model-based massing outputs, not for the technical operation of each phase.

Each phase adds one further source of modelling complexity while keeping the preceding conditions available for comparison. This makes it possible to attribute changes in model behaviour primarily to the modelling approach itself and not to changes in data, representation, or evaluation setup. Plausibility, control, and variation are analytically distinct while still forming a cumulative modelling sequence.

3.2 Experimental Conventions and Reproducibility

As machine-learning models are sensitive to variation in training and evaluation setup, which makes controlled conditions necessary, all experiments use standardised configuration templates, fixed dataset splits, and identical random seeds. Training configurations, evaluation procedures, and model outputs are documented so experiments can be retraced and compared across phases.

Evaluation combines quantitative metrics with comparative qualitative analysis. The quantitative component assesses geometric agreement between model outputs and reference data. Regulatory metrics, including GRZ, GFZ, and building height, are computed from modelled massing independently of the training objective, enabling regulatory characteristics to be examined separately from geometric overlap. This separation matters, as high geometric overlap does not necessarily imply regulatory alignment, while regulatory proximity may be achieved by geometrically different massing configurations.

Qualitative analysis interprets model outputs in relation to formal aspects such as street continuity, setback logic, consistency of building scale and spacing, and typological plausibility. For each phase, held-out test samples are inspected through 2D projection views and 3D context renderings, and selected examples illustrate recurrent strengths and characteristic failure modes.

Together, quantitative metrics and qualitative review support the interpretation of model behaviour across phases. Metrics indicate where modelled massing aligns with reference geometry or regulatory targets, while visual inspection helps show how such alignment is achieved spatially and where it breaks down. Similar numerical scores may correspond to different formal behaviours, and visually plausible outputs may still fail in terms of measurable density, height, or volumetric fit. The combined evaluation approach is used diagnostically, not to rank outputs by a single criterion or as evidence of architectural quality or planning admissibility.

3.3 Limits of the Research Design

As the thesis is grounded in Berlin as a single-city case study, its findings describe model behaviour within this specific urban context, the resulting dataset, and the associated regulatory setting. They do not claim transferability to other cities or planning regimes without further testing, nor do they extend to the general applicability of generative artificial intelligence in planning. The evaluation setup, in which existing LoD1 building volumes serve as ground truth, tests whether models can reproduce and steer observed massing patterns, not whether model outputs would be admissible or desirable in a forward-looking planning process.

Within the research-through-design approach adopted here, negative and partial results are also considered productive. When a single experiment or an entire phase does not meet its stated objective, the conditions under which it fails become part of the analysis. The phased structure is designed to make such boundaries visible without optimising toward a single best-performing system.

4 Data and Urban Context

To address the research questions of this thesis, a custom dataset was constructed to translate Berlin's urban fabric into a standardised computational representation for parcel-scale massing modelling. The workflow integrates CityGML LoD1 building volumes, cadastral parcel geometries and boundaries, the street network, and derived regulatory metrics for learning, conditioning, and evaluation. Developed with scalability in mind, the workflow could in principle be transferred to other cities or regions where comparable building, parcel, and infrastructure data are available, while the analysis itself remains grounded in Berlin.

The chosen computational representation and final dataset composition determine which spatial relations become measurable, which planning-relevant features can guide the models, and which urban conditions are simplified, underrepresented, or excluded altogether (Kitchin, 2014). The following sections explain why Berlin was selected as the geographical focus of the thesis, describe the source data and their transformation into the dataset, and discuss the abstractions introduced by this process. The resulting dataset provides a representation that is fixed across the three modelling phases developed in Chapter 5.

4.1 Urban Form Diversity and Data Basis

Berlin, serving as the geographical setting of this thesis, combines a diverse urban fabric, a coherent regulatory structure, and an open geospatial data environment suitable for developing and testing a machine-learning-based approach to urban massing modelling. The Geoportal Berlin (FIS-Broker) provides open geospatial data in a common coordinate reference system (EPSG 25833, UTM Zone 33N), including CityGML Level of Detail 1 (LoD1) building geometries, cadastral parcel layers, and infrastructure data maintained by the Senate Department for Urban Development and Housing.

Successive phases of urban development have produced a heterogeneous urban fabric in which spatial logics from different political regimes and planning paradigms coexist (Bentlin, 2019). This diversity is documented in the Berlin Environmental Atlas (*Umweltatlas Berlin*), whose 2020 layer *Stadtstruktur–Flächentypen differenziert* classifies block-level areas into 52 surface types (*Flächentypen*) grouped into six broader categories (Senatsverwaltung für Stadtentwicklung, Bauen und Wohnen, 2020). Within the residential categories, 17 surface types are distinguished, for which Table 4.1 reports the area-weighted mean values and 5 %–95 % percentile ranges for ground coverage ratio (GRZ), floor area ratio (GFZ), and storey count, together with one non-residential reference category (*Kerngebiet*).

Table 4.1 shows that Berlin's residential surface types span a wide quantitative range. GRZ varies from 0.1 in low-density allotment areas (Type 59) to 0.6 in dense *Gründerzeit* perimeter blocks (Type 1), GFZ reaches 3.1 in high-density cases, and mean storey counts range from 1.0 to 7.0. Large-panel housing estates (*Großsiedlungen*, Type 9) illustrate this differentiation by combining low ground coverage with comparatively high floor-area ratios and pronounced vertical extent, while variation within individual types remains substantial, as Type 2 shows with GFZ values between 1.5 and 3.4 across the 5th to 95th percentiles.

Type	Description	GRZ	GFZ	Storeys
1	Dense block development, enclosed courtyard (1870s–1918)	0.6 (0.5–0.7)	3.1 (2.4–3.8)	5.0 (4.5–5.6)
2	Closed block development, courtyard (1870s–1918)	0.5 (0.4–0.7)	2.5 (1.5–3.4)	4.8 (3.7–5.6)
3	Closed block development, garden courtyard (1870s–1918)	0.4 (0.3–0.6)	1.6 (0.7–2.4)	3.8 (2.7–4.8)
6	Mixed development, open courtyard, 2–4 storeys	0.3 (0.2–0.5)	0.9 (0.4–1.5)	2.8 (1.9–3.7)
10	Perimeter block with large courtyards (1920s–40s)	0.4 (0.2–0.5)	1.3 (0.6–2.2)	3.7 (2.6–5.0)
72	Parallel slab development (1920s–30s)	0.3 (0.1–0.4)	0.9 (0.3–1.6)	3.3 (2.0–4.3)
7	Cleared perimeter block, post-1945 infill	0.5 (0.3–0.7)	2.0 (1.0–2.9)	4.8 (3.3–6.1)
8	Heterogeneous mixed development, post-1945 infill	0.4 (0.2–0.7)	1.8 (0.7–2.9)	4.6 (2.7–6.3)
11	Free slab development (1950s–70s)	0.2 (0.1–0.3)	0.8 (0.4–1.5)	3.8 (2.3–5.6)
9	Large housing estate / point towers (1960s–90s)	0.2 (0.1–0.4)	1.4 (0.7–2.3)	7.0 (4.1–10.9)
73	Multi-storey housing, 1990s onward	0.3 (0.0–0.6)	1.1 (0.0–2.6)	4.0 (1.0–6.6)
22	Terraced and semi-detached houses with gardens	0.2 (0.1–0.3)	0.3 (0.1–0.7)	1.8 (1.0–2.9)
23	Detached single-family houses with gardens	0.2 (0.1–0.2)	0.2 (0.1–0.4)	1.5 (1.1–2.2)
21	Village mixed development (*Dörfliche Mischbebauung*)	0.2 (0.1–0.3)	0.3 (0.1–0.4)	1.5 (1.1–2.0)
24	Villas and *Stadtvillen*	0.2 (0.1–0.3)	0.4 (0.2–0.7)	2.1 (1.4–3.0)
25	Densification in single-family areas	0.2 (0.1–0.3)	0.5 (0.2–0.8)	2.3 (1.4–3.3)
59	Weekend houses / allotment-like areas	0.1 (0.0–0.2)	0.1 (0.0–0.2)	1.0 (0.0–1.3)
29	Core area (*Kerngebiet*)	0.6 (0.2–1.0)	2.9 (0.5–5.9)	5.0 (1.5–7.8)

Table 4.1: Berlin Urban Structure Metrics by Area Weighting

Once translated into the final dataset, this variation requires the model to learn not only numerical target values, but also the spatial relations through which similar regulatory metrics can take different urban forms.

4.2 Data Sources and Metric Definitions

The dataset integrates building volumes, cadastral parcels, and the street network to represent urban form as a relational configuration of built form, land subdivision, and public street space. These layers provide the basis for the spatial context channels C_0–C_3, introduced in Section 4.3, and define the spatial context from which parcel-scale massing is modelled.

The source for modelling Berlin's building geometry is the official CityGML LoD1 dataset, which represents buildings as extruded footprints with aggregated roof heights while omitting detailed roof shapes, façade articulation, and material attributes (Senatsverwaltung für Stadtentwicklung, Bauen und Wohnen, 2025a; Biljecki et al., 2016). Although more detailed datasets such as LoD2 are available, LoD1 is used because its aggregated height values provide a more consistent basis for GFZ calculation, height-based conditioning, and metric extraction. Incorporating LoD2's explicit roof geometry would introduce additional interpretative decisions and noise, while much of this detail would be lost at the downsampled training resolution of $2.0\,\mathrm{m}$.

Cadastral parcel geometries are sourced from Berlin's ALKIS dataset (*Flurstücke*) (Senatsverwaltung für Stadtentwicklung, Bauen und Wohnen, 2025b) and represent the official legal subdivision of the city. Each sample is centred on a single target parcel as the site for massing prediction. Parcels containing buildings that extend across multiple parcels are excluded to preserve unambiguous parcel-to-sample assignments, since shared property units cannot be reliably inferred from the available data.

As urban development is organised in relation to street structure, street network data are sourced from the *Detailnetz Berlin* dataset (Senatsverwaltung für Stadtentwicklung, Bauen und Wohnen, 2025c) to represent the public street structure within which parcels and buildings are embedded. This layer situates buildings and parcels within the surrounding block and public street structure. In the final dataset, centrelines are rasterised as buffer corridors with a total width of 8.0 m.

Based on these data layers, the thesis derives three quantitative parameters used throughout all experiments. The *Grundflächenzahl* (GRZ, ground coverage ratio) describes the proportion of a parcel covered by building footprints, corresponding to the definition in the BauNVO (2023), and is expressed here as

$$\text{GRZ} = \frac{A_{\text{footprint}}}{A_{\text{parcel}}} \tag{1}$$

where $A_{\text{footprint}}$ is the building footprint area and A_{parcel} the parcel area. In the voxel representation, the binary occupancy tensor $Y \in \{0,1\}^{D\times H\times W}$ is projected onto the horizontal plane to obtain a footprint mask $\mathbf{F} \in \{0,1\}^{H\times W}$, yielding

$$\text{GRZ} \approx \frac{\|\mathbf{F}\|_1 \cdot v^2}{A_{\text{parcel}}}, \quad \mathbf{F}_{h,w} = \bigvee_{d=0}^{D-1} Y_{d,h,w} \tag{2}$$

where $\|\mathbf{F}\|_1$ denotes the number of occupied cells and v^2 the horizontal voxel area.

The *Geschossflächenzahl* (GFZ, floor area ratio) describes total floor area relative to parcel size, corresponding to the definition in the BauNVO (2023), and is expressed here as

$$\text{GFZ} = \frac{\sum_{i=1}^{n_{\text{floors}}} A_{\text{floor},i}}{A_{\text{parcel}}} \tag{3}$$

In voxel form, GFZ is approximated by

$$\text{GFZ} \approx \frac{\|Y\|_1 \cdot v^3}{A_{\text{parcel}} \cdot h_{\text{storey}}} \tag{4}$$

where $\|Y\|_1$ is the number of occupied voxels, v the voxel size, and $h_{\text{storey}} = 3.0\,$m a fixed conversion factor. As LoD1 provides aggregated building heights but no explicit floor plates, GFZ is approximated as a floor-area equivalent and not reconstructed through legal full-storey classification. The fixed value of $3.0\,$m defines the assumed storey-height increment used to convert voxelised volume into a floor-area estimate. This is conservative for typologies such as *Gründerzeit* blocks, which often have higher storey heights, but avoids typology-specific height estimation that would require additional assumptions beyond the available data and the scope of this thesis.

Building height H_{height} is defined as the vertical extent of the building volume. In CityGML LoD1, this corresponds to

$$H_{\text{height}} = z_{\max} - z_{\min} \tag{5}$$

In voxel space, this quantity is estimated by the occupied vertical span

$$H_{\text{height}} = (z_{\max}^{\text{occ}} - z_{\min}^{\text{occ}} + 1) \cdot v = (z_{\text{hi}} - z_{\text{lo}}) \cdot v \tag{6}$$

where z_{lo} denotes the inclusive lower occupied index and z_{hi} the exclusive upper extrusion index under the half-open extrusion rule (Equation 9), so occupied voxels satisfy $z \in [z_{\text{lo}}, z_{\text{hi}})$ and $H_{\text{height}} = (z_{\text{hi}} - z_{\text{lo}}) \cdot v$. For evaluation, height is assigned a tolerance of one vertical voxel step. The resolution-dependent tolerances applied to GRZ, GFZ, and height are defined in Section 5.1.1.

Together, GRZ, GFZ, and height describe selected aspects of parcel-scale massing as comparable numerical descriptors. As discussed in Chapter 2, they do not exhaust planning judgement, but provide a shared target space for evaluation and conditioning. They serve as evaluation metrics across all three phases and as conditioning inputs in Phases II and III. Figure 4.1 illustrates how the three parameters are derived from parcel geometry and voxelised building volume.

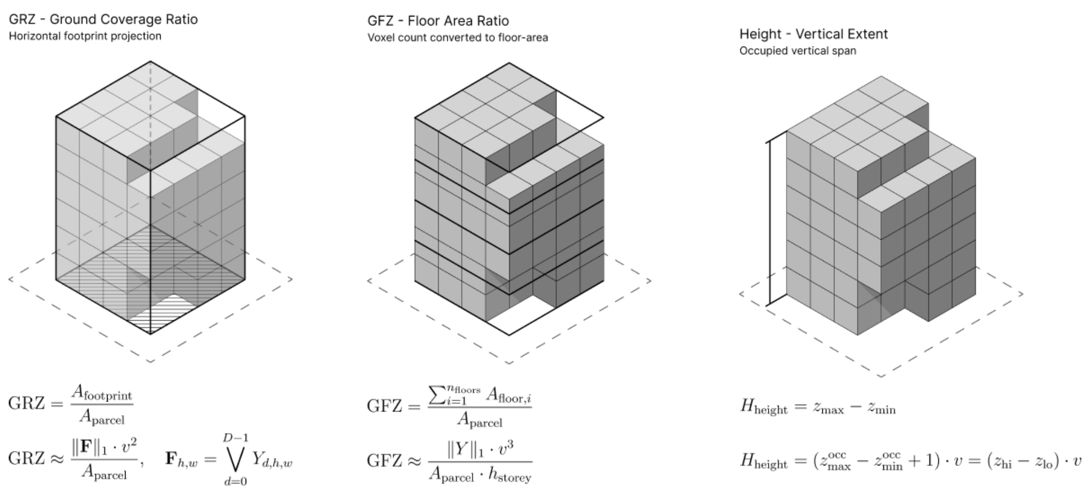

Figure 4.1: Derivation of GRZ, GFZ, and Height Metrics

4.3 Spatial Representation

Urban form is represented as a regular three-dimensional binary occupancy grid used across all phases. Each sample covers a fixed cubic extent of $L = 128\,\text{m}$ per axis in EPSG 25833 and is discretised at resolution $R = 256$, yielding $D = H = W = R$. The corresponding native voxel size is

$$v = \frac{L}{R} = \frac{128\,\text{m}}{256} = 0.5\,\text{m}. \tag{7}$$

Volumes are downsampled to $R_{\text{train}} = 64$ using stride-4 $4 \times 4 \times 4$ average pooling and re-binarised using a threshold of 0.5, yielding a training voxel size of $v_{\text{train}} = 2.0\,\text{m}$. Regulatory metrics are retained in both the native $0.5\,\text{m}$ representation and the downsampled $2.0\,\text{m}$ training grid. Native values $\mathbf{c}^{0.5\,\text{m}}$ are used for conditioning and reporting, while downsampled references $\mathbf{t}^{2.0\,\text{m}}$ are used to quantify the resolution shift introduced by training-grid discretisation. On the held-out test split, the mean offset $\mathbf{t}^{2.0\,\text{m}} - \mathbf{c}^{0.5\,\text{m}}$ is $+0.0067$ for GRZ, $+0.0467$ for GFZ, and $+0.56\,\text{m}$ for height, pointing to a small upward shift introduced by the coarser training grid. The tolerance-aware error definitions in Equations 18 and 19 absorb this shift for height and for GRZ at typical parcel sizes. For GFZ, the mean offset remains above the one-voxel tolerance throughout the observed parcel-size range and exceeds it by a factor of 6.5 at the median parcel area, so adjusted GFZ errors retain a resolution-induced component alongside model error. GRZ, GFZ, and building height are derived directly from occupancy counts using Equations 2, 4, and 6, allowing predicted and ground-truth volumes to be evaluated on the same discrete grid without geometric conversion.

Each sample encodes spatial context as a multi-channel tensor $\mathbf{X} \in \mathbb{R}^{C \times D \times H \times W}$, consisting of the target parcel mask (C_0), neighbouring building height context (C_1), parcel-boundary structure of the target and surrounding parcels (C_2), and street mask (C_3). Regulatory metrics are represented separately as a global vector $\mathbf{c} = (\text{GRZ}, \text{GFZ}, H_{\text{height}}) \in \mathbb{R}^3$. This separation keeps spatial context and regulatory targets distinct while providing a shared input format for all model variants. Figure 4.2 illustrates the four spatial input channels that form the context tensor.

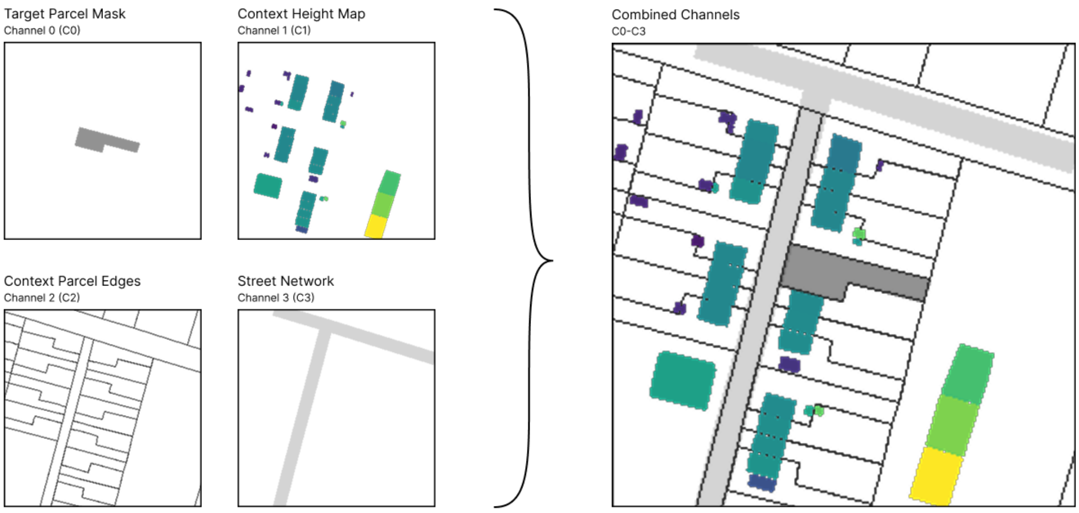

Figure 4.2: Parcel-Centred Context Channels for Model Input

Alternative 3D representations were considered in relation to the thesis' research questions, which require parcel-scale modelling, regulatory conditioning, and deterministic metric extraction within a shared representation. Implicit representations, such as Neural Radiance Fields or 3D Gaussian Splatting, are effective for visual reconstruction and view synthesis, but the quantities required here are not primarily visual surfaces. Footprint area, enclosed volume, height, and derived metrics would first require surface extraction, thresholding, or occupancy conversion, introducing approximation choices that make regulatory metrics less directly traceable to the resulting representation (Poole et al., 2022; Kerbl et al., 2023; Xiang et al., 2025; Liu et al., 2024).

Mesh-based representations provide explicit surface geometry and are common in architectural modelling, but they depend on stable topology, watertightness, and consistent mesh quality when used for volumetric evaluation. These conditions are difficult to guarantee for parcel-scale modelling across heterogeneous urban samples, especially when the source data consist of simplified LoD1 extrusions with limited vertex-level semantics. Deriving footprint area or enclosed volume from resulting meshes may require additional geometric validation and repair steps, while learning directly on meshes would introduce irregular connectivity that is less compatible with the convolutional architectures used in this thesis (Biljecki et al., 2016; Liu et al., 2024).

Point-cloud representations are effective for sparse geometric data and can encode surface samples without imposing a regular grid, but they do not directly define closed occupancy or interior volume. Footprint, volume, GFZ, and height would require reconstruction or inference before evaluation, while the irregular structure of point clouds further complicates direct comparison between modelled and reference massing. Because point clouds and meshes are often converted into regular voxel grids for convolutional processing, this thesis uses the voxel format directly as the primary representation (Qi et al., 2017).

Against this background, voxel grids are adopted as a well-established representation for three-dimensional convolutional models that encode space through explicit binary occupancy. Each cell is either occupied or empty, allowing parcel masks, neighbouring massing, street structure, predicted mass, and target occupancy to be represented on the same discrete grid. GRZ, GFZ, and height can then be computed directly from voxel counts without additional surface extraction or geometric reconstruction. Although the cubic cost of voxel grids limits resolution, this trade-off remains compatible with the LoD1 abstraction and parcel-scale focus of the thesis. Figure 4.3 summarises how the context tensor $\mathbf{X}$, regulatory vector $\mathbf{c}$, predicted occupancy $\hat{Y}$, and target occupancy Y relate to model prediction and metric extraction.

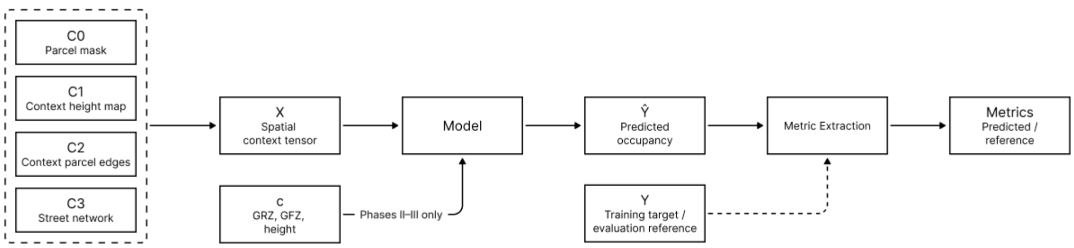

Figure 4.3: Voxel Modelling Workflow from GIS Data to Grid

4.4 Processing Pipeline

The source datasets are transformed into parcel-centred voxel samples through a sequence of operations within a fixed cubic sampling window of $L = 128$ m. Together with Berlin's cadastral structure and the contextual quality filter (Equation 10), this yields a dataset concentrated in small- to medium-sized parcels, consistent with the thesis's focus on infill situations but with implications for model generalisation discussed later in this chapter.

4.4.1 Dataset Construction and Sample Filtering

Parcel-centred candidate samples are constructed from a spatial index of CityGML buildings, represented by footprint polygons with vertical bounds $z_{\min}$ and $z_{\max}$. For each target parcel, a cubic sampling window with side length $L = 128$ m is centred on the parcel centroid, and all intersecting buildings are retrieved. Buildings overlapping the parcel by at least $5.0\,\mathrm{m}^2$ are classified as target parts, while smaller overlaps are ignored to avoid minor artefacts. Direct edge-adjacent neighbours are ranked by shared boundary length, and the two neighbours with the longest shared boundaries are retained for contextual comparison. Candidate samples with fewer than two such neighbours or with any building exceeding the maximum vertical extent $H_{\max}$ are excluded.

Contextual information is encoded in the input channels C_0–C_3. These channels are constructed in two dimensions and replicated along the vertical axis, as shown in Figure 4.2. For the neighbouring building height channel C_1, heights are normalised as

$$h_{\text{norm}} = \min\left(1, \frac{z_{\max} - z_{\min}}{H_{\max}}\right) \tag{8}$$

Neighbour footprints are weighted by h_{norm}, with $H_{\max} = L = 128$ m denoting the maximum vertical extent of the cubic sampling window. Where multiple footprints overlap in the raster, the highest height value is retained. Parcel and street geometries are rasterised using supersampling, and street centrelines are buffered by 4.0 m on each side, yielding a total corridor width of 8.0 m.

Building geometry is converted into voxel occupancy by rasterising footprints horizontally and discretising their LoD1 height values vertically. Each footprint is rasterised to a binary mask using 2×2 supersampling and a threshold of 0.5, then extruded between vertical indices z_{lo} and z_{hi},

$$z_{\text{lo}} = \left\lfloor \frac{z^{\text{part}}_{\min} - z^{\text{window}}_{\min}}{v} \right\rfloor, \quad z_{\text{hi}} = \left\lceil \frac{z^{\text{part}}_{\max} - z^{\text{window}}_{\min}}{v} \right\rceil \tag{9}$$

where $z^{\text{window}}_{\min}$ denotes the base elevation of the vertical window. The lower index is included and the upper index is excluded, so buildings occupy all voxel layers from z_{lo} up to but not including z_{hi}. This half-open convention makes the occupied height equal to $(z_{\text{hi}} - z_{\text{lo}}) \cdot v$.

Target and neighbouring building parts are aggregated into occupancy volumes Y and Y_{neigh} of dimensions $D \times H \times W$. During training, neighbourhood information is provided through the input tensor $\mathbf{X}$, while Y_{neigh} is retained for inspection, visualisation, and qualitative evaluation.

Candidate samples are excluded by three size-related filters. First, the mean value of channel C_0 is used as the parcel-window coverage ratio. As C_0 is a binary parcel mask replicated through the voxel grid, its mean corresponds to the fraction of the fixed sampling window occupied by the target parcel. Samples are removed when this fraction exceeds 0.60. Second, the mean value of the binary target footprint mask measures the fraction of the horizontal window covered by the target building footprint, with samples removed above 0.30. Third, the floor-area equivalent derived in Equation 4 must lie between 50 and 20,000 m^2. Together, these filters remove extremely small, extremely large, or window-dominating cases in which the parcel or target building would leave too little surrounding context to construct meaningful input channels and support contextual comparison.

Each remaining candidate sample must pass a final contextual filter, in which the target parcel is compared with the two direct edge-adjacent neighbours sharing the longest parcel boundaries. For both GRZ and GFZ, the target passes this filter only if its value remains within a combined relative and absolute tolerance of each neighbour, defined as

$$|m_{\text{target}} - m_{\text{neighbour}}| \leq \max(\tau_{\text{rel}} \cdot m_{\text{neighbour}}, \tau_{\text{abs}}), \tag{10}$$

where $m \in \{\text{GRZ}, \text{GFZ}\}$, $\tau_{\text{rel}} = 0.20$, and $\tau_{\text{abs}} = 0.05$. These thresholds retain parcels with density values comparable to their two direct edge-adjacent reference neighbours while excluding isolated outliers, thereby prioritising dataset-internal contextual consistency over maximum dataset size.

Figure 4.4 summarises the complete construction and filtering pipeline. Candidate samples that pass all stages are retained in the final dataset. Each retained sample is stored as a compressed NumPy archive (NPZ) containing $\mathbf{X} \in \mathbb{R}^{4 \times D \times H \times W}$, Y, Y_{neigh}, and metadata. These metadata combine parcel and grid identifiers, spatial bounds, derived regulatory metrics, geometric quantities, neighbourhood descriptors, and street-related measures. They are also written as human-readable JSON, enabling filtering, analysis, and conditional training without repeated voxel processing. A representative metadata JSON record is provided in Appendix A.

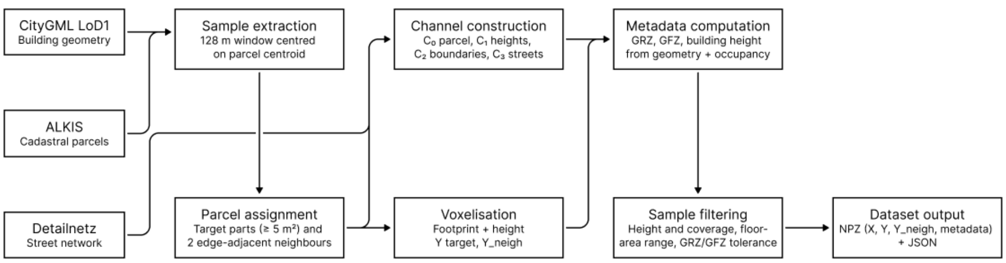

Figure 4.4: Simplified Parcel-Centred Sample Processing Pipeline

4.4.2 Split and Distribution Profile

After processing all cadastral parcels available in the source dataset, only samples that passed the construction and filtering logic described in Section 4.4.1 were retained. To match the thesis focus on infill situations and preserve sufficient surrounding context around each parcel within the fixed 128 m sampling window, Bucket A (100–500 m^2) was randomly sampled to 8,000 cases and Bucket B (500–1,500 m^2) to 5,000 cases using the fixed seed 42, while Bucket C (1,500–5,000 m^2) retains all 164 available larger-parcel cases. The final dataset comprises 13,164 samples. Table 4.2 reports the resulting bucket counts and split assignments. Figure 4.5 visualises their relative composition.

The dataset is partitioned into training (70%), validation (15%), and test (15%) subsets using a fixed manifest generated with seed 42 by shuffling the full sample list once and assigning samples to the three subsets. All model phases and experiments operate on this fixed manifest, ensuring comparability while keeping the test samples unseen during training. The manifest is defined at the parcel-sample level, not by spatial blocks, so neighbouring parcels may occur in different subsets and partially share their fixed 128 m context window. Figure 4.6 shows the spatial distribution across Berlin.

Bucket	Total	Train	Validation	Test
A (100–500 m^2)	8,000	5,578	1,210	1,212
B (500–1,500 m^2)	5,000	3,530	741	729
C (1,500–5,000 m^2)	164	107	24	33
Total	**13,164**	**9,215**	**1,975**	**1,974**

Table 4.2: Dataset Split Counts by Bucket

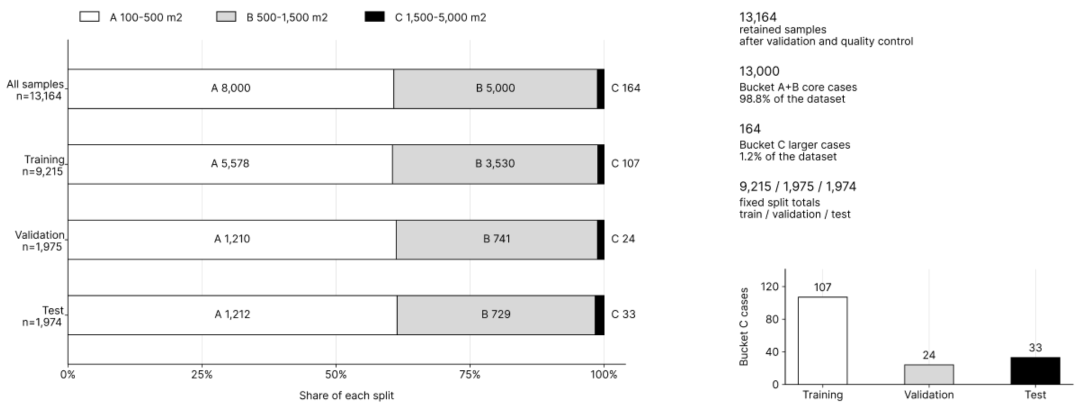

Figure 4.5: Dataset Composition by Parcel-Size Bucket

Table 4.3 summarises the distribution of key regulatory metrics and geometric attributes across all 13,164 samples. GRZ has a median of 0.31 and an interquartile range of 0.23–0.52, suggesting a predominance of moderate-density cases. GFZ is strongly right-skewed, with a median of 0.86 and a mean of 1.81, reflecting a smaller number of high-density multi-storey cases in the upper tail. Building height ranges from 3.0 to 34.5 m, parcel area spans 100–4,900 m^2 with a concentration in Bucket A, and neighbour counts vary from 2 to 228, covering both peripheral and dense inner-city contexts. Figure 4.7 visualises these distributional patterns.

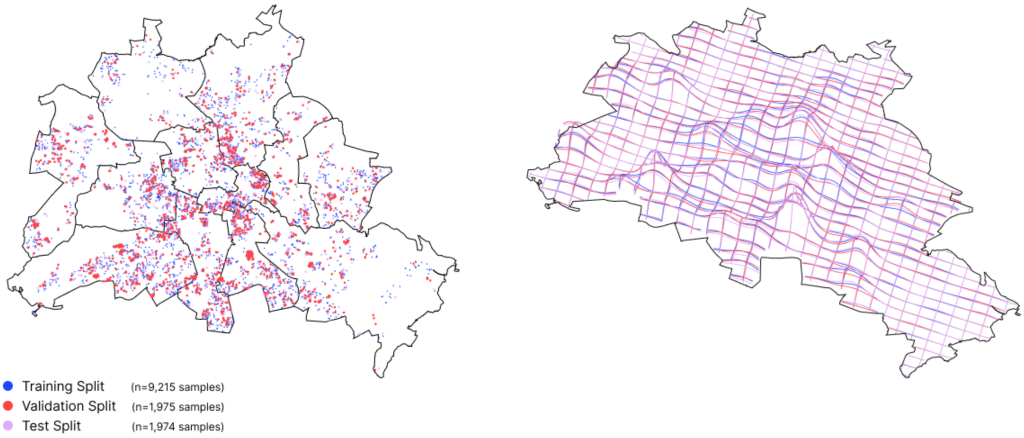

Figure 4.6: Spatial Distribution and Density of Retained Samples

Metric	Mean	Median	Std	Min	Max	Q25	Q75
GRZ	0.38	0.31	0.20	0.06	1.00	0.23	0.52
GFZ	1.81	0.86	1.78	0.08	9.99	0.56	3.12
Target height [m]	12.5	9.5	6.3	3.0	34.5	8.0	19.5
Parcel area [m^2]	482	372	355	100	4,900	204	684
Target footprint [m^2]	201	89	209	31	1,749	61	292
Neighbours (count)	48	43	24	2	228	32	56
Mean neighbour height [m]	10.0	7.4	5.7	1.2	25.9	5.6	15.6

Table 4.3: Dataset Summary Statistics

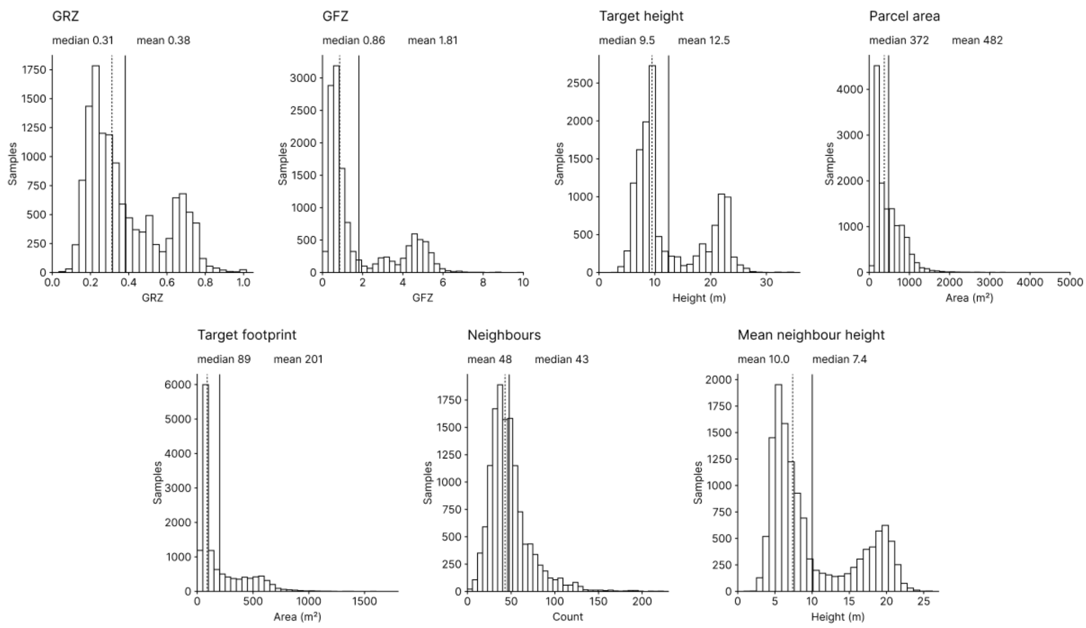

Figure 4.7: Metric Distributions Across Retained Samples

4.5 Dataset Limitations

Urban form is continuous, heterogeneous, and shaped by material, typological, architectural, social, and historical differentiation. Any digital representation necessarily reduces this complexity by selecting certain properties and omitting others. Here, CityGML LoD1 and voxelisation translate buildings into discrete occupancy grids (Figure 4.8) derived from extruded footprints with aggregated heights, omitting architectural detail and social dimensions. As a result, the machine-learning models learn recurring patterns in discretised occupancy and spatial context, while continuous geometry and socio-spatial processes remain outside the representation.

While necessary, this reduction also defines what the models can learn, particularly since the underlying cadastral and building datasets were produced for administrative purposes and not for machine-learning applications (Biljecki et al., 2016; Scott, 1998; Kitchin, 2014). Regulatory metrics enter the models as numerical descriptors of density and massing, not as full representations of the legal, social, or functional conditions that shape planning decisions. The outputs can thus be assessed in terms of volumetric fit and regulatory indicators, but not as complete accounts of urban form.

Additionally, small and medium parcels dominate the dataset, while large parcels are sparse, as shown in Table 4.2. This concentration is reflected in the distributions of GFZ, building height, and footprint area reported in Table 4.3 and Figure 4.7, where compact parcels and moderate-density configurations occupy most of the sample space, while denser or larger-footprint configurations appear mainly in the upper tails. As a result, the models encounter compact parcels with comparatively small buildings far more often during training than atypical or large-scale massing configurations, and their behaviour on larger sites should be interpreted with greater caution.

The filtering logic defined in Equation 10 narrows the dataset further by excluding parcels that diverge strongly from two direct edge-adjacent reference neighbours in GRZ and GFZ. This yields internally consistent cases while removing outliers, typological exceptions, and ambiguous situations that would otherwise broaden the range of contextual conditions represented in the data. This narrowing is a deliberate choice, as the available computational budget led the thesis to prioritise data quality and contextual consistency over maximum diversity. A larger and more diverse dataset would be desirable for broader generalisation, but would also require substantially greater computational resources for effective training. This imbalance is reinforced by the fixed 128 m spatial window, which reduces the available context as parcel size increases. Expanding the window would increase memory demands or require a coarser spatial resolution, further reducing the fidelity of GRZ and GFZ calculations. As a result, the models can be expected to learn conservative patterns of contextual fit and to reproduce surrounding density conditions, with a focus on small-to-medium infill situations where local form relations are central.

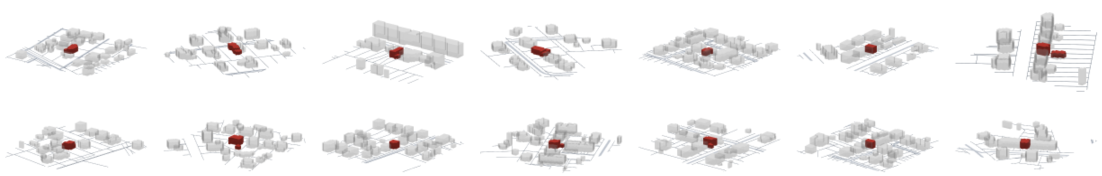

Figure 4.8: Representative Samples from the Voxelised Dataset

5 Iterative Research: Methods and Model Evolution

Building on the dataset and shared spatial representation developed in Chapter 4, the thesis proceeds through a three-phase experimental sequence. Phase I tests whether the surrounding urban fabric of a target parcel alone can support plausible parcel-scale massing. Phase II introduces GRZ, GFZ, and building height as conditioning variables for regulatory steering. Phase III shifts from single-output prediction to the sampling of multiple variants through a two-stage latent diffusion pipeline. Across these phases, the analysis examines how changes in model formulation affect the relation between context, regulatory targets, and resulting massing.

5.1 Shared Setup and Evaluation

Before the individual phases are introduced, this section defines the common experimental setup used to make their results comparable. All phases use the same spatial representation, dataset split, and metric definitions unless stated otherwise. This shared setup ensures that differences in performance and behaviour can be attributed primarily to the model architecture, conditioning strategy, and sampling procedure.

5.1.1 Unified Evaluation Framework

Across the three phases, the thesis reports 45 exploratory training runs and four final reruns, resulting in 49 model-training runs. Of the exploratory runs, five re-execute phase-specific reference configurations as within-group baselines, so the series covers 40 distinct exploratory configurations. Evaluation-only analyses such as Phase II TTA sensitivity, threshold diagnostics, direct non-TTA inference, and isolated control are excluded from this count. Model selection uses the phase-specific validation criteria described below on the shared validation split of 1,975 samples, while the held-out test split of 1,974 samples is reserved for final cross-phase comparison. Compact run labels beginning with P1, P2, P3A, and P3B denote Phase I, Phase II, Phase III Stage A, and Phase III Stage B respectively. The corresponding identifiers and filenames are indexed in the appendix, with complete training configurations available in the accompanying code repository.

In Phases I and II, checkpoints are selected by the highest validation Intersection over Union (IoU), computed with the run-specific inference setting, including threshold calibration and, where enabled, test-time augmentation. IoU is the primary voxel-level overlap metric between predicted and target occupancy. In Phase III, Stage A is selected by the lowest validation reconstruction-plus-KL loss (Equation (35)), and Stage B by the lowest validation denoising loss (Equation (42)). Full-pipeline geometric metrics for Phase III are computed separately after latent sampling, VAE decoding, and threshold calibration on the validation split. Cross-phase comparisons and regulatory accuracy tests are reported only for the selected checkpoint of each final model. IoU is defined as

$$\text{IoU}(Y_{\text{pred}}, Y_{\text{gt}}) = \frac{|Y_{\text{pred}} \cap Y_{\text{gt}}|}{|Y_{\text{pred}} \cup Y_{\text{gt}}|} \tag{11}$$

Dice complements IoU by combining precision and recall into a single overlap score. Precision penalises excessive predicted mass, while recall penalises missing target mass, so a high Dice score requires both few false positives and few false negatives. Let TP, FP, and FN denote true positive, false positive, and false negative occupied voxels in the binarised prediction, and let $\varepsilon = 10^{-6}$ denote a numerical stability constant. Precision, recall, and Dice are defined by

$$\text{Precision} = \frac{TP}{TP + FP + \varepsilon} \tag{12}$$

$$\text{Recall} = \frac{TP}{TP + FN + \varepsilon} \tag{13}$$

$$\text{Dice} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall} + \varepsilon} \tag{14}$$

This evaluation Dice metric differs from the Dice term used during training in Section 5.1.2 and Equation 22, which is defined on continuous probabilities and serves as a differentiable overlap objective during optimisation.

Volumetric agreement is evaluated with three measures, ΔVol, VE, and ZPE. The relative volumetric error ΔVol compares predicted and ground-truth occupied voxel counts after normalisation by the ground-truth volume,

$$\Delta\text{Vol}(Y_{\text{pred}}, Y_{\text{gt}}) = \frac{\big|\|Y_{\text{pred}}\|_1 - \|Y_{\text{gt}}\|_1\big|}{\max(\|Y_{\text{gt}}\|_1,\ 1)} \tag{15}$$

which keeps the measure comparable across parcels of different size. The absolute volume error VE reports the same discrepancy in raw voxels. For a batch of B samples with binarised predictions $Y_{\text{pred},b} \in \{0,1\}^{D \times H \times W}$ and ground-truth occupancies $Y_{\text{gt},b}$, it is defined as

$$\text{VE} = \frac{1}{B} \sum_{b=1}^{B} \big|\ \|Y_{\text{pred},b}\|_1 - \|Y_{\text{gt},b}\|_1\big| \tag{16}$$

where $\|\cdot\|_1$ denotes the number of occupied voxels in the sample. The mean absolute vertical profile error ZPE measures slice-wise deviation in the vertical mass distribution and captures whether building mass is placed at the correct height levels even when total volume is similar.

$$\text{ZPE} = \mathbb{E}_{b,d}\left[\left|\sum_{h,w} Y_{\text{pred},b,d,h,w} - \sum_{h,w} Y_{\text{gt},b,d,h,w}\right|\right] \tag{17}$$

In the ZPE formulation, each term denotes the absolute difference in occupied voxel count at vertical slice d for sample b. The metric is reported at the 2.0 m voxel resolution used in the experiments.

For evaluation in the reported thesis experiments, each model output is first passed through a sigmoid activation, converting voxel logits into occupancy probabilities in $[0, 1]$, and then binarised using the threshold that maximises validation IoU over the grid $\{0.40, 0.45, 0.50, 0.55, 0.60\}$. The selected threshold is applied unchanged to the test split. Additional threshold sensitivity analyses are reported in the Phase I experiment section and for the final Phase II model in Section 5.3.5.

The model's ability to match regulatory target values is evaluated by applying the same voxel-counting rules to predictions and ground truth. GRZ and GFZ follow Equations 2 and 4, while height follows the discrete occupancy-span definition in Equation 6. For predicted volumes, occupancy is not clipped to the target parcel mask. As a result, GRZ and GFZ can also reflect off-parcel over-prediction. In extreme cases, predicted GRZ may exceed 1.0 when the predicted occupied footprint extends beyond the parcel area.

As predictions and targets are represented on a discrete voxel grid, their measured GRZ, GFZ, and height values are limited by voxel resolution, and small deviations may arise from discretisation alone. Resolution-dependent tolerance bands account for this discretisation limit by separating quantisation effects from model error. For parcel area A_{parcel}, voxel dimension v, and assumed storey height h_{storey}, the absolute resolution tolerances are

$$\tau^{\text{res}}_{\text{GRZ}} = \frac{v^2}{A_{\text{parcel}}}, \quad \tau^{\text{res}}_{\text{GFZ}} = \frac{v^3}{A_{\text{parcel}} \cdot h_{\text{storey}}}, \quad \tau^{\text{res}}_{\text{height}} = v \tag{18}$$

where each tolerance describes the smallest change that can be resolved on the evaluation grid. GRZ changes by one horizontal footprint cell relative to the parcel area, GFZ by one occupied voxel converted through the assumed storey height, and height by one vertical voxel step. GRZ and GFZ tolerances scale inversely with parcel area. The height tolerance $\tau^{\text{res}}_{\text{height}} = v$ is a one-slice comparison tolerance on the shared 2.0 m grid.

Regulatory deviations are summarised by the adjusted relative error

$$\varepsilon_{\text{adj}}(p,\, r) = \frac{\max\big(0,\ |p - r| - \tau^{\text{res}}\big)}{\max(|r|,\, \varepsilon)} \tag{19}$$

where p denotes the prediction, r the reference target, τ^{res} the corresponding tolerance from Equation 18, and $\varepsilon = 10^{-6}$. Only deviations beyond this tolerance contribute to the reported error after clipping at zero.

Controllability in Phases II and III is evaluated by scaling all three conditioning inputs to -10% and $+10\%$ of their native ground-truth values $\mathbf{c}^{0.5\,\mathrm{m}}$, keeping the comparison near the filtered training distribution in Equation 10. Larger offsets are avoided in the main tables to prevent target response from being conflated with out-of-distribution behaviour, although the final Phase II qualitative panel also visualises a wider $\pm 20\%$ target-scaling experiment as a supplementary illustration. For the final Phase II checkpoint, an additional single-variable sensitivity analysis scales one conditioning input at a time while keeping the other two fixed, with results interpreted in light of the coupled structure of GRZ, GFZ, and height. No isolated single-variable analysis is reported for Phase III, as the joint target-scaling experiment already shows insufficient occupancy stability for finer variable-specific diagnosis.

5.1.2 Shared Training Setup

The following training conditions are shared across phases and experiments, unless noted otherwise.

Dataset and split. All experiments use the same voxelised dataset of 13,164 samples and the same fixed 70–15–15 split manifest generated with seed 42, which yields 9,215 training samples, 1,975 validation samples, and 1,974 test samples. The held-out test split evaluates unseen samples under the same spatially mixed sampling regime, not strict spatial out-of-area generalisation.

Input resolution and downsampling. All samples are downsampled from the native 256^3 resolution to 64^3 using stride-4 average pooling, with target masks re-binarised at 0.5 after pooling, yielding an effective voxel dimension of $v_{\text{train}} = 128\,\text{m}/64 = 2.0\,\text{m}$. Training loaders apply random horizontal 90° rotations and flips to input and target tensors, while validation and test loaders disable augmentation.

Optimiser and learning rate schedule. All runs use the AdamW optimiser (Loshchilov & Hutter, 2019) with weight decay 0.01, cosine learning-rate decay, and gradient clipping at a maximum norm of 1.0. Initial learning rates are phase-specific and reported in the respective sections.

Gradient accumulation and effective batch size. Because the 3D tensors limit the physical batch size, gradients are accumulated over several batches before each optimiser step. The resulting effective batch size varies slightly by phase and is reported in the respective sections.

Reproducibility and training strategy. All training runs use a fixed random seed of 42. Each phase starts with exploratory 10-epoch runs, followed by a 50-epoch rerun of the strongest setup. In Phase III, Stage A (VAE) and Stage B (diffusion) are rerun separately. Checkpoint selection follows Section 5.1.1.

Conditioning normalisation in conditioned models. In Phase II and Stage B of Phase III, the model receives a regulatory target vector $\mathbf{c}^{0.5\,\mathrm{m}} \in \mathbb{R}^3$ alongside the spatial input, containing GRZ, GFZ, and target height. This native vector provides the conditioning values and the regulatory reference throughout the experiments. Given the different scales and units of the components, each dimension is z-score normalised using training-split statistics.

$$\tilde{\mathbf{c}} = \frac{\mathbf{c} - \boldsymbol{\mu}}{\boldsymbol{\sigma} + 10^{-8}} \tag{20}$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ denote the per-dimension mean and standard deviation of the training split and are reused unchanged for validation, test, and inference.

Training objective in Phases I and II. Phases I and II use a combined voxel-space objective consisting of weighted BCE, soft Dice loss, and an additional volume regulariser. Together, these terms address class imbalance, geometric overlap, and total-volume consistency. As empty voxels far outnumber occupied voxels in the 64^3 occupancy grids, occupied voxels are given a higher positive class weight, addressing a known difficulty in CNN-based learning under strong class imbalance (Buda et al., 2018).

The positive class weight w_+ is computed per batch from the ratio of negative to positive voxels, with N_{pos} clamped to 1 and the resulting weight capped at 50.

$$w_+ = \min\left(\frac{N_{\text{neg}}}{\max(N_{\text{pos}},\ 1)},\ 50\right) \tag{21}$$

where $N_{\text{pos}} = \sum_i y_i$ and $N_{\text{neg}} = \sum_i(1 - y_i)$ are the per-batch voxel counts. The cap limits unstable gradients in very sparse parcels.

The BCE term evaluates voxel-wise occupancy errors. Its voxel-wise contributions are scaled by w_+ and further modulated by a surface-weight map controlled by ω_s. The boundary mask $\mathcal{B} = \text{dilate}(Y) - \text{erode}(Y)$ is derived from the binary ground truth Y. Boundary voxels receive weight $w_i^{\text{surf}} = 1 + (\omega_s - 1) \cdot \mathcal{B}_i$, while non-boundary voxels keep weight 1.0. This increases the contribution of building edges without changing the ground-truth geometry.

A soft Dice term complements the voxel-wise BCE term with a global overlap signal on continuous sigmoid probabilities $\hat{p}_i = \sigma(\hat{y}_i)$, using a small numerical stabilisation constant ε, and is defined by

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_i \hat{p}_i\, y_i + \varepsilon}{\sum_i \hat{p}_i + \sum_i y_i + \varepsilon} \tag{22}$$

The BCE and Dice terms are combined with equal weight to form the primary loss.

$$\mathcal{L}_{\text{primary}} = \tfrac{1}{2}\mathcal{L}_{\text{BCE}} + \tfrac{1}{2}\mathcal{L}_{\text{Dice}} \tag{23}$$

A differentiable volume regulariser $\mathcal{R}_{\text{vol}}$ adds a global mass signal by penalising the relative mismatch between predicted and ground-truth voxel counts.

$$\mathcal{R}_{\text{vol}} = \frac{1}{B}\sum_{b=1}^{B} \frac{\left|\sum_i \hat{p}_{b,i} - \sum_i y_{b,i}\right|}{\max\left(\sum_i y_{b,i},\ 1\right)} \tag{24}$$

The resulting total loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{primary}} + \lambda_{\text{vol}}\, \mathcal{R}_{\text{vol}} \tag{25}$$

The regularisation coefficient λ_{vol} is phase-specific. In Phase III, this voxel-space objective is replaced by the Stage A VAE objective and the Stage B diffusion objective introduced in Section 5.4.

5.2 Phase I: Unconditional 3D U-Net

Phase I establishes the basis for all subsequent experimental phases by testing whether the surrounding urban fabric of a target parcel, represented through the four spatial context channels (C_0–C_3), contains enough information to support plausible parcel-scale massing prediction without regulatory input. A positive result is necessary for the subsequent phases, since failure at this stage would suggest that the selected representation and spatial context are insufficient to support the more complex modelling tasks developed later in the thesis.

5.2.1 Objective and Architecture

The model developed in Phase I is implemented as a three-dimensional U-Net for voxel-based occupancy prediction. Its skip connections combine coarse contextual features from deeper layers with higher-resolution spatial detail from earlier encoder stages, allowing the model to infer plausible building mass from the surrounding urban fabric while preserving parcel-level geometric detail in the voxel grid. The architecture described below corresponds to the reference run `P1.01`. The final Phase I architecture is selected through the experiment sequence and described in Subsection 5.2.4.

DoubleConv3D block. The DoubleConv3D block is used to extract and refine local volumetric features within each resolution level of the U-Net. It applies two convolution–normalisation–activation stages while preserving spatial resolution, so that the feature representation is enriched before downsampling in the encoder or after feature fusion in the decoder. The first stage produces an intermediate representation $\mathbf{h}_1$,

$$\mathbf{h}_1 = \text{SiLU}(\text{GN}(\text{Conv}_1(\mathbf{u}))) \tag{26}$$

and the second stage yields the final block output $\mathbf{h}_2$,

$$\mathbf{h}_2 = \text{SiLU}(\text{GN}(\text{Conv}_2(\mathbf{h}_1))) \tag{27}$$

where $\mathbf{u}$ denotes the block input feature map, and Conv_1 and Conv_2 denote $3 \times 3 \times 3$ convolutions with padding 1. This preserves spatial dimensions and keeps the features aligned for the skip connections. GN denotes Group Normalisation (Wu & He, 2018), and SiLU denotes the Sigmoid Linear Unit activation (Ramachandran et al., 2017). Figure 5.1 summarises the block used throughout Phase I.

Group Normalisation is used instead of Batch Normalisation, since three-dimensional volumetric tensors constrain the feasible batch size under the available memory budget. By normalising within channel groups, not across the batch, it provides more stable feature scaling in low-batch training (Wu & He, 2018). SiLU is used instead of ReLU, as its smooth, non-zero negative branch avoids hard activation cut-offs and supports more stable gradient flow in this low-batch volumetric setting (Ramachandran et al., 2017).

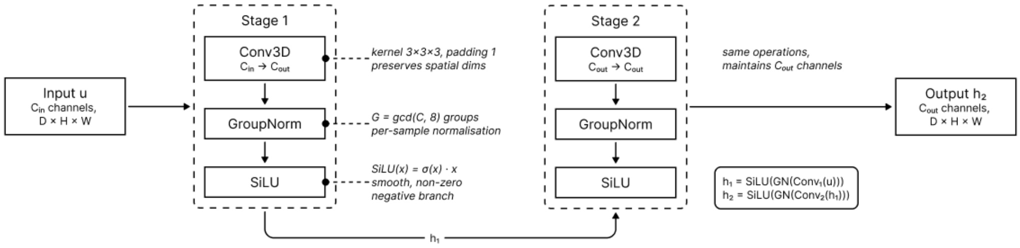

Figure 5.1: Phase I DoubleConv3D Building Block

Encoder. The encoder receives the four spatial context channels and compresses the 64^3 input through repeated DoubleConv3D blocks and stride-2 convolutional downsampling. Each block output is retained as a skip connection for the decoder. Using stride-2 convolutions instead of max-pooling keeps downsampling learnable and allows the network to adapt which spatial information is preserved at each scale.

Bottleneck. The bottleneck operates at the lowest spatial resolution and aggregates broader spatial context into a compact feature representation. A Dropout3d layer follows the bottleneck block and regularises the representation by dropping entire feature channels instead of individual activations (Tompson et al., 2015).

Decoder. The decoder mirrors the encoder through transposed convolutions, skip concatenation, and DoubleConv3D fusion blocks. At each level, the upsampled feature map is aligned to the corresponding skip connection by padding when necessary, concatenated, and refined by a DoubleConv3D block. A final output head maps the restored full-resolution representation to one occupancy logit per voxel. Sigmoid activation and thresholding at the calibrated decision boundary yield the final building-mass prediction.

Training objective. In the architecture shown in Figure 5.2, the shared BCE–soft-Dice objective with volume regularisation produces one context-derived prediction per input.

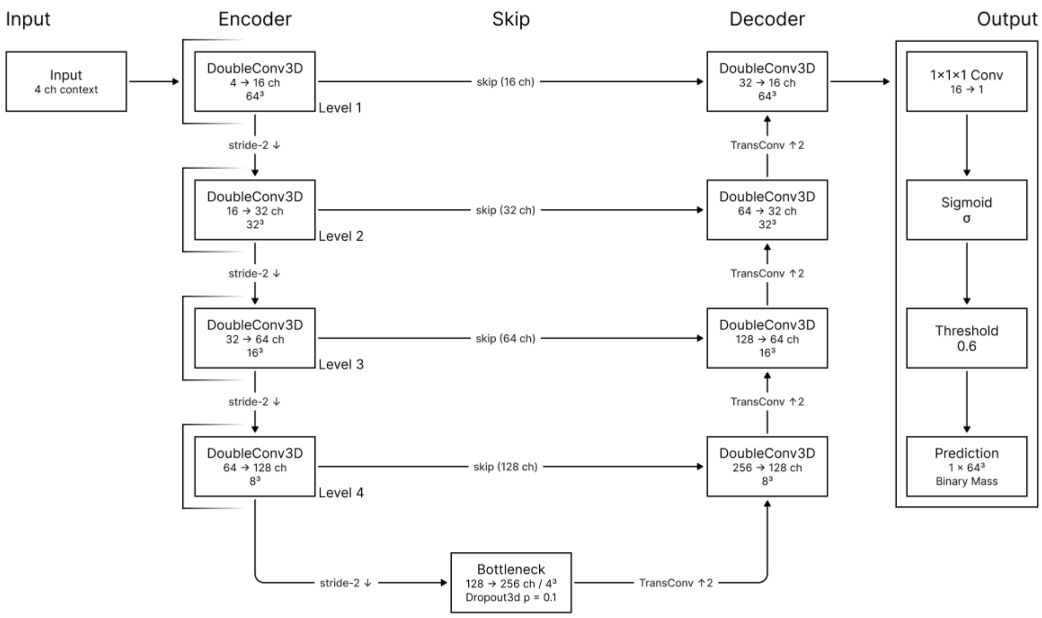

Figure 5.2: Phase I Reference U-Net Architecture

5.2.2 Experiment Setup

Phase I consists of seven experiment groups and twelve 10-epoch runs. Unless noted otherwise, all runs follow the shared training setup in Section 5.1.2 and the evaluation protocol in Section 5.1.1. Phase I defaults are an initial learning rate of 7×10^{-4}, a physical batch size of 2, gradient accumulation over 2 steps, and an effective batch size of 4. All runs use the shared 64^3 representation. Full configuration files are available in the accompanying repository and indexed in the appendix. The strongest configuration is rerun for 50 epochs to produce the final Phase I model.

P1.01: Baseline. The baseline is the reference configuration, namely a four-level three-dimensional U-Net with 16 base channels, no coordinate channels, and the default loss balance $\omega_s = 1.0$ and $\lambda_{\text{vol}} = 0.1$. This run is labelled `P1.01`.

P1.02: Network capacity. Two variants test sensitivity to model size. A more compact configuration reduces the network to 12 base channels at depth 3 under the run `P1.02a`, while a wider configuration increases the base width to 24 channels at depth 4 under the run `P1.02b`. Both runs exclude coordinate channels and retain the baseline loss setup.

P1.03: Coordinate channels. The baseline architecture is extended by appending three normalised spatial coordinate maps to the input, expanding it from 4 to 7 channels. The run `P1.03` tests whether explicit grid position adds useful signal beyond the four spatial context channels, since building mass may depend partly on its location relative to parcel boundaries and surrounding context.

P1.04: Surface loss weighting. Surface boundary emphasis is tested at $\omega_s = 1.5$ in two variants, namely `P1.04a` without coordinate channels and `P1.04b` with them. The experiment evaluates whether stronger boundary weighting sharpens the predicted envelope and how it interacts with coordinate encoding.

P1.05: Slim capacity and coordinate interaction. A reduced architecture with 12 base channels and depth 4 is evaluated without coordinate channels in `P1.05a` and with coordinate channels in `P1.05b`. The setup assesses whether explicit coordinate channels can compensate for reduced model capacity.

P1.06: Loss balance. The role of the volume regulariser is isolated at the baseline capacity. The run `P1.06a` removes the volume regulariser by setting $\lambda_{\text{vol}} = 0.0$, while `P1.06b` combines a surface weighting of $\omega_s = 1.5$ with a stronger volume regulariser of $\lambda_{\text{vol}} = 0.2$. The comparison tests whether the two loss components are complementary or produce diminishing returns.

P1.07: Threshold strategy. Experiment 7 compares automatic threshold calibration with a fixed threshold of 0.5. The run `P1.07a` evaluates the baseline geometry alone, while `P1.07b` evaluates the baseline augmented with coordinate channels. The pair tests whether automatic threshold search improves performance over a fixed baseline.

5.2.3 Experiment Results

Phase I shows that unconditional context-based volumetric prediction is feasible within the selected voxel representation and at the 64^3 training resolution. Table 5.1 reports the full validation metrics for all Phase I runs, with only small differences between architectural variants.

Experiment	Run	IoU	Dice	Prec.	Recall	VE [voxel]	ZPE [voxel]
Baseline	`P1.01`	0.7215	0.8383	0.8157	0.8621	55.59	1.2681
Capacity	`P1.02a`	0.7262	0.8414	0.8090	0.8764	58.65	1.2802
Capacity	`P1.02b`	0.7250	0.8406	0.8201	0.8621	55.00	1.2359
Coordinates	`P1.03`	0.7168	0.8350	0.8123	0.8591	55.81	1.2432
Surface	`P1.04a`	0.7233	0.8394	0.8248	0.8546	52.39	1.2469
Surface	`P1.04b`	0.7216	0.8383	0.8046	0.8749	60.89	1.3426
Slim	`P1.05a`	0.7255	0.8409	0.8139	0.8698	55.63	1.2551
Slim + coords	`P1.05b`	0.7196	0.8370	0.8226	0.8518	51.84	1.2166
Loss balance	`P1.06a`	0.7238	0.8398	0.7995	0.8844	66.22	1.4044
Loss balance	`P1.06b`	0.7132	0.8326	0.8146	0.8514	54.33	1.2560
Manual thresh.	`P1.07a`	0.7182	0.8360	0.7964	0.8796	66.81	1.3993
Manual thresh.	`P1.07b`	0.7153	0.8340	0.8009	0.8700	60.91	1.3214

Table 5.1: Phase I Validation Metrics by Experiment

The baseline run `P1.01` reaches a validation IoU of 0.7215, establishing a strong reference point and suggesting that the four spatial context channels already contain sufficient contextual information to support plausible massing prediction without regulatory conditioning.

The capacity runs provide the strongest validation results in Phase I. The compact three-level variant `P1.02a` reaches the highest validation IoU overall at 0.7262, followed by `P1.05a` at 0.7255 and `P1.02b` at 0.7250. The narrow spread of about 0.0012 in IoU indicates a small cluster of competitive compact-capacity variants. This suggests that the task is not primarily limited by model width at this resolution. The selected voxel representation and spatial context channels already provide a strong signal that can be captured by relatively compact U-Net variants.

The coordinate experiment `P1.03` reaches 0.7168 validation IoU and is below the non-coordinate baseline. The same pattern appears in the combined variants, where `P1.04b` stays below `P1.04a` and `P1.05b` stays below `P1.05a`. Coordinate channels provide no clear benefit in Phase I and appear to add complexity without strengthening the context-derived prior. This is consistent with the parcel-centred setup, where relative geometry, parcel boundaries, neighbouring massing, and street structure already encode most of the spatial information needed for unconditional prediction.

The loss-focused experiments show a trade-off between overlap and mass consistency. Surface boundary weighting without coordinate channels yields a modest gain over the baseline, with `P1.04a` reaching 0.7233 validation IoU. Removing the volume regulariser keeps IoU relatively high at 0.7238 but increases absolute volume error to 66.22 voxels, while the combined high-surface/high-volume setting performs weakest in IoU at 0.7132.

The threshold-sensitivity runs support the validation-based calibration strategy used for the Phase I rerun and the later phases. At a fixed threshold of 0.5, `P1.07a` reaches 0.7182 validation IoU, below the automatically calibrated baseline at 0.7215, while the coordinate-augmented variant falls further to 0.7153. This shows that the optimal occupancy boundary is not necessarily located at the default probability threshold, making validation-based threshold calibration appropriate for sparse voxel prediction.

Overall, Phase I shows that spatial context alone supports plausible unconditional prediction with strong volumetric overlap. Moderate capacity reduction provides the clearest gain, while coordinate encoding and stronger loss weighting do not outperform the compact-capacity variants. Accordingly, `P1.02a` advances to the final 50-epoch rerun. Figure 5.3 also shows that most variants converge within a narrow ten-epoch performance band.

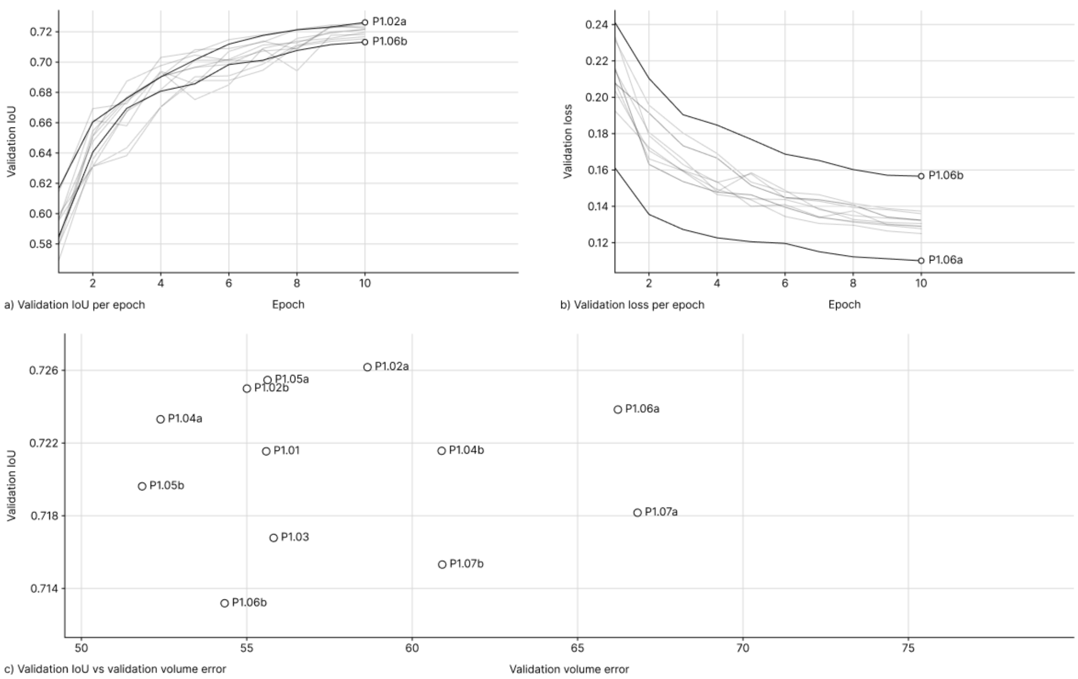

Figure 5.3: Phase I Validation Performance Across Variants

5.2.4 Final Model and Analysis

The final Phase I model is the 50-epoch rerun `P1-Final`. It retains the compact unconditional architecture shown in Figure 5.4, a depth-3 three-dimensional U-Net with base width 12, four contextual input channels, no coordinate channels, surface weighting $\omega_s = 1.0$, and volume regularisation $\lambda_{\text{vol}} = 0.1$. The selected checkpoint comes from epoch 47, chosen by peak validation IoU, and the calibrated threshold remains $\hat{t} = 0.60$.

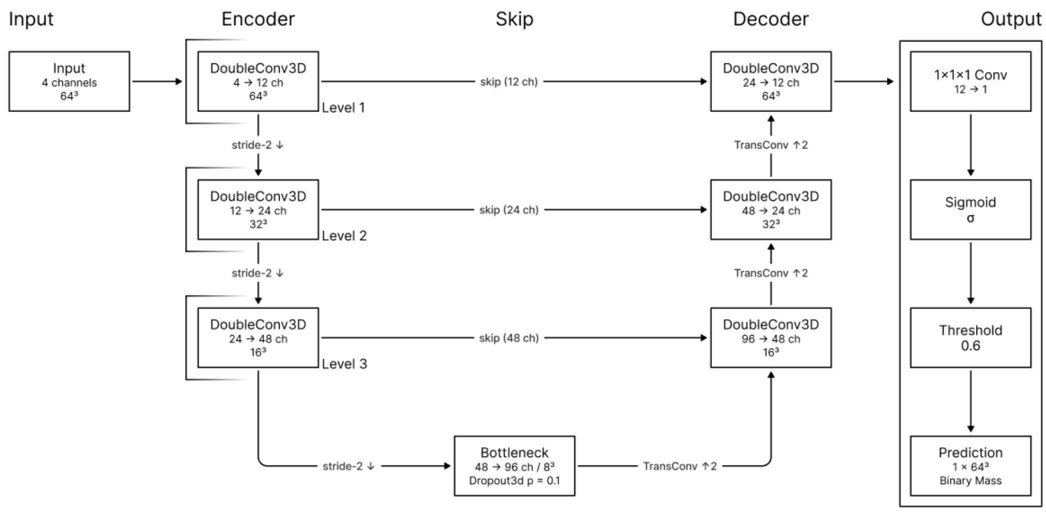

Figure 5.4: Phase I Final U-Net Architecture

Table 5.2 reports the segmentation metrics for the selected epoch-47 checkpoint. Validation metrics are used for checkpoint selection and threshold calibration, while test metrics report held-out generalisation.

Split	IoU	Dice	Prec.	Recall	ΔVol	VE [voxel]	ZPE [voxel]	Thresh.
Validation	0.7590	0.8630	0.8426	0.8844	0.1013	41.63	1.0343	0.60
Test	0.7564	0.8613	0.8414	0.8822	0.0990	43.16	1.0657	0.60

Table 5.2: Phase I Final Segmentation Metrics

Relative to the 10-epoch configuration `P1.02a`, validation IoU rises from 0.7262 to 0.7590 (+3.28 percentage points), and Dice rises from 0.8414 to 0.8630 (+2.16 percentage points). Absolute volume error drops from 58.65 to 41.63 voxels, and ZPE falls from 1.2802 to 1.0343. The rerun shows that the compact depth-3 variant benefits from longer optimisation in the unconditional setting. Test performance stays close to validation, with an IoU gap of only 0.0026, which suggests stable generalisation to unseen samples.

The qualitative analysis supports the quantitative results, with parcel placement, footprint extent, and vertical massing usually aligning with the immediate neighbourhood context and producing coherent infill envelopes. The precision–recall imbalance indicates mild over-prediction, as uncertain predictions more often extend beyond the ground-truth volume. Figure 5.5 shows a representative excerpt, with residual errors mainly visible along boundaries and finer footprint deviations.

The qualitative sample figures in this and the following phases are read consistently. The three-dimensional view shows the thresholded predicted or sampled target volume in blue and the surrounding neighbourhood in grey. The two-dimensional prediction panels show unthresholded probability maps, where lighter values indicate higher predicted occupancy, intermediate grey values indicate diffuse or uncertain activation, and black indicates negligible predicted occupancy.

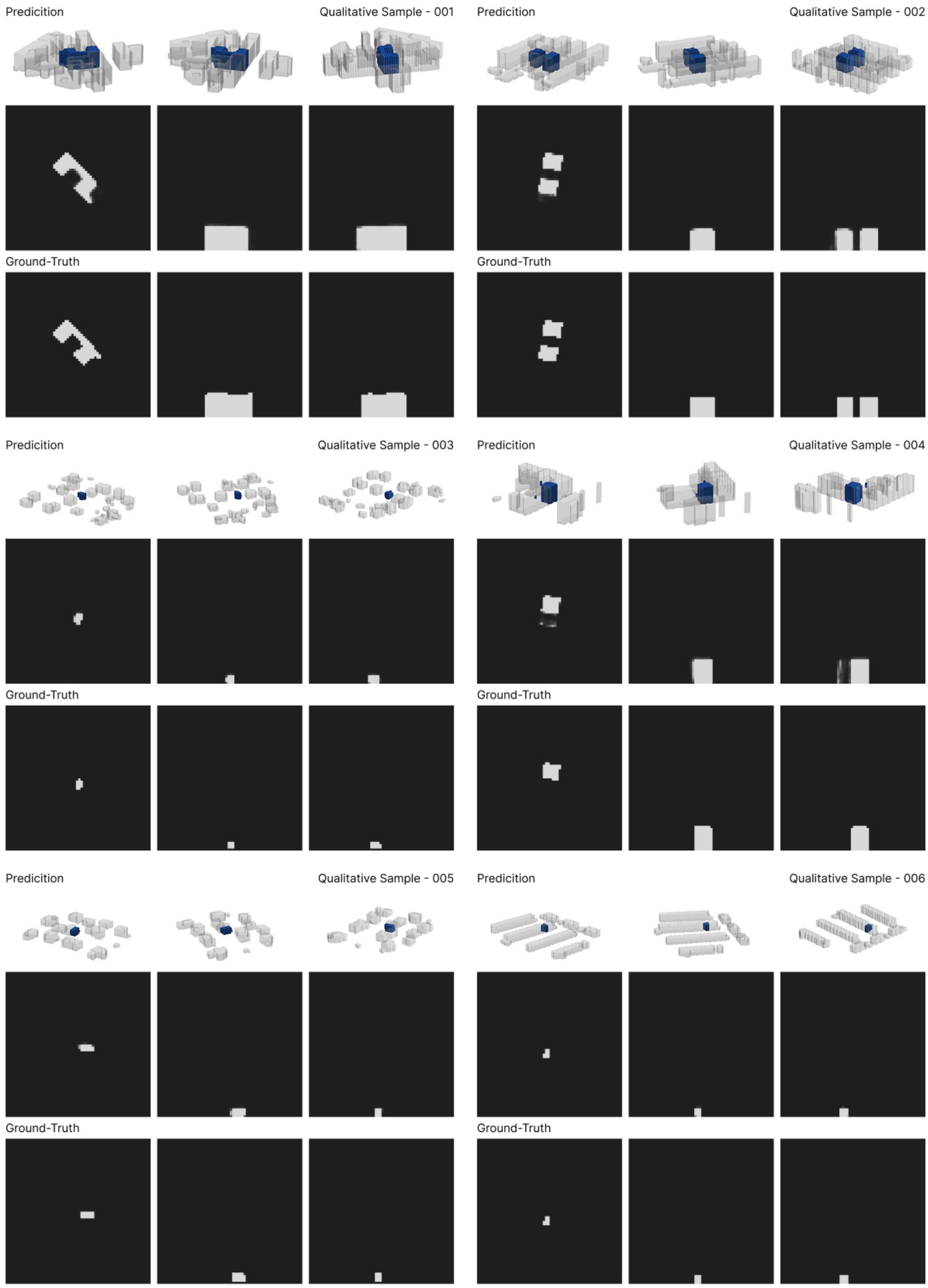

Figure 5.5: Phase I Predictions and Ground Truth

Because the unconditional Phase I model has no conditioning pathway, regulatory evaluation is limited to the native reference-target setting $\mathbf{c}^{0.5\,\mathrm{m}}$. It nevertheless provides an informative baseline, with mean adjusted relative error below 0.10 for all three regulatory metrics on the held-out test split ($n = 1{,}974$) and close to zero for height. Table 5.3 suggests that a substantial share of the regulatory signal is already latent in the spatial context.

Metric	Mean pred.	Mean ref. $\mathbf{c}^{0.5\,\mathrm{m}}$	Mean $\varepsilon_{\mathrm{adj}}$	Median $\varepsilon_{\mathrm{adj}}$
GRZ	0.3905	0.3765	0.0814	0.0446
GFZ	1.9306	1.8238	0.0932	0.0657
Height [m]	13.37	12.52	0.0065	0.0000

Table 5.3: Phase I Regulatory Errors by Metric

Building height shows the strongest implicit agreement, with a median adjusted error of zero, meaning that for at least half of the test parcels the predicted height remains within the tolerance band induced by the 2.0 m voxel resolution and plausibly reflects the local vertical regime encoded in the neighbour-height field. GRZ and GFZ also stay close to their parcel-specific targets, suggesting that footprint extent and overall density are partly recoverable from context alone. Figure 5.6 summarises this agreement across all three metrics.

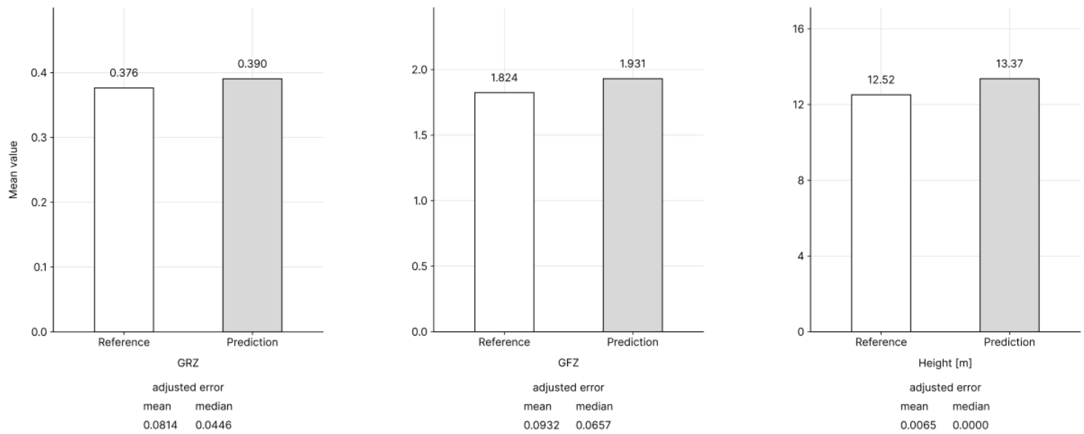

Figure 5.6: Phase I Regulatory Summary Metrics

These results establish Phase I as a viable basis for Phases II and III by demonstrating that the voxel-based representation and the 64^3 training resolution are sufficient to support plausible parcel-scale massing prediction. The model learns to recover much of the density and height structure embedded in the dataset from spatial context alone, suggesting that the voxel representation captures relevant urban-morphological information in a form that can be learned by a compact 3D U-Net and supports the basic potential of context-based volumetric learning for early-stage massing prediction. However, the Phase I model produces only a single estimate for a fixed input tensor $\mathbf{X}$ and cannot respond to changes in GRZ, GFZ, or target height. This limitation motivates Phase II, which retains the encoder–decoder logic of Phase I while introducing GRZ, GFZ, and height as conditioning inputs.

5.3 Phase II: Conditional 3D U-Net

Phase II extends the Phase I architecture by adding regulatory conditioning through GRZ, GFZ, and building height. To make the model responsive to parcel-specific targets, these values are supplied alongside the spatial context and injected into the network through Feature-wise Linear Modulation (FiLM; Perez et al., 2017). The architecture retains the Phase I encoder–decoder logic while replacing the unconditioned DoubleConv3D blocks introduced in Phase I with FiLM-augmented variants.

5.3.1 Objective and Architecture

The Phase II reference model `P2.01a` builds on the Phase I U-Net architecture by replacing all DoubleConv3D blocks with FiLMDoubleConv blocks, which condition the feature maps on the regulatory target vector at every encoder, bottleneck, and decoder level. The paired run `P2.01b` keeps the architecture otherwise unchanged and adds coordinate channels to isolate their contribution. All other hyperparameters follow the shared setup unless varied in the experiment sequence below.

Conditioning input. The conditioning vector $\mathbf{c} = (\text{GRZ}_{\text{target}}, \text{GFZ}_{\text{target}}, H_{\text{height,target}}) \in \mathbb{R}^3$ contains parcel-specific targets for ground coverage, floor area ratio, and building height, derived from metadata computed at the native $0.5\,$m voxel resolution. Each component is z-score normalised with training-split statistics according to Equation (20), producing $\tilde{\mathbf{c}} \in \mathbb{R}^3$. The default setup uses the full vector, while reduced variants $\{\text{GRZ}, \text{GFZ}\}$, $\{\text{GRZ}, \text{H}\}$, and $\{\text{GFZ}\}$ are evaluated in the conditioning-channel comparison reported in Section 5.3.2.

Conditioning dropout. During training, the Phase II model can randomly mask the normalised conditioning input to reduce reliance on the regulatory vector. The mechanism borrows the training-time dropout logic of classifier-free guidance (Ho & Salimans, 2022), but no guidance is applied at inference, so the model predicts directly from the full conditioning vector without conditional–unconditional mixing. For each sample b, a binary mask $m_b \in \{0, 1\}$ is drawn from a Bernoulli distribution and multiplied by the normalised conditioning vector,

$$\tilde{\mathbf{c}}_b^{\text{train}} = m_b \cdot \tilde{\mathbf{c}}_b, \quad m_b \sim \text{Bernoulli}(1 - p_{\text{drop}}) \tag{28}$$

When $m_b = 0$, the FiLM pathway receives a zero vector, corresponding to the training-set mean in normalised conditioning space, so the model relies more strongly on spatial context. The dropout rate p_{drop} is treated as a hyperparameter, with $p_{\text{drop}} = 0$ disabling conditioning dropout entirely.

FiLM module. Feature-wise Linear Modulation (FiLM; Perez et al., 2017) conditions intermediate feature maps through per-channel scale and shift parameters derived from $\tilde{\mathbf{c}}$. This allows the conditioning vector to amplify, suppress, or offset learned feature channels according to the target GRZ, GFZ, and height values. GRZ, GFZ, and height are parcel-level targets, so the same modulation is broadcast across all voxel positions in a feature map.

For each sample b, a two-layer multilayer perceptron (MLP) with a SiLU nonlinearity transforms the normalised conditioning vector $\tilde{\mathbf{c}}_b \in \mathbb{R}^{d_{\text{cond}}}$ into channel-wise FiLM parameters for the current feature map. For a feature map with C channels, the MLP outputs $2C$ values, which are split into a scale vector $\boldsymbol{\gamma}_b \in \mathbb{R}^C$ and a shift vector $\boldsymbol{\beta}_b \in \mathbb{R}^C$. The scale vector modulates channel strength through $(1 + \boldsymbol{\gamma}_b)$, while the shift vector adjusts the activation level.

$$[\boldsymbol{\gamma}_b;\, \boldsymbol{\beta}_b] = W_2 \cdot \text{SiLU}(W_1 \tilde{\mathbf{c}}_b + \mathbf{b}_1) + \mathbf{b}_2 \tag{29}$$

where $W_1 \in \mathbb{R}^{2C \times d_{\text{cond}}}$ and $W_2 \in \mathbb{R}^{2C \times 2C}$. Across a batch, these vectors are stacked to tensors of shape (B, C), reshaped to $(B, C, 1, 1, 1)$, and broadcast over the feature map $\mathbf{f} \in \mathbb{R}^{B \times C \times D \times H \times W}$. FiLM is then applied as

$$\text{FiLM}(\mathbf{f}_b,\, \tilde{\mathbf{c}}_b) = \mathbf{f}_b \odot (1 + \boldsymbol{\gamma}_b) + \boldsymbol{\beta}_b \tag{30}$$

The residual scale term $(1 + \boldsymbol{\gamma}_b)$ preserves an identity case when $\boldsymbol{\gamma}_b = \mathbf{0}$ and $\boldsymbol{\beta}_b = \mathbf{0}$. FiLM can then let parcel-level targets reweight learned spatial features without altering the spatial topology of the feature map.

FiLMDoubleConv block. The FiLMDoubleConv block extends the Phase I DoubleConv3D block by applying FiLM (Equations (29) and (30)) after each convolution–normalisation–activation stage. The normalised regulatory vector modulates the feature channels after each convolutional stage, allowing GRZ, GFZ, and height to influence how spatial features are weighted. Here, $\mathbf{u}$ denotes the block input feature map, yielding

$$\mathbf{h}_1 = \text{FiLM}(\text{SiLU}(\text{GN}(\text{Conv}_1(\mathbf{u}))),\ \tilde{\mathbf{c}}) \tag{31}$$

$$\mathbf{h}_2 = \text{FiLM}(\text{SiLU}(\text{GN}(\text{Conv}_2(\mathbf{h}_1))),\ \tilde{\mathbf{c}}) \tag{32}$$

Figures 5.7 and 5.8 show how the same FiLM adapter modulates the intermediate representation $\mathbf{h}_1$ and final output $\mathbf{h}_2$ after the two convolutional stages while preserving the Phase I convolutional structure.

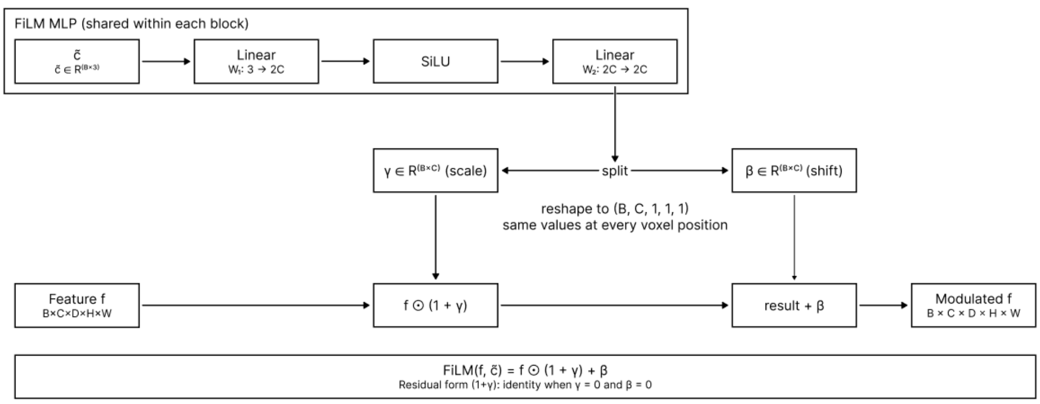

Figure 5.7: Phase II FiLM Scale-and-Shift Modulation

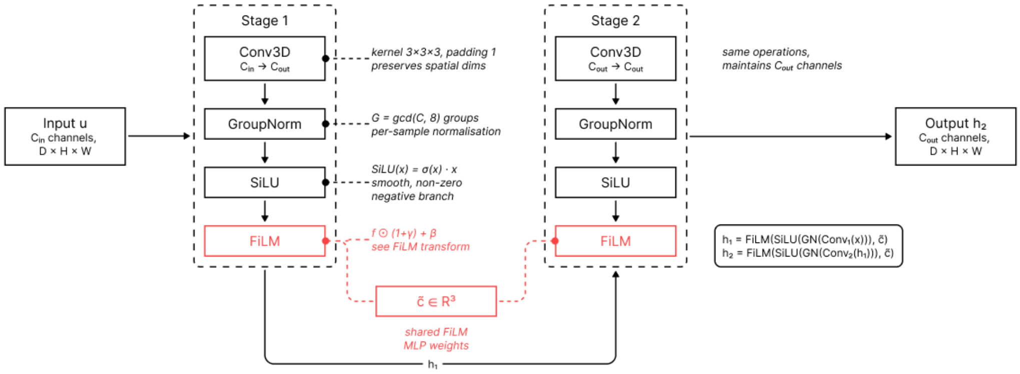

Figure 5.8: Phase II FiLMDoubleConv Building Block

Encoder–decoder structure. Figure 5.9 shows the Phase I encoder–decoder with every DoubleConv3D block, including the bottleneck, replaced by a FiLMDoubleConv block that injects the regulatory signal throughout the network. It compresses the 64^3 input through stride-2 downsampling, restores resolution through transposed convolutions and skip concatenation, and maps the final decoder features to one occupancy logit per voxel.

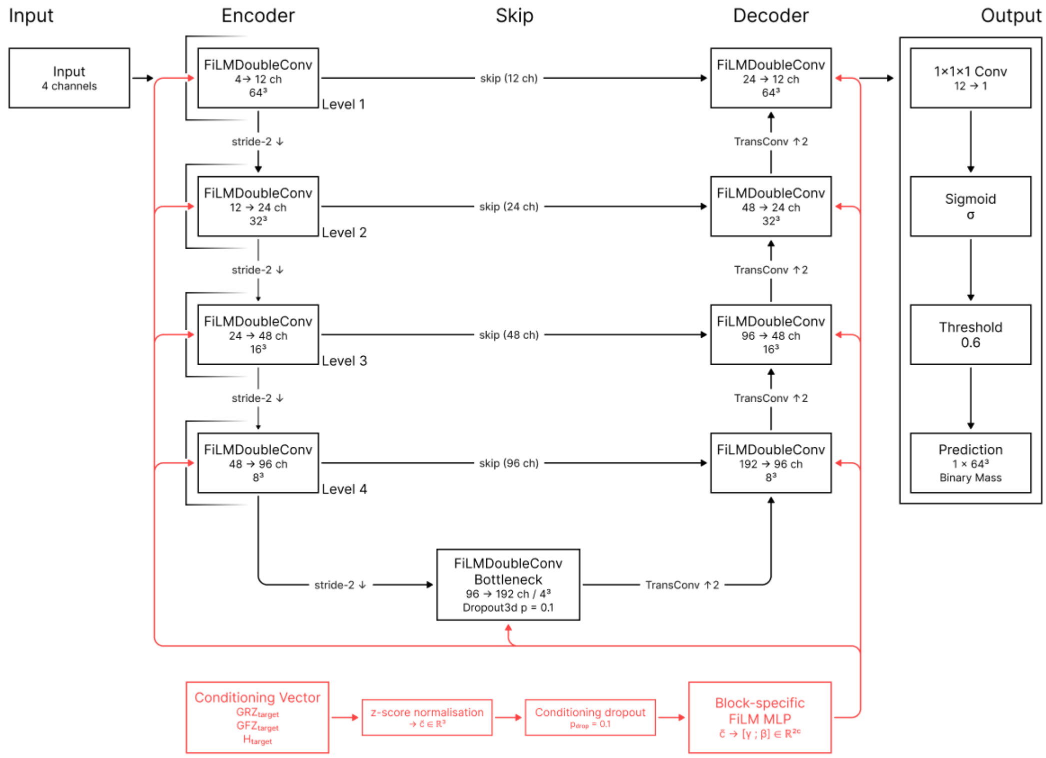

Figure 5.9: Phase II Reference Conditional U-Net

Training objective. Phase II uses the same BCE–soft-Dice objective with volume regularisation defined in Section 5.1.2. The conditioning signal changes the architecture but not the voxel-space training objective. Surface weighting ω_s, the volume-regularisation coefficient λ_{vol}, and the conditioning-dropout rate p_{drop} are treated as hyperparameters whose final values are selected through the experiment sequence below.

5.3.2 Experiment Setup

Phase II consists of five training experiment groups and one evaluation-time sensitivity group, covering fourteen 10-epoch training runs and two evaluation-only entries. Two of these runs, `P2.03c` and `P2.04a`, re-execute the reference configuration `P2.01a` within their experiment groups and reproduce its checkpoint and metrics under the fixed seed. Unless noted otherwise, all runs follow the shared training setup in Section 5.1.2 and the unified evaluation protocol in Section 5.1.1. Phase II defaults are an initial learning rate of 7×10^{-4}, a physical batch size of 2, gradient accumulation over 2 steps, and an effective batch size of 4. All runs operate on the shared 64^3 representation. Full run configurations are available in the accompanying repository and indexed in the appendix.

P2.01: Reference and coordinate sensitivity. The first experiment group establishes the Phase II reference setup while testing the effect of coordinate channels. Both runs use the depth-4 conditioned U-Net with 12 base channels, full regulatory conditioning $\{\text{GRZ}, \text{GFZ}, H\}$, conditioning dropout $p_{\text{drop}} = 0.1$, and the reference loss balance. `P2.01a` defines the coordinate-free reference, while `P2.01b` adds the three coordinate maps to the otherwise unchanged architecture. The pair tests whether FiLM-based conditioning changes the coordinate-channel pattern observed in Phase I.

P2.02: Conditioning-channel comparison. Starting from the coordinate-free reference, `P2.02a` removes target height, `P2.02b` removes GFZ, and `P2.02c` reduces the conditioning signal to GFZ alone. The group tests whether the full three-dimensional conditioning vector is necessary or whether reduced regulatory input retains most of the achievable control.

P2.03: Conditioning dropout. Architecture and loss terms are fixed at the compact reference while only p_{drop} varies. The runs `P2.03a`, `P2.03b`, and `P2.03c` test $p_{\text{drop}} \in \{0.0, 0.05, 0.10\}$, where `P2.03c` corresponds to the reference setting and re-executes the configuration of `P2.01a`. The comparison evaluates whether conditioning dropout improves robustness or whether direct use of the full regulatory vector is sufficient.

P2.04: Capacity transfer. The capacity group tests whether the Phase I capacity pattern transfers to the conditioned model. `P2.04a` re-executes the 12-base-channel, depth-4 reference configuration of `P2.01a`, `P2.04b` reuses the best Phase I architecture, and `P2.04c` tests a wider 16-channel depth-4 variant. All three runs exclude coordinate channels, isolating capacity from the coordinate-channel variable.

P2.05: Loss balance. The loss-balance group revisits the most relevant Phase I loss variants in the conditioned setting. `P2.05a` tests surface emphasis at $\omega_s = 1.5$ with the default volume regulariser, `P2.05b` removes the volume regulariser ($\lambda_{\text{vol}} = 0.0$), and `P2.05c` combines surface emphasis with no volume regularisation.

P2.06: Evaluation-time sensitivity. The evaluation-time sensitivity group compares two inference settings for the same trained coordinate-free reference model. `P2.06a` uses direct inference, passing the voxel grid through the model once. `P2.06b` applies rotational TTA over 0°, 90°, 180°, and 270°, rotates the predictions back into the original orientation, and averages them before thresholding. The comparison tests whether rotational averaging improves validation metrics without retraining or changing the learned model.

5.3.3 Experiment Results

The Phase II experiments show that FiLM-based regulatory conditioning moves the 10-epoch validation results into a stronger band than the Phase I experiment series, although the leading variants are tightly clustered. The coordinate-free reference configuration is re-executed as the baseline of the dropout and capacity groups and reappears as the direct-inference entry of the evaluation-time sensitivity group, so identical values occur more than once in the overview. Table 5.4 summarises the validation metrics for all Phase II training runs and evaluation-time sensitivity entries.

Experiment	Run	IoU	Dice	Prec.	Recall	VE [voxel]	ZPE [voxel]
Reference	`P2.01a`	0.7412	0.8513	0.8209	0.8841	45.57	1.0306
Reference	`P2.01b`	0.7450	0.8538	0.8309	0.8781	42.36	0.9859
Channels	`P2.02a`	0.7429	0.8525	0.8228	0.8843	42.49	1.0495
Channels	`P2.02b`	0.7433	0.8527	0.8155	0.8935	50.89	1.0891
Channels	`P2.02c`	0.7241	0.8400	0.8141	0.8676	43.33	1.1647
Dropout	`P2.03a`	0.7487	0.8563	0.8291	0.8853	41.55	0.9822
Dropout	`P2.03b`	0.7447	0.8537	0.8261	0.8831	41.45	0.9676
Dropout	`P2.03c`	0.7412	0.8513	0.8209	0.8841	45.57	1.0306
Capacity	`P2.04a`	0.7412	0.8513	0.8209	0.8841	45.57	1.0306
Capacity	`P2.04b`	0.7437	0.8530	0.8230	0.8853	45.52	1.0291
Capacity	`P2.04c`	0.7434	0.8528	0.8236	0.8842	44.65	1.0121
Loss balance	`P2.05a`	0.7426	0.8523	0.8129	0.8956	53.43	1.1217
Loss balance	`P2.05b`	0.7353	0.8475	0.7887	0.9158	72.47	1.3613
Loss balance	`P2.05c`	0.7302	0.8441	0.7788	0.9213	80.96	1.4845
Eval sens.	`P2.06a`	0.7412	0.8513	0.8209	0.8841	45.57	1.0306
Eval sens.	`P2.06b`	0.7444	0.8535	0.8213	0.8884	46.35	1.0378

Table 5.4: Phase II Validation Metrics by Experiment

The no-coordinate reference `P2.01a` reaches a validation IoU of 0.7412, showing that the compact U-Net backbone remains effective in the FiLM-conditioned setting. The paired coordinate variant `P2.01b` raises IoU to 0.7450 and lowers volume and z-profile errors. In contrast to Phase I, where coordinate channels did not improve the unconditional model, their modest gain here suggests that explicit positional information may become more useful once the network also receives parcel-level regulatory targets.

Removing conditioning dropout produces the best-performing 10-epoch run, with `P2.03a` reaching the highest validation IoU in the series at 0.7487, ahead of `P2.03b` at 0.7447 and `P2.03c` at 0.7412. Relative to the full-conditioning reference, the no-dropout run reduces absolute volume error from 45.57 to 41.55 voxels and z-profile error from 1.0306 to 0.9822. This suggests that forcing the model to ignore the regulatory vector during training weakens the learned conditioning response in Phase II.

The experiment group focused on conditioning-channel composition shows that two-component regulatory subsets stay competitive, while GFZ-only conditioning is insufficient. `P2.02b` reaches a validation IoU of 0.7433 and `P2.02a` reaches 0.7429, whereas `P2.02c` drops to 0.7241. This suggests that the model benefits from complementary conditioning information, while GFZ alone is too compressed to distinguish between different combinations of footprint extent and vertical massing. The final rerun retains the full GRZ/GFZ/height vector.

Capacity changes stay within a narrow performance band, with P2.04b reaching a validation IoU of 0.7437 and P2.04c reaching 0.7434, both only slightly above the 12-channel depth-4 reference at 0.7412. Their volume and z-profile errors are also close, so changing depth or width shifts the error profile only marginally and does not outperform the compact backbone variants.

Loss-focused changes expose a trade-off between overlap quality and mass control, as surface emphasis raises validation IoU from 0.7412 to 0.7426 while simultaneously increasing volume and z-profile errors to 53.43 voxels and 1.1217. Removing the volume regulariser degrades the massing profile more strongly, with P2.05b falling to 0.7353 and P2.05c to 0.7302, while absolute volume error rises to 72.47 and 80.96 voxels.

The evaluation-time sensitivity group affects inference only and is considered separately from the training variants. Using the same weights with rotational TTA raises validation IoU from 0.7412 to 0.7444 and recall from 0.8841 to 0.8884, pointing to a modest inference-time gain without affecting training-run selection. For this reason, the improvement is treated as an evaluation setting, not a model change.

Overall, the Phase II experiments show that FiLM-based regulatory conditioning improves the compact U-Net relative to the Phase I 10-epoch experiment series. The best training result comes from removing conditioning dropout, while capacity changes, coordinate channels, loss-balance changes, and reduced regulatory vectors remain competitive in parts but do not outperform the full GRZ/GFZ/height no-dropout setup. Rotational TTA provides a modest additional inference-time gain. Figure 5.10 illustrates that the leading runs are tightly clustered, and P2.03a is selected for the final 50-epoch rerun.

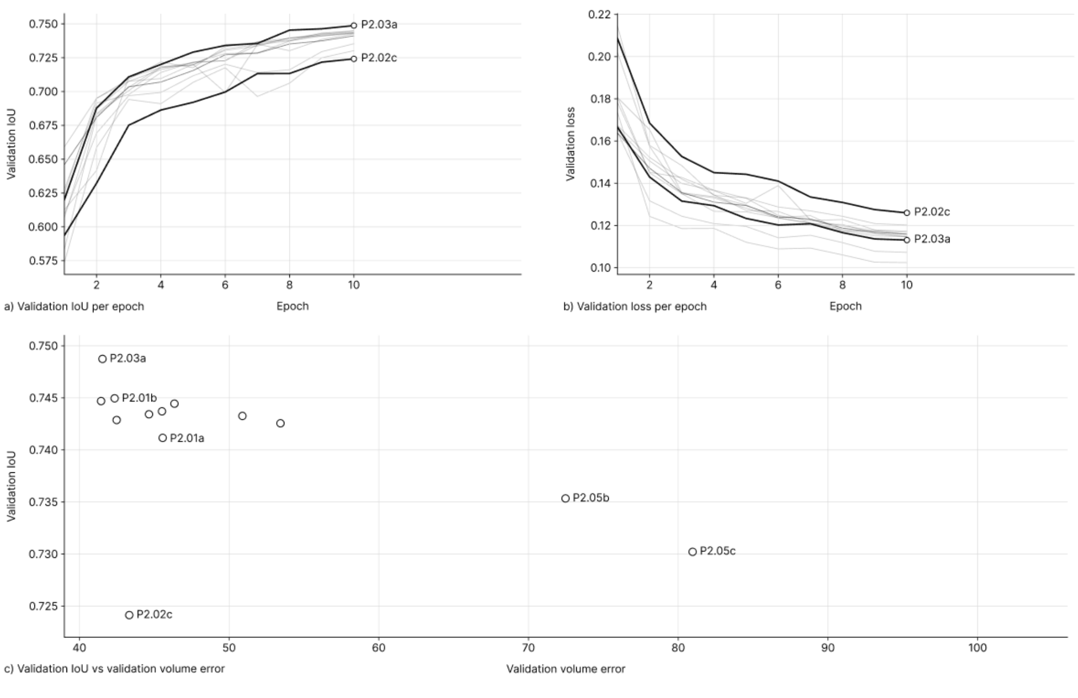

Figure 5.10: Phase II Validation Performance Across Variants

5.3.4 Final Model and Analysis

The final Phase II model is the 50-epoch rerun `P2-Final` and is summarised in Figure 5.11. It is a depth-4 conditional 3D U-Net with base width 12, full GRZ/GFZ/height conditioning, no coordinate channels, no conditioning dropout ($p_{\text{drop}} = 0.0$), surface weighting $\omega_s = 1.0$, and volume regularisation $\lambda_{\text{vol}} = 0.1$. The final checkpoint is the epoch-45 checkpoint selected by highest validation IoU and uses the validation-calibrated threshold $\hat{t} = 0.60$. Table 5.5 reports TTA-based validation metrics together with test metrics from direct non-TTA inference and rotational TTA, while the quantitative regulatory and controllability analyses use direct non-TTA inference to measure the model's conditioning response.

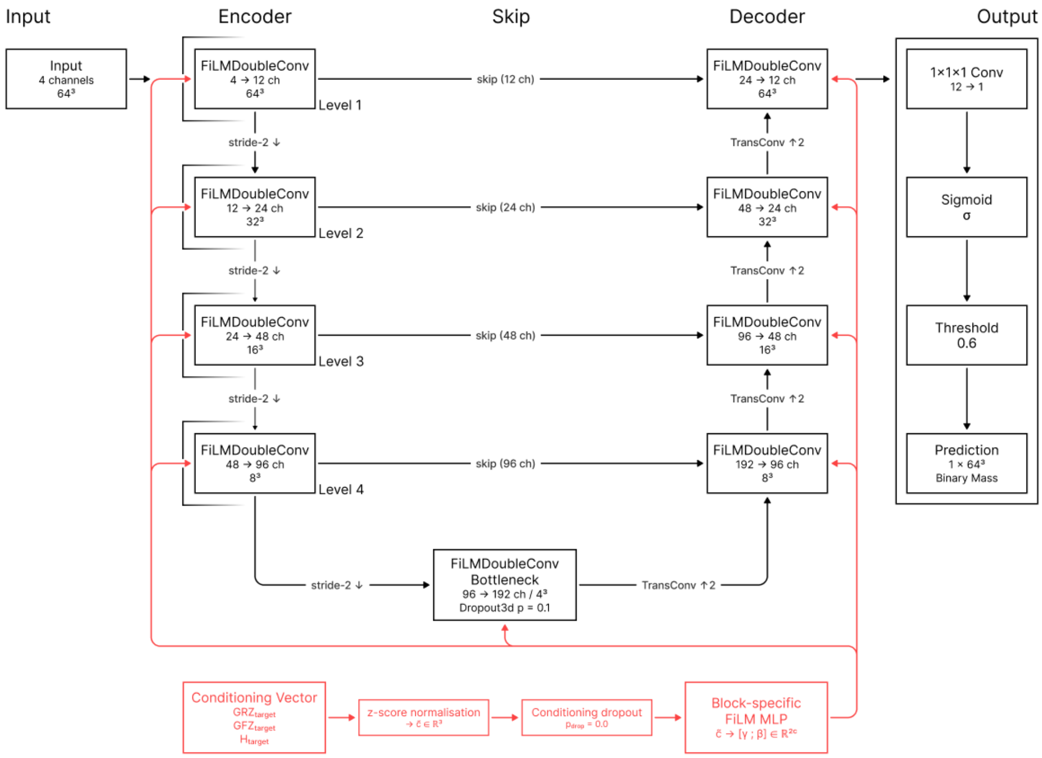

Figure 5.11: Phase II Final Conditional U-Net

Split	Inference	IoU	Dice	Prec.	Recall	ΔVol	VE [voxel]	ZPE [voxel]	Thresh.
Validation	rotational TTA	0.7819	0.8776	0.8583	0.8979	0.0741	31.15	0.8159	0.60
Test	non-TTA	0.7754	0.8735	0.8530	0.8950	0.0743	33.04	0.8554	0.60
Test	rotational TTA	0.7800	0.8764	0.8542	0.8998	0.0728	33.74	0.8640	0.60

Table 5.5: Phase II Final Segmentation Metrics

Relative to the 10-epoch run, validation IoU rises from 0.7487 to 0.7819 (+3.32 percentage points), while VE falls from 41.55 to 31.15 voxels (-10.40 voxels) and ZPE from 0.9822 to 0.8159 (-0.1663 voxels). The TTA-based validation–test IoU gap remains small at 0.0019, supporting stable generalisation to unseen parcels. The direct non-TTA test pass reaches an IoU of 0.7754 and a Dice score of 0.8735, with 33.04 voxels VE and 0.8554 ZPE. Compared with this direct pass, rotational TTA raises test IoU to 0.7800 (+0.46 percentage points), Dice to 0.8764 (+0.29 percentage points), and recall from 0.8950 to 0.8998, while VE and ZPE are slightly higher.

As shown in Figure 5.12, the qualitative analysis of the held-out test samples supports the quantitative results. Parcel placement, footprint extent, and vertical massing usually align with the neighbourhood context while responding to parcel-specific regulatory targets. Residual uncertainty more often appears as mild over-prediction than underfilling.

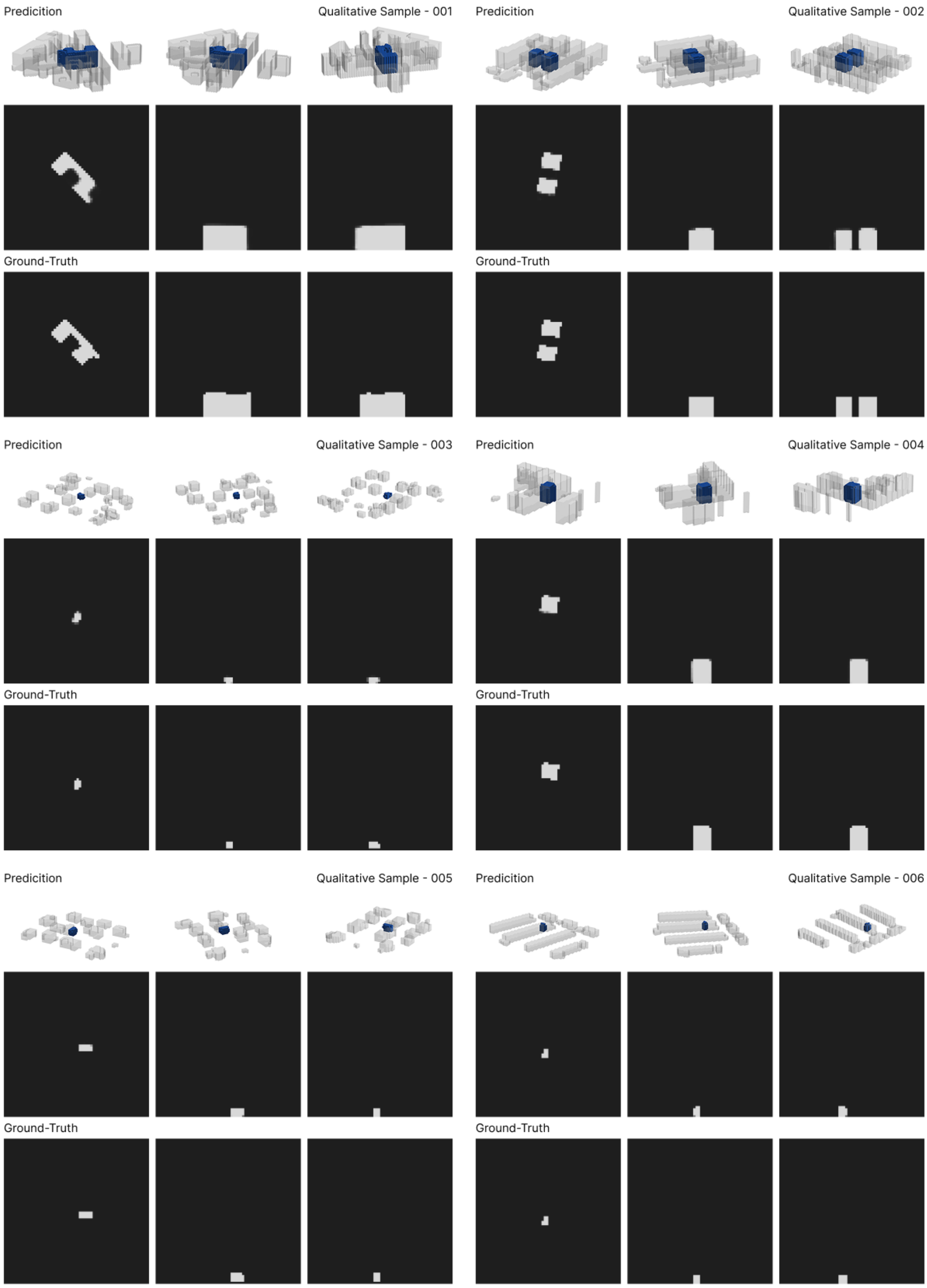

Figure 5.12: Phase II Predictions and Ground Truth

Regulatory evaluation uses the normalised conditioning vector derived from the native ground-truth regulatory targets and a direct non-TTA pass through the final checkpoint. Table 5.6 compares the mean predicted regulatory values with the corresponding native reference targets and reports the adjusted relative error ε_{adj}. Lower values denote closer agreement after accounting for the resolution tolerance introduced in Section 5.1.1. On the held-out test split, mean adjusted relative error stays below 0.08 for GRZ, GFZ, and height, demonstrating close target alignment under native conditioning.

Metric	Mean pred.	Mean ref. $\mathbf{c}^{0.5\,\mathrm{m}}$	Mean ε_{adj}	Median ε_{adj}
GRZ	0.3908	0.3765	0.0662	0.0349
GFZ	1.9320	1.8238	0.0706	0.0521
Height [m]	13.44	12.52	0.0187	0.0000

Table 5.6: Phase II Regulatory Errors by Metric

The strongest agreement appears in building height, where a median adjusted error of zero indicates that at least half of the test parcels fall within the one-slice comparison tolerance on the 2.0 m evaluation grid. GRZ and GFZ also remain close to their parcel-specific targets, with mean adjusted errors of 0.0662 and 0.0706. This suggests that conditioning improves direct density control alongside geometric overlap, although the mean predicted values remain slightly above the native references for all three metrics. A substantial share of these offsets reflects the resolution shift measured in Section 4.3, which accounts for $+0.0067$ of the $+0.0143$ GRZ difference, $+0.0467$ of the $+0.1082$ GFZ difference, and $+0.56$ m of the $+0.92$ m height difference in Table 5.7. The remaining model-specific component is correspondingly smaller than the raw mean differences suggest. Figure 5.13 summarises this regulatory agreement.

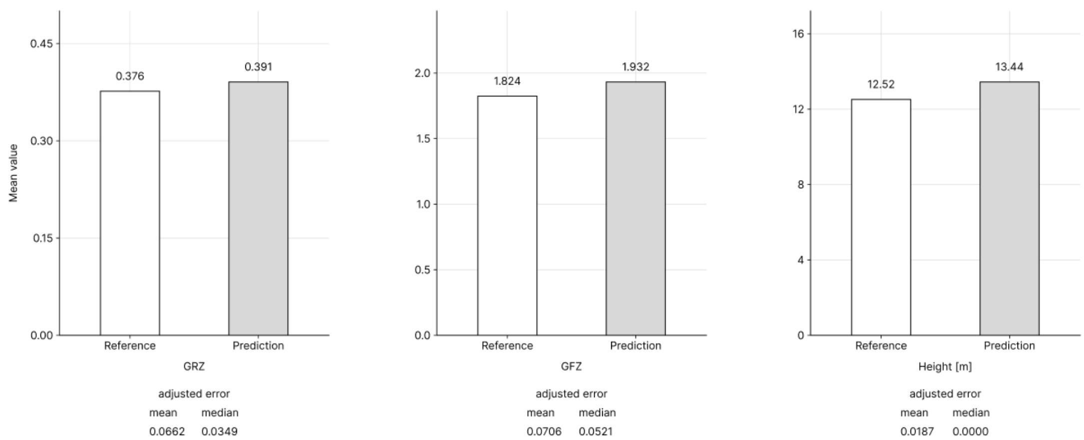

Figure 5.13: Phase II Regulatory Summary Metrics

With the parcel context fixed, Table 5.7 reports the model response under joint -10% and $+10\%$ scaling of GRZ, GFZ, and height, where *Pred.* denotes the mean predicted value, *Cond.* the scaled conditioning target, and Δ their difference. Mean predicted values decrease under -10% and increase under $+10\%$ conditioning across all three metrics, revealing a monotonic response to the regulatory target vector. The strongest response appears for GFZ, whose predicted range of 0.3768 closely matches the conditioning-target span of 0.3648. GRZ and height follow more conservatively, suggesting a target response moderated by the learned spatial prior. Figure 5.14 visualises selected samples across the wider -20% to $+20\%$ conditioning sweep.

	GRZ			GFZ			Height		
Scenario	Pred.	Cond.	Δ	Pred.	Cond.	Δ	Pred.	Cond.	Δ (m)
-10% **cond.**	0.3806	0.3388	0.0418	1.7443	1.6414	0.1029	12.72	11.27	1.45
Native GT	0.3908	0.3765	0.0143	1.9320	1.8238	0.1082	13.44	12.52	0.92
$+10\%$ **cond.**	0.4070	0.4141	-0.0071	2.1211	2.0062	0.1149	14.36	13.77	0.59

Table 5.7: Phase II Response to Joint Target Scaling

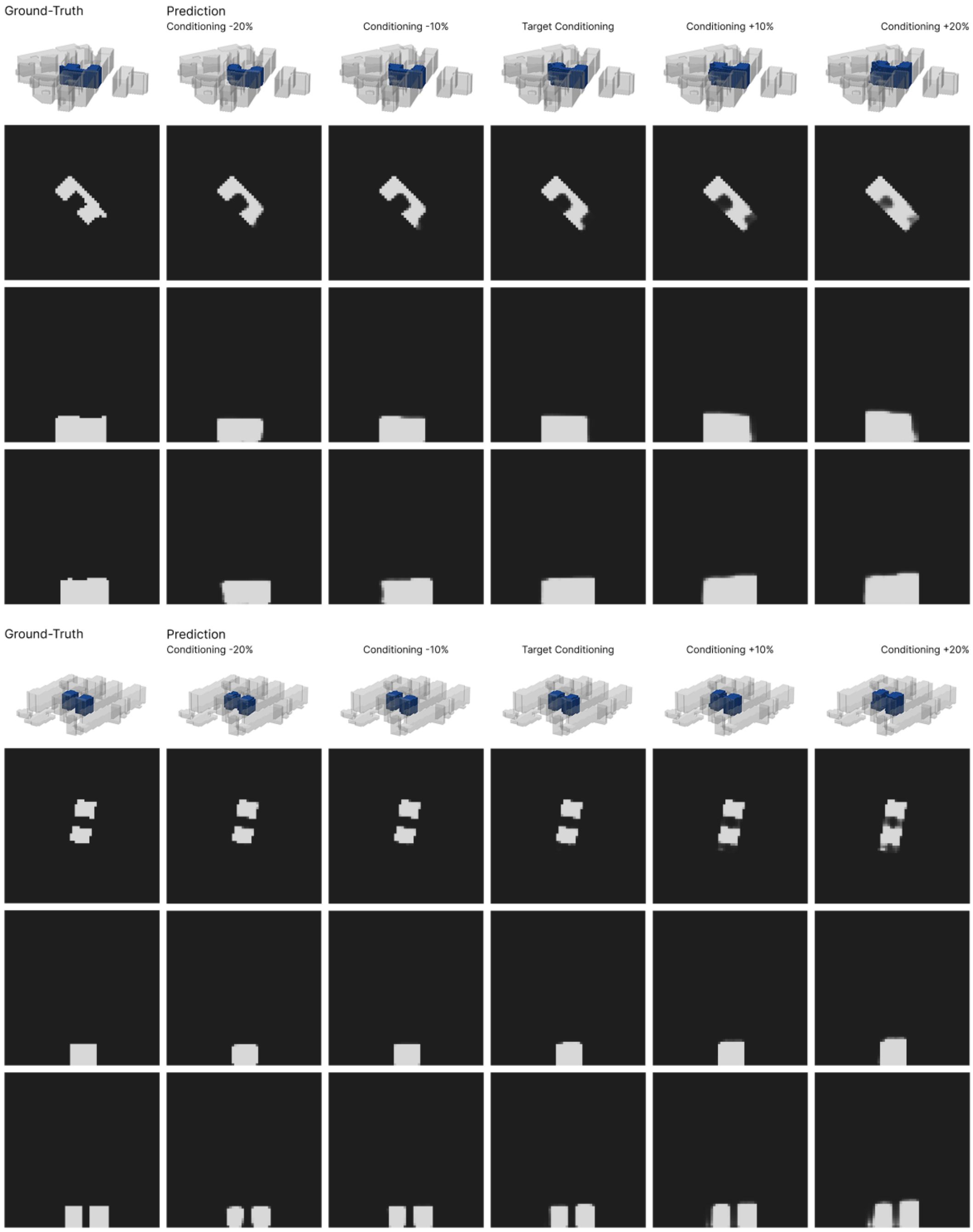

Figure 5.14: Phase II Target-Conditioning Sample Sweep

To examine variable-specific sensitivity, the final checkpoint is reevaluated through isolated -10% and $+10\%$ changes, where one conditioning value is varied while the other two are fixed. As Table 5.8 shows, all three variables respond monotonically, while GFZ and height capture 65.4% and 61.2% of the requested span, compared with 34.0% for GRZ. Figure 5.15 compares these isolated responses with the coupled target sweep. The lower GRZ response is plausible because footprint extent is strongly constrained by parcel geometry and surrounding context.

Controlled input	Pred. -10%	Pred. GT	Pred. $+10\%$	Target span	Pred. span	Span captured
GRZ	0.3814	0.3908	0.4070	0.0753	0.0256	34.0%
GFZ	1.8179	1.9320	2.0565	0.3648	0.2386	65.4%
Height [m]	12.76	13.44	14.30	2.50	1.53	61.2%

Table 5.8: Phase II Isolated Target Sensitivity

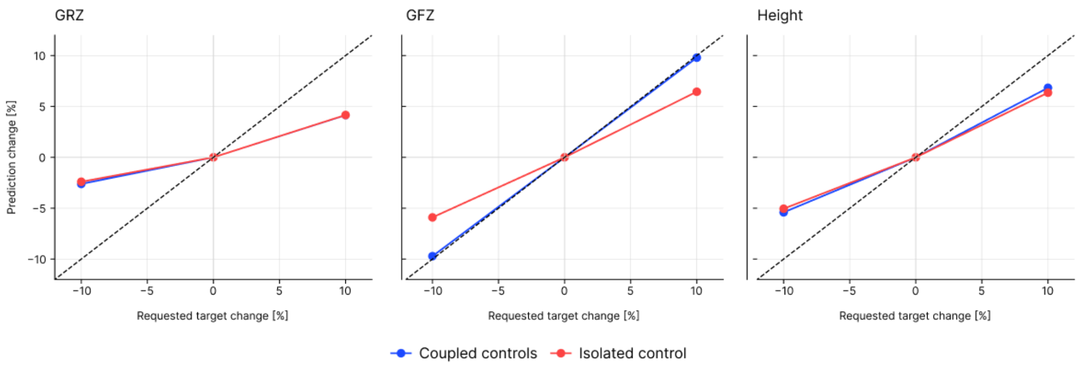

Figure 5.15: Phase II Coupled and Isolated Target Response

Taken together, the Phase II results show reliable regulatory steering for a fixed spatial context and conditioning vector $(\mathbf{X}, \tilde{\mathbf{c}})$. The model returns a single target-aligned prediction, but its response differs across GRZ, GFZ, and height. This reflects the coupled structure of the regulatory metrics, as GFZ corresponds approximately to GRZ multiplied by the number of full storeys, which is itself tied to building height through the assumed storey height.

5.3.5 Supplementary Threshold Sensitivity Analysis

This supplementary analysis examines whether stricter evaluation-time binarisation reduces the mild over-prediction of the final Phase II checkpoint. The regular comparative evaluation is fixed at the shared validation-based threshold of $\hat{t} = 0.60$. The analysis is used only as a diagnostic on the held-out test split and does not affect model selection or any within-phase or cross-phase comparison. Segmentation metrics are reevaluated under the retained rotational-TTA protocol, while the supplementary regulatory analysis uses the same threshold values through the direct non-TTA regulatory pipeline.

Threshold	GRZ rel. adj.	GFZ rel. adj.	H rel. adj.	GRZ abs.	GFZ abs.	H abs. [m]
0.60	0.0662	0.0706	0.0187	0.0361	0.1528	0.9914
0.65	0.0646	0.0687	0.0186	0.0351	0.1448	0.9823
0.70	0.0632	0.0671	0.0186	0.0342	0.1369	0.9747
0.75	0.0620	0.0659	0.0186	0.0333	0.1291	0.9666
0.80	**0.0609**	**0.0653**	**0.0186**	**0.0324**	**0.1221**	**0.9605**

Table 5.9: Phase II Threshold Regulatory Errors

Threshold	IoU	Dice	Prec.	Recall	ΔVol	VE [voxel]	ZPE [voxel]
0.60	0.7800	0.8764	0.8542	**0.8998**	0.0728	33.74	0.8640
0.65	0.7804	0.8766	0.8579	0.8962	**0.0715**	31.50	0.8372
0.70	**0.7805**	**0.8767**	0.8618	0.8922	**0.0715**	29.44	0.8145
0.75	0.7802	0.8766	0.8657	0.8877	0.0717	27.68	0.7945
0.80	0.7797	0.8762	**0.8702**	0.8824	0.0733	**26.17**	**0.7796**

Table 5.10: Phase II Threshold Segmentation Metrics

Table 5.10 shows precision rising from 0.8542 to 0.8702 and recall falling from 0.8998 to 0.8824 across $\hat{t} = 0.60$–0.80. Test IoU reaches a local maximum of 0.7805 at $\hat{t} = 0.70$ before declining slightly to 0.7797 at $\hat{t} = 0.80$. VE and ZPE decrease more consistently, falling from 33.74 to 26.17 voxels and from 0.8640 to 0.7796 across the threshold range. Figure 5.16 shows that higher thresholds trim outer voxels in denser samples while leaving placement, orientation, and typological character unchanged.

Regulatory agreement under native ground-truth conditioning $\mathbf{c}^{0.5\,\mathrm{m}}$ also improves moderately with stricter thresholds. GRZ and GFZ adjusted relative errors fall from 0.0662 to 0.0609 and from 0.0706 to 0.0653, while height remains almost unchanged at approximately 0.0186–0.0187. Absolute errors decline in parallel (Table 5.9).

The threshold comparison shows a small divergence between overlap quality, which peaks near $\hat{t} = 0.70$, and regulatory closeness, which is strongest at $\hat{t} = 0.80$. This suggests that the final Phase II checkpoint is slightly permissive at the shared threshold.

Overall, the Phase II results show that the model preserves parcel-scale plausibility while adding regulatory control. However, for a fixed context and conditioning vector, it still produces a single prediction. This motivates Phase III, which keeps the same context and regulatory framing but tests controlled variation through a latent diffusion pipeline.

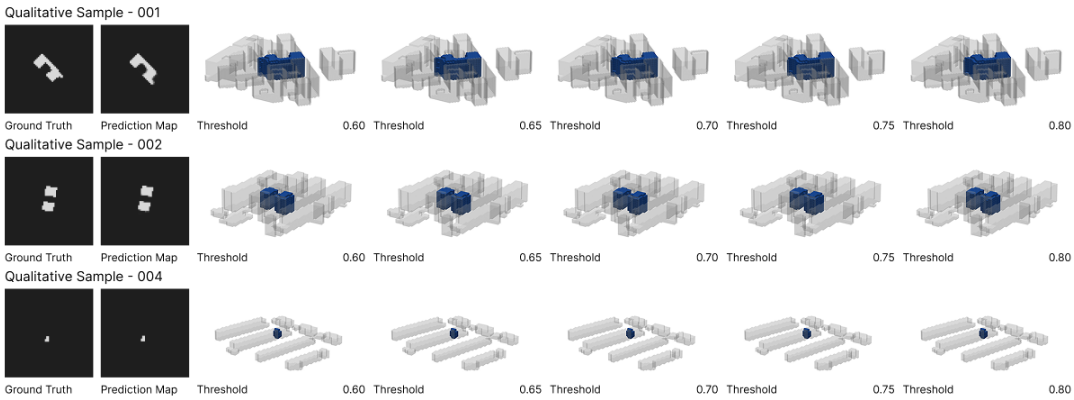

Figure 5.16: Phase II Predictions Across Thresholds

5.4 Phase III: Latent Diffusion Model

Phase III shifts from deterministic target-conditioned prediction to the sampling of diverse variants. It tests whether a two-stage latent diffusion pipeline can produce multiple massing variants under identical GRZ, GFZ, height, and context conditions. The phase replaces direct voxel-space prediction with a variational autoencoder (VAE), which encodes binary building volumes into a continuous latent representation, and a conditional denoising diffusion model, which samples from this latent space under spatial and regulatory conditioning. This combination makes multi-output sampling tractable within the available computational budget. Unlike Phases I and II, the VAE and diffusion stages are selected by their respective validation losses, with decoded geometry evaluated afterward under the shared protocol.

5.4.1 Objective and Architecture

To extend target-conditioned prediction toward conditional sampling, the model represents a distribution over plausible building volumes for a fixed spatial context $\mathbf{X}$ and regulatory vector $\tilde{\mathbf{c}}$. Here, $Y \in \{0,1\}^{64\times64\times64}$ denotes the target occupancy volume, $\mathbf{X} \in \mathbb{R}^{C_X\times64\times64\times64}$ the spatial context tensor, and $\tilde{\mathbf{c}} \in \mathbb{R}^3$ the normalised regulatory conditioning vector from Equation (20). Conditional sampling is factorised into a latent representation $\mathbf{z}$, a decoder that maps latent codes back to voxel space, and a conditional latent model that samples $\mathbf{z}$ under spatial and regulatory conditioning,

$$p(Y \mid \mathbf{X}, \tilde{\mathbf{c}}) = \int p_\theta(Y \mid \mathbf{z})\, p_\psi(\mathbf{z} \mid \mathbf{X}, \tilde{\mathbf{c}})\, d\mathbf{z}. \tag{33}$$

Here, p_θ is implemented by the VAE decoder and p_ψ by the conditional diffusion model. Training proceeds in two stages, with Stage A learning the latent representation from target building volumes and Stage B learning conditional sampling in the fixed latent space.

Stage A – Latent Representation Learning. Stage A trains a three-dimensional variational autoencoder on binary target occupancy volumes and maps them to a continuous latent code $\mathbf{z}$, which defines the representation space for Stage B.

VAE encoder–decoder. The VAE maps the binary occupancy volume Y to an encoder-predicted Gaussian posterior $q_\phi(\mathbf{z} \mid Y)$ over latent codes, following the VAE formulation of Kingma and Welling (2013). The encoder uses DoubleConv3D blocks and stride-2 convolutions without skip connections, so relevant information must pass through the latent bottleneck. The chosen latent resolution R_z sets both compression strength and encoder–decoder depth, with the number of stages defined as $d_{\mathsf{VAE}} = \log_2(64/R_z) + 1$.

Parallel $1 \times 1 \times 1$ heads produce $\boldsymbol{\mu}$ and $\log \boldsymbol{\sigma}^2$ on a C_z-channel latent grid. The decoder mirrors the encoder through DoubleConv3D blocks and transposed convolutions back to 64^3, ending in an occupancy-logit head, as summarised in Figure 5.17. Sigmoid activation and binarisation are applied during inference.

Differentiable latent sampling. During VAE training, latent samples are drawn from the predicted posterior using the reparameterised sampling rule

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}). \tag{34}$$

Here, $\boldsymbol{\sigma} = \exp(0.5 \cdot \log \boldsymbol{\sigma}^2)$ and $\odot$ denotes elementwise multiplication. This keeps sampling differentiable and preserves gradient flow during VAE optimisation. Stage B diffusion training samples latents from the frozen VAE encoder in the same way, while diffusion validation uses posterior mean latents $\mathbf{z} = \boldsymbol{\mu}$ to reduce stochastic variation in checkpoint comparison.

VAE training objective. Stage A optimises a variational reconstruction objective that combines occupancy reconstruction with regularisation toward a standard Gaussian latent prior. The objective is defined as

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \lambda_{\text{KL}}\, \mathcal{L}_{\text{KL}}. \tag{35}$$

The reconstruction term $\mathcal{L}_{\text{recon}}$ measures how closely the decoded occupancy logits match the ground-truth volume. It is computed as voxelwise binary cross-entropy with logits and averaged over all voxel elements. The KL term $\mathcal{L}_{\text{KL}}$ regularises the latent space by penalising deviations of the encoder-predicted posterior $q_\phi(\mathbf{z} \mid Y)$ from the standard Gaussian prior $\mathcal{N}(0, \mathbf{I})$. This latent divergence is summed over latent elements and scaled by the number of voxel elements in the input volume Y.

$$\mathcal{L}_{\text{recon}} = \frac{1}{N_x} \sum_{i=1}^{N_x} \ell_{\text{BCE}}(\hat{Y}_i, Y_i), \tag{36}$$

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2N_x} \sum_j \left(1 + \log \sigma_j^2 - \mu_j^2 - \exp(\log \sigma_j^2)\right), \tag{37}$$

where $N_x = \mathrm{numel}(Y)$ for the current batch. The coefficient λ_{KL} controls the trade-off between reconstruction fidelity and latent regularity, with smaller values favouring accurate reconstruction and larger values enforcing stronger regularisation.

To prevent the KL term from dominating too early, λ_{KL} can be linearly increased over the first E_{anneal} epochs,

$$\lambda_{\text{KL},e} = \begin{cases} \lambda_{\text{KL}}^{\max}, & E_{\text{anneal}} = 0, \\ \min\left(1, \dfrac{e+1}{E_{\text{anneal}}}\right) \lambda_{\text{KL}}^{\max}, & E_{\text{anneal}} > 0. \end{cases} \tag{38}$$

where e is the zero-based epoch index and $\lambda_{\text{KL}}^{\max}$ is the target KL coefficient. During annealed runs, λ_{KL} in Equation 35 is replaced by the epoch-dependent value $\lambda_{\text{KL},e}$.

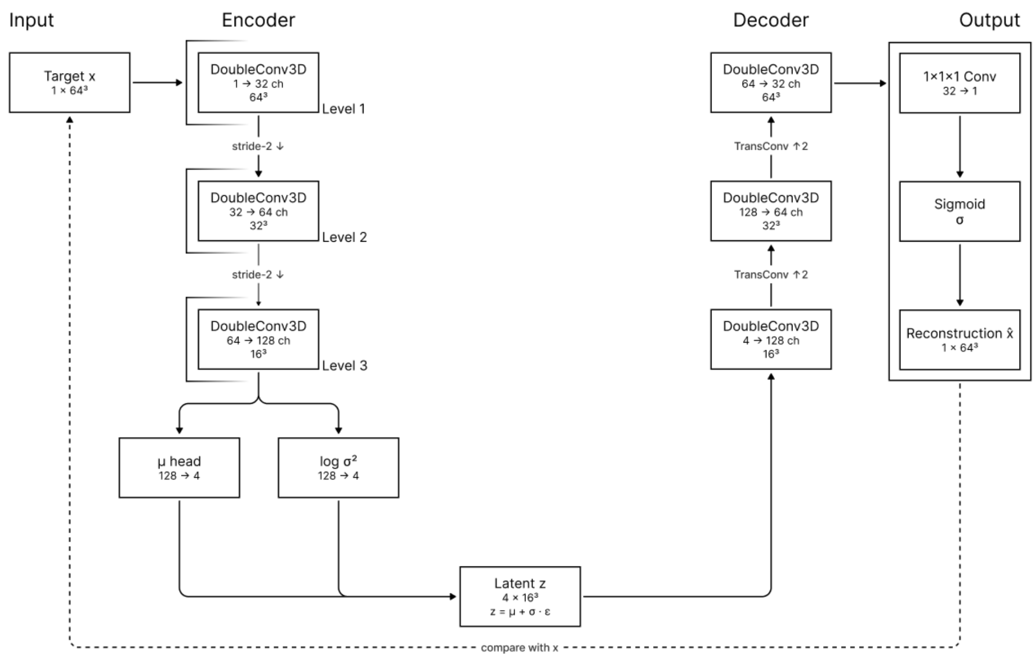

Figure 5.17: Phase III Stage A Reference VAE Architecture

Stage B – Conditional Latent Diffusion. Stage B trains a conditional denoising diffusion model in the 32^3 latent representation selected in the Stage A comparison and learned by Stage A. It receives a noisy latent tensor, diffusion timestep, spatial context tensor, and normalised regulatory vector.

Noise schedule and forward process. The forward process adds Gaussian noise to a clean latent code $\mathbf{z}_0$ over a finite sequence of timesteps $t \in \{0, \ldots, T-1\}$. The Phase III experiments vary the noise schedule and timestep horizon, comparing cosine and linear schedules with 1000-step and 500-step variants. For a given schedule, the cumulative coefficients are defined as $\bar{\alpha}_t = \prod_{s=0}^{t}(1-\beta_s)$, with $\bar{\alpha}_{-1} = 1$ used implicitly at the final DDIM step. Here, β_s denotes the diffusion noise-schedule coefficient and is unrelated to the FiLM shift vector $\boldsymbol{\beta}_b$ introduced in Equations (29) and (30). The noisy latent is sampled as

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{z}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}). \tag{39}$$

During training, each batch samples t and $\boldsymbol{\epsilon}$, constructs $\mathbf{z}_t$, and trains the network to recover the added noise. Stage B uses the frozen Stage A VAE for latent encoding and decoding throughout training, validation, and sampling. Training uses reparameterised encoder samples, validation uses posterior mean latents $\mathbf{z}_0 = \boldsymbol{\mu}$, and DDIM sampling decodes sampled latent codes through the frozen Stage A decoder.

DoubleConvCond block. The block shown in Figure 5.18 extends Phase II's FiLMDoubleConv with timestep conditioning. Spatial context instead enters through channel-wise concatenation with the noisy latent before the encoder and propagates through the encoder–decoder hierarchy.

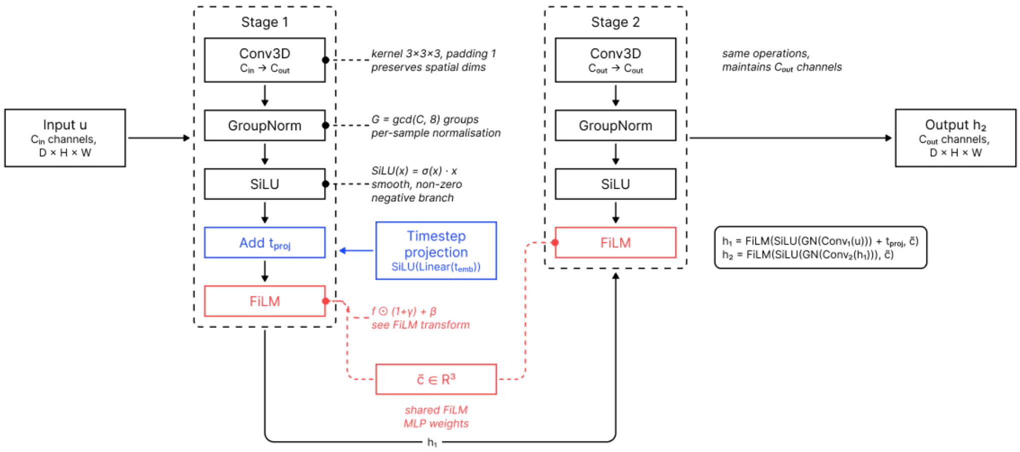

Figure 5.18: Phase III Conditional Convolution Block

The timestep is encoded through a sinusoidal embedding and a two-layer MLP, producing $\mathbf{t}_{\text{emb}} \in \mathbb{R}^{256}$. This global timestep embedding is shared across the network, while each block learns its own projection to the local channel width C, allowing the same timestep to affect different feature scales differently. The projected embedding is reshaped and broadcast over the spatial dimensions before being added to the feature map. With $\mathbf{u}$ as block input, the first stage combines local feature extraction, timestep injection, and regulatory FiLM modulation,

$$\mathbf{h}_1 = \text{FiLM}(\text{SiLU}(\text{GN}(\text{Conv}_1(\mathbf{u}))) + \text{SiLU}(\text{Linear}(\mathbf{t}_{\text{emb}})),\ \tilde{\mathbf{c}})\,, \tag{40}$$

The second stage applies convolution, normalisation, activation, and FiLM modulation without an additional timestep projection. The timestep information is carried forward through $\mathbf{h}_1$, yielding

$$\mathbf{h}_2 = \text{FiLM}(\text{SiLU}(\text{GN}(\text{Conv}_2(\mathbf{h}_1))),\ \tilde{\mathbf{c}})\,. \tag{41}$$

Together, the two stages allow each block to combine local latent features from the convolutional path, the current diffusion timestep, and the regulatory conditioning vector $\tilde{\mathbf{c}}$. FiLM modulation is applied after both convolutional stages, so the regulatory signal affects both the intermediate representation $\mathbf{h}_1$ and the final block output $\mathbf{h}_2$. This lets the diffusion U-Net adapt its denoising operation to both the current noise level and the requested regulatory target.

DiffusionUNet3D architecture. Figure 5.19 summarises the Phase III Stage B reference configuration, a three-dimensional U-Net operating on the fixed 32^3 latent representation learned by Stage A with $C_z = 4$ latent channels. It receives the noisy latent tensor $\mathbf{z}_t$, timestep t, regulatory vector $\tilde{\mathbf{c}}$, and spatial context tensor $\mathbf{X}$. The spatial context is resized to the latent resolution by trilinear interpolation when necessary and concatenated channel-wise with the noisy latent before the encoder.

The encoder reduces resolution from 32^3 to 4^3 through three stride-2 DoubleConvCond levels, followed by a conditioned bottleneck. The decoder mirrors this path with transposed-convolution upsampling, skip concatenation, and DoubleConvCond fusion blocks. Timestep conditioning and FiLM modulation remain active throughout the encoder, bottleneck, and decoder. A final $1 \times 1 \times 1$ convolution outputs the predicted noise $\hat{\boldsymbol{\epsilon}}$ with the same spatial and channel dimensions as the input latent. The first-stage input width is $C_z + C_X$, giving 8 input channels in the coordinate-free setting ($C_z = 4$, $C_X = 4$) and 11 input channels when coordinate maps are added ($C_X = 7$).

The Stage B model omits bottleneck Dropout3d. Conditioning dropout regularises training, while EMA-smoothed weights stabilise sampling and evaluation.

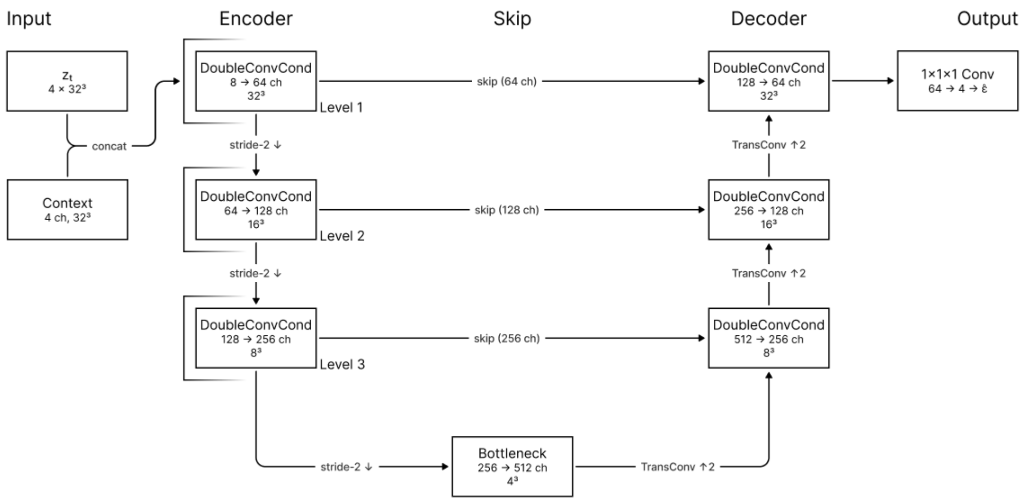

Figure 5.19: Phase III Stage B Reference Diffusion U-Net

Diffusion training objective. The diffusion model predicts the noise added to a latent code and minimises the mean squared error between true and predicted noise following the ϵ-parameterisation of Ho et al. (2020),

$$\mathcal{L}_{\text{diff}} = \frac{1}{N_z} \sum_{i=1}^{N_z} \left(\epsilon_i - \hat{\epsilon}_{\psi,i}(\mathbf{z}_t, t, \mathbf{X}, \tilde{\mathbf{c}})\right)^2, \quad N_z = B\, C_z\, R_z^3. \tag{42}$$

Here, the loss is averaged over all latent elements in the current batch. Unlike the BCE–soft-Dice objective in Phases I and II, this objective operates entirely in latent space, so decoded voxel-space metrics are computed only in a separate evaluation pass.

Mean-conditional conditioning dropout. Stage B applies a per-sample Bernoulli mask to the full regulatory vector during training,

$$\tilde{\mathbf{c}}_b^{\text{train}} = m_b \cdot \tilde{\mathbf{c}}_b, \quad m_b \sim \text{Bernoulli}(1 - p_{\text{drop}}), \tag{43}$$

so the network learns denoising under both the requested regulatory vector and the training-set mean regulatory condition. Z-score normalisation maps the training-set mean to $\tilde{\mathbf{c}} = \mathbf{0}$, so the dropped branch is not semantically unconditional. For this reason, the guidance rule is interpreted as mean-conditional guidance, not true classifier-free guidance,

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_{\text{mean}} + s\left(\boldsymbol{\epsilon}_{\text{cond}} - \boldsymbol{\epsilon}_{\text{mean}}\right), \tag{44}$$

where $\boldsymbol{\epsilon}_{\text{mean}}$ is the prediction obtained with $\tilde{\mathbf{c}} = \mathbf{0}$, which corresponds to the training-set mean under z-score normalisation, and $\boldsymbol{\epsilon}_{\text{cond}}$ is the prediction obtained with the requested regulatory vector. The guidance parameter $s \geq 1$ controls the strength of guidance. Larger values extrapolate away from the mean regulatory condition toward the requested target vector and can reduce sample diversity. Unless stated otherwise, Stage B evaluations use $s = 2.0$.

Exponential moving average. Stage B maintains an exponential moving average (EMA) of the diffusion parameters with decay rate $\rho_{\text{EMA}} = 0.999$. This keeps a smoothed copy of the model weights across training updates and is used to stabilise sampled outputs. EMA weights are used for inference and evaluation when available.

Inference and sampling. Sampling starts from a latent tensor sampled from $\mathcal{N}(0, \mathbf{I})$ at the highest sampled diffusion index. DDIM sampling is then applied with $S = 100$ steps and $\eta = 0$, following Song et al. (2020). The sampling steps are chosen approximately uniformly over the timestep range $0, \ldots, T-1$ and traversed in reverse order, so denoising proceeds from high to low noise and ends at $t = 0$.

After the reverse process reaches $t = 0$, the implementation applies the implicit final DDIM step with $\bar{\alpha}_{-1} = 1$. At each step, the model predicts the noise component and updates the latent toward a less noisy state. With $\eta = 0$, this trajectory is deterministic for fixed initial noise, spatial context $\mathbf{X}$, and regulatory vector $\tilde{\mathbf{c}}$. The estimated clean latent $\hat{\mathbf{z}}_0$ is decoded by the frozen VAE into occupancy logits, mapped through a sigmoid, and thresholded to obtain the building-mass prediction. Figure 5.20 illustrates this pipeline.

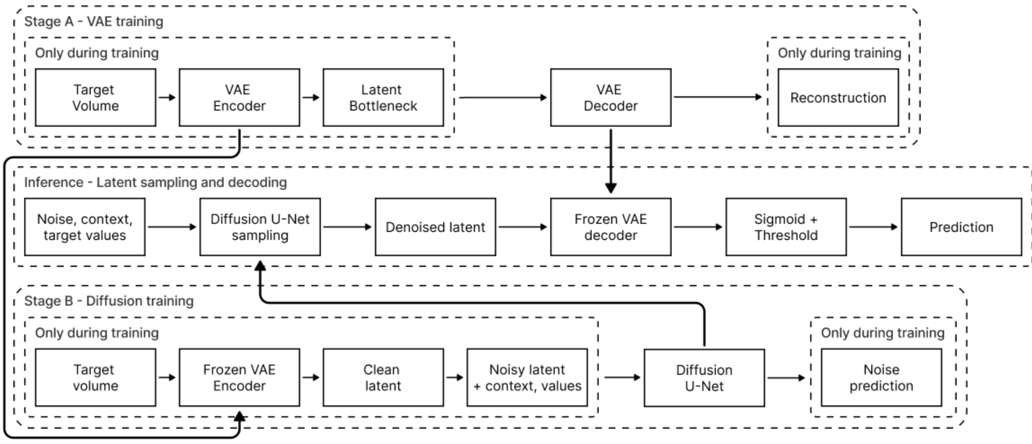

Figure 5.20: Simplified Phase III Training and Inference Pipelines

5.4.2 Experiment Setup

Phase III comprises six experiment groups with nineteen 10-epoch experiment runs and two 50-epoch reruns. Three of the Stage B runs, `P3B.04a`, `P3B.05a`, and `P3B.06a`, re-execute the Stage B reference configuration `P3B.03a` within their experiment groups and reproduce its checkpoint and validation loss under the fixed seed. Unless noted otherwise, all runs follow the shared setup in Section 5.1.2 and the unified evaluation protocol in Section 5.1.1. The 10-epoch Stage A experiments train the VAE on target volumes Y with a learning rate of 3×10^{-4} and an effective batch size of 8, while the retained 50-epoch Stage A rerun uses 2×10^{-4} and an effective batch size of 4. Stage B uses the selected Stage A VAE as a frozen encoder–decoder and trains the diffusion model with $\mathbf{X}$ and $\tilde{\mathbf{c}}$ using a learning rate of 2×10^{-4} and an effective batch size of 8. Stage A is selected by minimum validation $\mathcal{L}_{\text{VAE}}$, Stage B by minimum validation $\mathcal{L}_{\text{diff}}$, and IoU is computed only in the post-training geometric evaluation.

P3A.01: Stage A latent compression and KL regularisation. The first Stage A group varies the latent bottleneck. `P3A.01a` defines the 16^3 four-channel reference with KL coefficient $\lambda_{\text{KL}} = 3 \times 10^{-4}$, `P3A.01b` and `P3A.01d` vary latent resolution, and `P3A.01c` increases this coefficient to $\lambda_{\text{KL}} = 1 \times 10^{-3}$.

P3A.02: Stage A VAE capacity. The second Stage A group fixes the bottleneck at 16^3 and varies encoder–decoder capacity. `P3A.02a` defines the reference, `P3A.02b` reduces base width, `P3A.02c` increases it, and `P3A.02d` tests additional latent channels. The selected VAE configuration is rerun for 50 epochs as `P3A-Final` and frozen for all Stage B experiments.

P3B.03: Stage B diffusion schedule and horizon. The first Stage B group evaluates schedule and horizon under the frozen VAE. `P3B.03a` defines the cosine 1000-step reference, `P3B.03b` replaces the schedule with a linear one, and `P3B.03c` shortens the horizon to 500 steps. DDIM sampling with 100 steps, $\eta = 0$, and guidance scale 2.0 remains fixed.

P3B.04: Stage B diffusion U-Net capacity. The capacity group tests the denoising backbone. `P3B.04a` re-executes the reference configuration, `P3B.04b` trades width for depth, and `P3B.04c` widens the network at the same depth.

P3B.05: Stage B context encoding sensitivity. Coordinate sensitivity is revisited in the stochastic setting. `P3B.05a` re-executes the coordinate-free reference configuration, while `P3B.05b` appends the three normalised coordinate channels.

P3B.06: Stage B mean-conditional conditioning dropout. The conditioning-dropout runs test $p_{\text{drop}} \in \{0.02, 0.05, 0.10\}$ using `P3B.06a`, `P3B.06b`, and `P3B.06c`, where `P3B.06a` corresponds to the reference setting of $p_{\text{drop}} = 0.02$ and re-executes `P3B.03a`. The range remains low, reflecting the Phase II result that stronger masking did not improve performance, while Stage B still requires non-zero dropout to learn the mean-conditional branch used in guided sampling.

5.4.3 Experiment Results

The Phase III experiments evaluate how effectively the Stage A VAE compresses target building volumes into a stable latent representation and how reliably the Stage B diffusion model learns conditional denoising within this frozen latent representation. The reported values in Table 5.11 are stage-specific validation losses used for checkpoint selection. They are not directly comparable between Stage A and Stage B, as $\mathcal{L}_{\text{VAE}}$ and $\mathcal{L}_{\text{diff}}$ measure different objectives. The Stage B reference configuration is re-executed as the baseline of the capacity, context-encoding, and conditioning-dropout groups, so its validation loss of 0.2552 appears four times in the overview.

Stage – Experiment	Run	Best val. loss
Stage A – Compression	`P3A.01a`	6.07×10^{-5}
Stage A – Compression	`P3A.01b`	2.11×10^{-4}
Stage A – Compression	`P3A.01c`	6.36×10^{-5}
Stage A – Compression	`P3A.01d`	1.88×10^{-6}
Stage A – Capacity	`P3A.02a`	5.65×10^{-5}
Stage A – Capacity	`P3A.02b`	8.58×10^{-5}
Stage A – Capacity	`P3A.02c`	1.68×10^{-5}
Stage A – Capacity	`P3A.02d`	1.91×10^{-5}
Stage B – Schedule	`P3B.03a`	0.2552
Stage B – Schedule	`P3B.03b`	0.1300
Stage B – Schedule	`P3B.03c`	0.2594
Stage B – Capacity	`P3B.04a`	0.2552
Stage B – Capacity	`P3B.04b`	0.2736
Stage B – Capacity	`P3B.04c`	0.2583
Stage B – Coords	`P3B.05a`	0.2552
Stage B – Coords	`P3B.05b`	0.2684
Stage B – Cond. dropout	`P3B.06a`	0.2552
Stage B – Cond. dropout	`P3B.06b`	0.2572
Stage B – Cond. dropout	`P3B.06c`	0.2570

Table 5.11: Phase III Validation Losses by Experiment. Runs `P3B.04a`, `P3B.05a`, and `P3B.06a` re-execute the reference configuration `P3B.03a`.

Stage A results. Latent compression has the strongest effect under the combined VAE validation objective. The expanded 32^3 configuration `P3A.01d` achieves the lowest validation $\mathcal{L}_{\text{VAE}}$ at 1.88×10^{-6}, substantially below the 16^3 reference at 6.07×10^{-5} and the 8^3 variant at 2.11×10^{-4}. This suggests that excessive spatial compression removes information needed to reconstruct sparse and sharply bounded occupancy volumes. Increasing λ_{KL} to 10^{-3} does not improve the 16^3 reference, implying that stronger latent regularisation is less useful than preserving sufficient spatial detail in this setting.

Capacity changes provide a secondary refinement within the fixed 16^3 bottleneck. The widest configuration `P3A.02c` reaches 1.68×10^{-5}, followed by `P3A.02d` at 1.91×10^{-5}, both outperforming the base32 and base24 variants with four latent channels. This suggests that additional encoder–decoder capacity and latent-channel capacity can improve reconstruction within a given bottleneck, but do not compensate for the information loss introduced by a smaller latent grid. Even the strongest 16^3 capacity variant remains about an order of magnitude above the selected 32^3 configuration, showing that validation $\mathcal{L}_{\text{VAE}}$ is highly sensitive to latent resolution within the tested experiment set. Under the Stage A selection rule, `P3A.01d` advances to the 50-epoch rerun `P3A-Final`, whose final diagnostics are reported in the following final-model analysis and whose checkpoint is frozen for all Stage B experiments.

Stage B results. With the selected Stage A checkpoint fixed, the Stage B experiments compare the denoising schedule, U-Net capacity, context encoding, and conditioning dropout within a shared latent representation and decoder. The clearest effect comes from the noise schedule, where the linear schedule `P3B.03b` reaches a validation $\mathcal{L}_{\text{diff}}$ of 0.1300, roughly halving the cosine-reference loss of 0.2552. The shortened cosine variant remains close to the reference at 0.2594, suggesting that the linear schedule provides a more tractable denoising target in the learned latent representation used here.

Other Stage B modifications do not improve on the reference setting. Capacity variants remain above the cosine reference, the coordinate-augmented run again underperforms the coordinate-free setup, and higher conditioning-dropout rates do not reduce validation loss. The remaining Stage B loss is not substantially reduced by additional diffusion U-Net capacity, explicit coordinate encoding, or stronger conditioning dropout in this experiment set. Under the Stage B selection rule, the linear 1000-step schedule is carried into the 50-epoch final rerun `P3B-Final`, together with the retained reference settings for capacity, context encoding, and conditioning dropout.

Phase III selects a two-stage configuration for final evaluation. Stage A retains the less-compressed 32^3 bottleneck, while stronger KL regularisation and additional 16^3 capacity remain insufficient under validation $\mathcal{L}_{\text{VAE}}$. Stage B retains the linear 1000-step schedule, while coordinate channels, increased diffusion U-Net capacity, and higher conditioning-dropout rates do not yield competitive gains. Figure 5.21 shows the corresponding Stage A and Stage B training trajectories before the final reruns are analysed below.

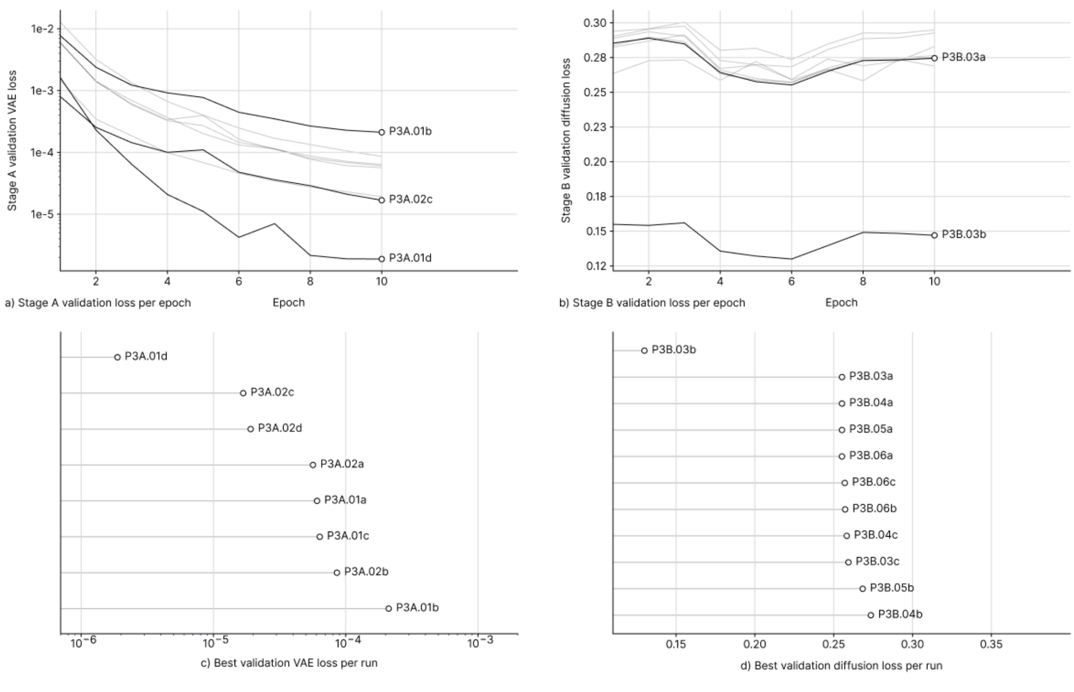

Figure 5.21: Phase III Training Curves for Selected Runs

5.4.4 Final Model and Analysis

Stage A – Final VAE. The final Stage A model is the 50-epoch rerun `P3A-Final`. It uses latent resolution 32^3, latent channel count $C_z = 4$, base width 32, KL coefficient $\lambda_{\text{KL}} = 3 \times 10^{-4}$ with KL annealing over the first 10 epochs, learning rate 2×10^{-4}, and effective batch size 4. The selected checkpoint is the epoch-48 checkpoint, where validation $\mathcal{L}_{\text{VAE}}$ reaches its minimum.

The encoder compresses the 64^3 binary input volume through a single stride-2 downsampling stage into a latent tensor of shape 4×32^3. Separate $1 \times 1 \times 1$ heads parameterise $\boldsymbol{\mu}$ and $\log \boldsymbol{\sigma}^2$, and the decoder mirrors this reduced-compression path through one upsampling stage, as shown in Figure 5.22.

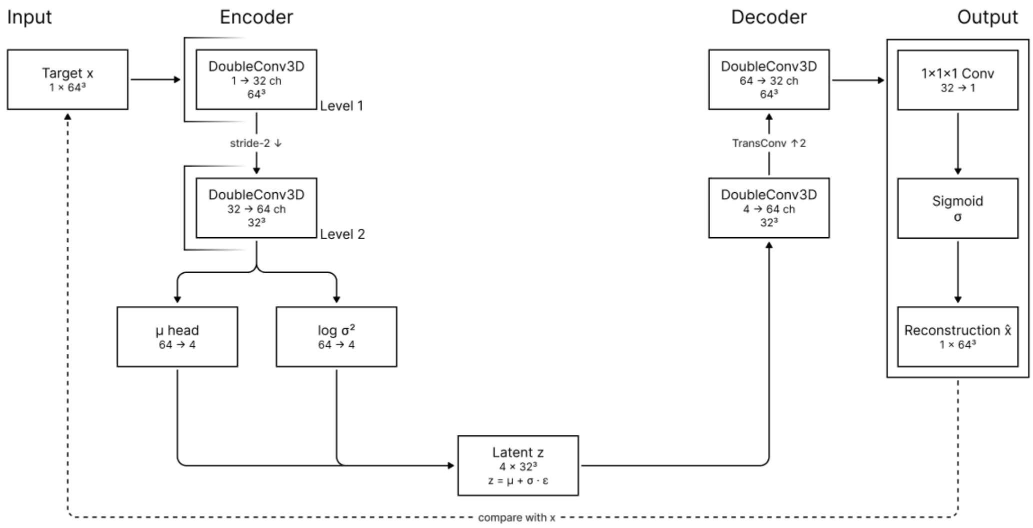

Figure 5.22: Phase III Final VAE Architecture

Run	Best ep.	Best val. $\mathcal{L}_{\text{VAE}}$	Test $\mathcal{L}_{\text{recon}}$	Test $\mathcal{L}_{\text{KL}}$	IoU	Dice	Prec.	Recall	$\triangle$Vol	VE [voxel]	ZPE [voxel]
Final VAE	48	8.03×10^{-7}	7.62×10^{-8}	2.62×10^{-3}	0.99998	0.99999	0.99999	0.99999	1.09×10^{-5}	0.0066	0.00010

Table 5.12: Phase III Final VAE Reconstruction Metrics

Table 5.12 reports Stage A diagnostics beyond the aggregate validation loss used for checkpoint selection. The reconstruction and KL components are evaluated separately on the test set. Single reparameterised forward-pass reconstruction yields near-perfect occupancy overlap, with IoU 0.99998, Dice 0.99999, and very low volume and z-profile errors.

However, this reconstruction quality is achieved with a selected Stage A representation that is only weakly compressed, suggesting that the voxelised building volumes do not admit strong spatial compression without substantial reconstruction loss in the tested VAE setup. Based on occupancy reconstruction, the final VAE provides a high-fidelity but only moderately compressed latent interface for Stage B, whose usefulness for conditional sampling is evaluated separately in the frozen Stage B diffusion pipeline.

Stage B – Final Diffusion Model. The final Phase III Stage B model is the 50-epoch rerun `P3B-Final`. Figure 5.23 summarises the selected DiffusionUNet3D architecture with a linear 1000-step diffusion schedule, base width 64, depth 3, no coordinate channels, full GRZ/GFZ/height conditioning, mean-conditional conditioning dropout $p_{\text{drop}} = 0.02$, DDIM sampling with $S = 100$ and $\eta = 0$ over approximately uniformly spaced reverse-time steps, guidance scale $s = 2.0$, and the frozen Stage A VAE checkpoint from `P3A-Final`. In this final Stage B rerun, EMA weights are available from epoch 5 onward, and the final checkpoint is selected at epoch 32, where validation $\mathcal{L}_{\text{diff}}$ reaches its minimum.

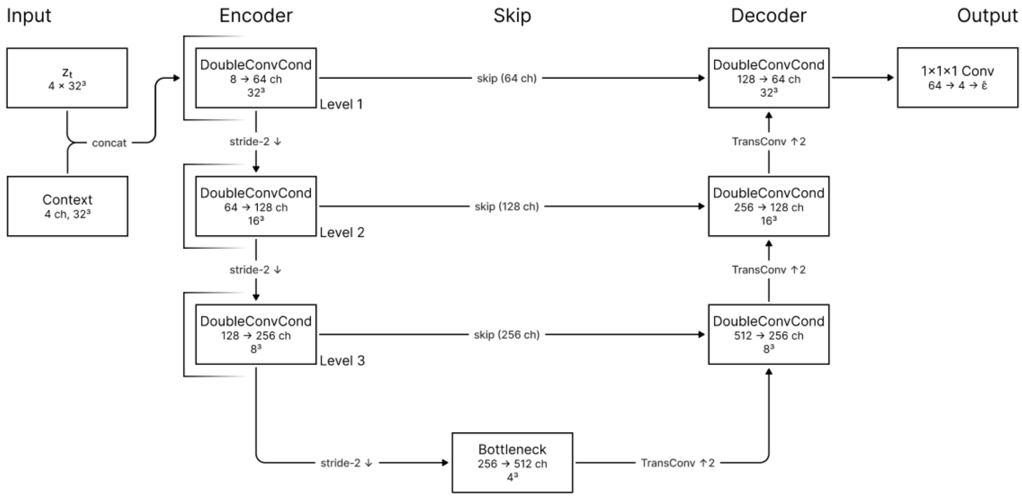

Figure 5.23: Phase III Final Diffusion U-Net

Run	Best epoch	Best val. $\mathcal{L}_{\text{diff}}$	Last val. $\mathcal{L}_{\text{diff}}$	Last train. $\mathcal{L}_{\text{diff}}$
`P3B-Final`	32	0.1265	0.1489	0.2783

Table 5.13: Phase III Final Diffusion Training Metrics

Split	IoU	Dice	Prec.	Recall	$\triangle$Vol	VE [voxel]	ZPE [voxel]	Thresh.
Validation	0.0651	0.1223	0.0678	0.6215	41.6451	3474.45	54.7517	0.60
Test	0.0691	0.1292	0.0722	0.6164	37.9921	3376.98	53.2471	0.60

Table 5.14: Phase III Final Segmentation Metrics

Tables 5.13 and 5.14 show that the longer Stage B rerun yields only a modest denoising gain over the best 10-epoch diffusion configuration. Validation $\mathcal{L}_{\text{diff}}$ decreases from 0.1300 in `P3B.03b` to 0.1265 in the final rerun, an improvement of about 2.7%, but this does not translate into decoded geometric performance. Validation IoU remains at 0.0651 and test IoU at 0.0691, while precision remains low, recall remains high, and both volume and z-profile errors remain large. The small validation–test gap of 0.0040 IoU indicates that the weakness of the final Stage B model is consistent across splits.

Figure 5.24 shows the mismatch between denoising loss and decoded geometry. Some outputs retain weak activation near the target centroid and coarse vertical correspondence, but footprints remain soft and boundary artefacts persist. Weaker cases become diffuse and fragmented. The model localises a rough building prior but not a coherent parcel-bounded volume.

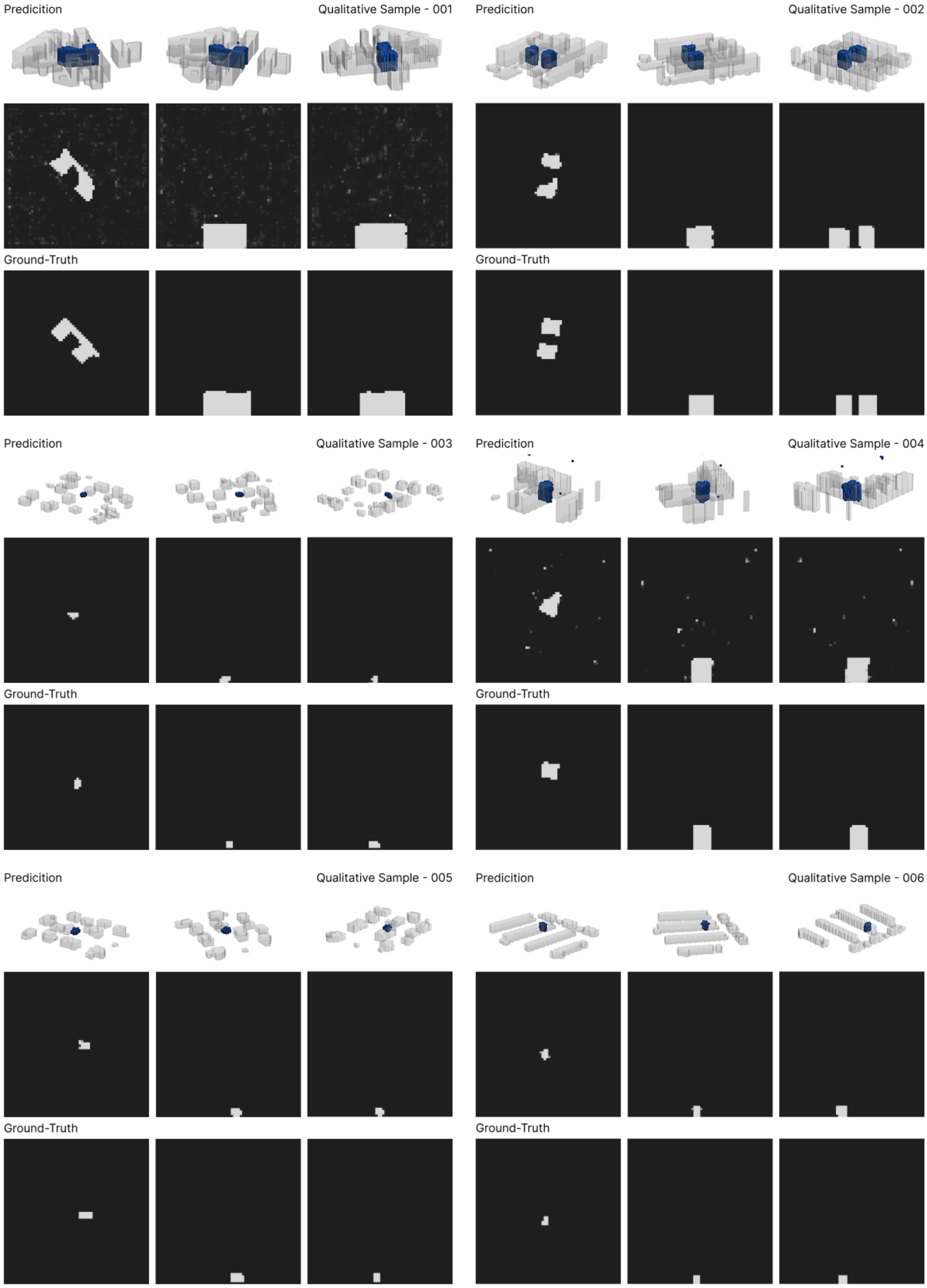

Figure 5.24: Phase III Samples and Ground Truth

Regulatory evaluation is applied on the held-out test split under conditioning derived from the native ground-truth target tuple $\mathbf{c}^{0.5\,\mathrm{m}}$.

Metric	Mean pred.	Mean cond. $\mathbf{c}^{0.5\,\mathrm{m}}$	Mean $\varepsilon_{\mathrm{adj}}$	Median $\varepsilon_{\mathrm{adj}}$
GRZ	5.6042	0.3765	17.8517	0.2992
GFZ	31.4725	1.8238	38.1178	0.3042
Height [m]	42.75	12.52	3.0069	0.0000

Table 5.15: Phase III Regulatory Errors by Metric

Figure 5.25 summarises the resulting metric-level errors. Mean predicted GRZ and GFZ strongly exceed their conditioning targets, with GRZ rising to 5.6042 against 0.3765 and GFZ to 31.4725 against 1.8238. Mean predicted height is also inflated at 42.75 m against a conditioning-target mean of 12.52 m. Median adjusted GRZ and GFZ errors remain around 0.30, while mean errors rise to 17.8517 and 38.1178, which points to a subset of catastrophic failures dominating the aggregate error profile. The zero median height error indicates that at least half of the samples remain within the one-slice height tolerance, while the inflated mean height is driven by failed samples with excessive vertical activation.

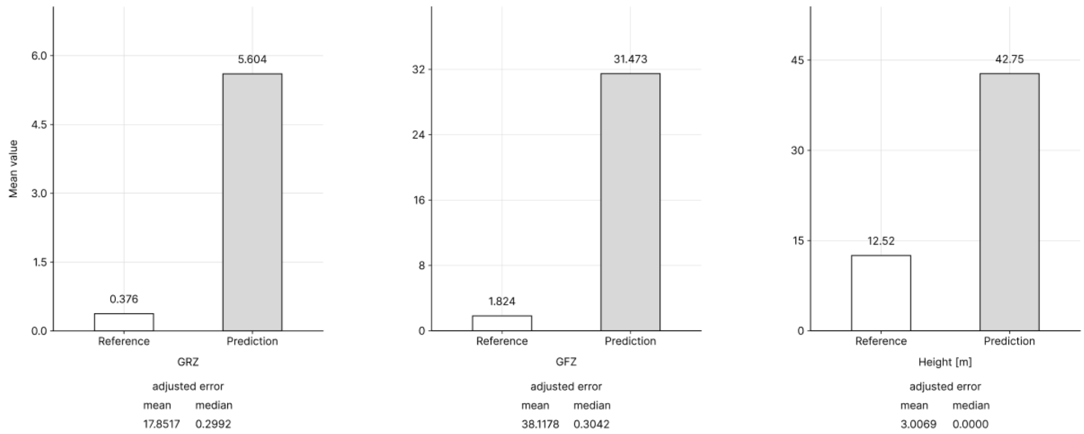

Figure 5.25: Phase III Regulatory Summary Metrics

To keep stochastic sampling comparable across conditions, each parcel reuses the same initial diffusion noise for the native ground-truth, -10%, and $+10\%$ conditioning scenarios shown in Figure 5.26. GRZ and GFZ move opposite to the intended conditioning trend, with GRZ falling from 5.7360 to 5.3919 and GFZ from 32.4953 to 29.6207 as conditioning increases from -10% to $+10\%$. Height changes only slightly and remains inflated by roughly 30 m relative to the conditioning-target mean. This instability dominates the conditioning signal, so no isolated controllability test is reported for Phase III.

The final Phase III model can sample multiple outputs, but the observed variation appears mainly as instability in the occupancy field and not as reliable control over plausible massing alternatives. Although the frozen Stage A checkpoint provides a low-loss reconstruction space, the combined latent-diffusion pipeline does not yet support robust conditional sampling. Phase III remains conceptually relevant, but technically unresolved under the tested configuration.

	GRZ			GFZ			Height		
Scenario	Pred.	Cond.	Δ	Pred.	Cond.	Δ	Pred.	Cond.	Δ (m)
-10% **cond.**	5.7360	0.3388	5.3972	32.4953	1.6414	30.8539	42.74	11.27	31.47
Native GT	5.6042	0.3765	5.2277	31.4725	1.8238	29.6487	42.75	12.52	30.23
$+10\%$ **cond.**	5.3919	0.4141	4.9777	29.6207	2.0062	27.6145	44.99	13.77	31.22

Table 5.16: Phase III Response to Joint Target Scaling

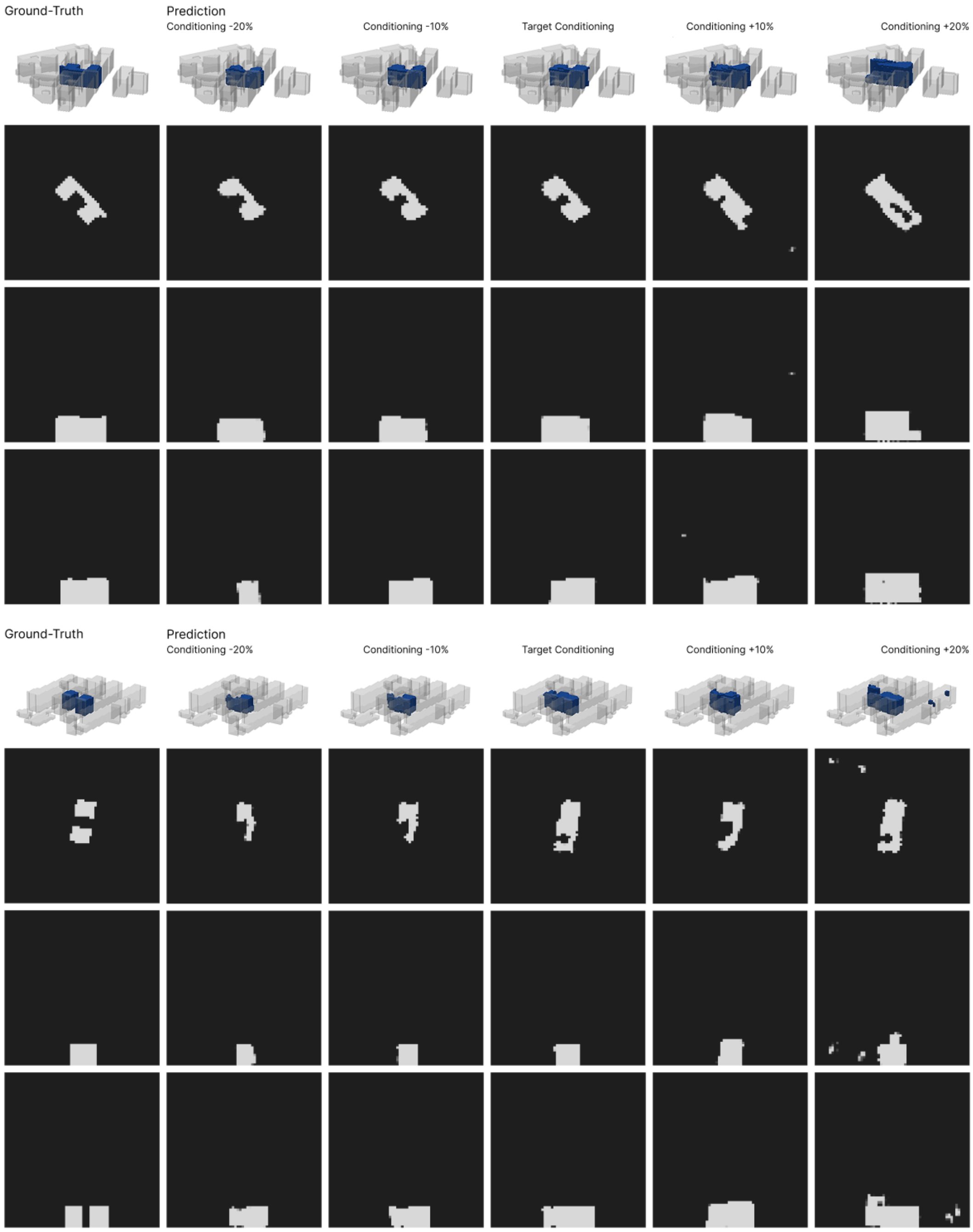

Figure 5.26: Phase III Conditioning Sweep Samples

5.5 Cross-Phase Comparative Evaluation

The final models of the three phases can now be compared across geometry, regulatory accuracy, controllability, and qualitative contextual fit. Where necessary, the comparison distinguishes model behaviour from evaluation-time settings such as thresholding, test-time augmentation, and sampling. The section concludes by summarising the main trade-offs that shape the subsequent discussion.

5.5.1 Quantitative Comparison

The cross-phase comparison uses direct non-TTA inference and the shared validation-calibrated threshold $\hat{t} = 0.60$ for all three phases. The comparison first examines geometric and regulatory performance under unperturbed targets, before turning to the controllability analysis.

Phase	IoU	Dice	Prec.	Recall	ΔVol	VE [voxel]	ZPE [voxel]
I	0.7564	0.8613	0.8414	0.8822	0.0990	43.16	1.0657
II	**0.7754**	**0.8735**	**0.8530**	**0.8950**	**0.0743**	**33.04**	**0.8554**
III	0.0691	0.1292	0.0722	0.6164	37.9921	3376.98	53.2471

Table 5.17: Cross-Phase Segmentation Metrics

Phase	GRZ rel. adj.	GFZ rel. adj.	H rel. adj.
I	0.0814	0.0932	0.0065
II	0.0662	0.0706	0.0187
III	17.8517	38.1178	3.0069

Table 5.18: Cross-Phase Regulatory Errors

Table 5.17 reports an IoU of 0.7564 for context-only Phase I. Phase II improves the geometric metrics, raising IoU to 0.7754 and reducing volume error from 43.16 to 33.04 voxels, while both precision and recall increase. Its strongest configuration removes conditioning dropout entirely, suggesting that the conditioned U-Net benefits from full exposure to the regulatory targets during training. The final Phase II rotational-TTA pass adds another 0.0046 IoU as a phase-specific evaluation-time refinement (Table 5.5), while the supplementary threshold analysis in Section 5.3.5 suggests that the shared threshold of $\hat{t} = 0.60$ is slightly permissive, with overlap quality peaking near $\hat{t} = 0.70$.

Phase III performs substantially worse than both U-Net phases, with IoU falling to 0.0691, precision to 0.0722, and recall remaining at 0.6164. This turns the mild over-prediction pattern of Phases I and II into diffuse mass activation. Although Stage A reaches a validation $\mathcal{L}_{\text{VAE}}$ of 8.03×10^{-7} and near-perfect occupancy-overlap diagnostics under single reparameterised reconstruction, Stage B does not translate this latent representation into stable conditional sampling. The linear noise schedule reduces validation denoising loss from 0.2552 to 0.1300, but decoded geometric performance remains weak. The training trajectories reinforce this contrast, with both U-Net phases improving late in their 50-epoch reruns, while Stage B of Phase III reaches its best validation loss at epoch 32 and degrades afterward.

Table 5.18 and Figure 5.27 follow the same ranking. Phase I approximates parcel targets from context alone, with mean adjusted errors of 0.0814 for GRZ, 0.0932 for GFZ, and 0.0065 for height. Phase II lowers GRZ and GFZ errors to 0.0662 and 0.0706, while height rises slightly to 0.0187 but retains a median adjusted error of zero, suggesting that conditioning mainly improves density-related alignment. Phase III loses regulatory fidelity, with mean errors of 17.8517 for GRZ and 38.1178 for GFZ, while the corresponding medians remain near 0.30, consistent with severe failure cases dominating the aggregate error profile.

	GRZ		GFZ		Height [m]	
Phase	Δ target	Δ pred.	Δ target	Δ pred.	Δ target	Δ pred.
II	+0.0753	+0.0264	+0.3648	+0.3768	+2.50	+1.64
III	+0.0753	-0.3441	+0.3648	-2.8746	+2.50	+2.25

Table 5.19: Cross-Phase Conditioning Response

Table 5.19 shows a monotonic Phase II response, with GFZ nearly matching the requested span ($+0.3768$ vs. $+0.3648$), while GRZ and height respond more conservatively. The isolated sensitivity analysis confirms that this response is variable-specific, with GFZ and height capturing 65.4% and 61.2% of the requested span, while GRZ captures only 34.0%. The weaker GRZ response reflects the coupling between GRZ, GFZ, and height, and suggests that footprint coverage remains more strongly anchored in the context-derived prior. The Phase II conditioning-channel experiments support this interpretation, as two-variable subsets remain competitive, whereas GFZ alone lowers validation IoU by approximately 0.017 relative to the full three-variable reference.

In contrast, Phase III does not preserve directional control, as GRZ and GFZ shift opposite to the conditioning targets while height changes only weakly from an already inflated prediction level. Predicted GRZ reaches 5.60 against a conditioning-target mean of 0.38, with values above 1.0 indicating occupancy beyond the parcel boundary.

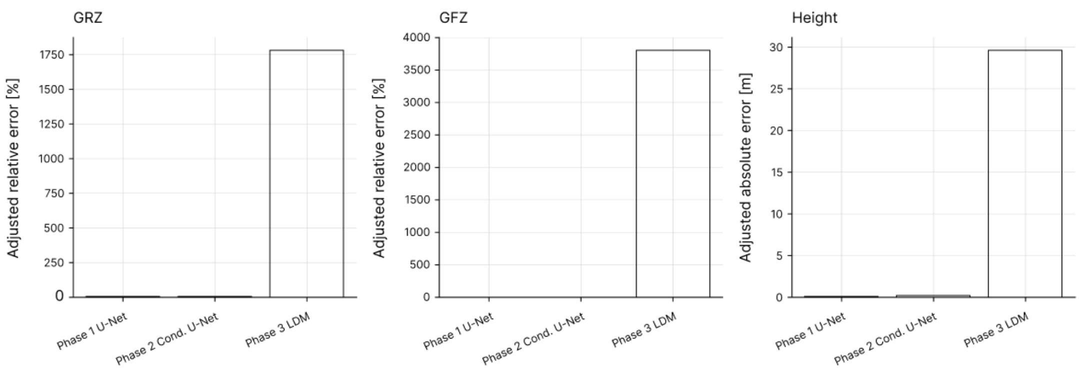

Figure 5.27: Final Regulatory Errors Across Phases

5.5.2 Qualitative Comparison

Figure 5.28 compares three selected held-out test parcels across the three final models and the ground truth, supporting visual comparison of contextual fit, footprint extent, vertical massing, and characteristic failure modes. The three parcels are the same as those shown in the per-phase qualitative samples, allowing the unconditional, conditioned, and diffusion models to be read side by side. Additional within-phase sample figures, controllability panels, and variant panels extend this qualitative view.

In Phase I and II, the predicted mass remains anchored to the parcel position, aligns with the street and neighbour structure, and forms a parcel-scale infill envelope. Phase I already captures centroid position, overall height, and broad footprint logic, but often smooths or slightly extends the envelope at its edges. Phase II preserves this contextual plausibility while resolving the envelope more tightly around parcel-specific targets, a difference that stays subtle in the three-dimensional views but becomes clearer in the footprint projections, where the conditioned model produces sharper parcel-aligned outlines.

The Phase III column departs from this pattern across all three parcels, though to differing degrees. The first two parcels retain a weak central concentration and coarse height correspondence, while the third case, a tall volume set among equally tall neighbouring slabs, loses the parcel-bounded envelope and scatters occupancy across the surrounding space, including isolated voxels detached above the building line. This visual gradient follows the regulatory profile, where moderate median errors coincide with a subset of strongly failing parcels.

Figure 5.29 isolates the variation behaviour by sampling several outputs for two parcels under identical context and regulatory targets. The variants differ strongly from one another, yet none forms a coherent parcel-bounded volume. For the first parcel, one variant collapses into a diffuse cloud of activation that fills much of the sampling window, while the others stay weakly localised around the target centroid with fragmented footprints. For the second parcel, the outputs remain sparse and fragmented, again accompanied by isolated voxels above the site. The pre-threshold probability maps make this directly visible, with low-confidence mass spread across the plane and only faint concentration at the parcel, so the variation appears as sampling instability spanning diffuse overactivation and sparse fragmentation, instead of a set of distinct but plausible massing alternatives.

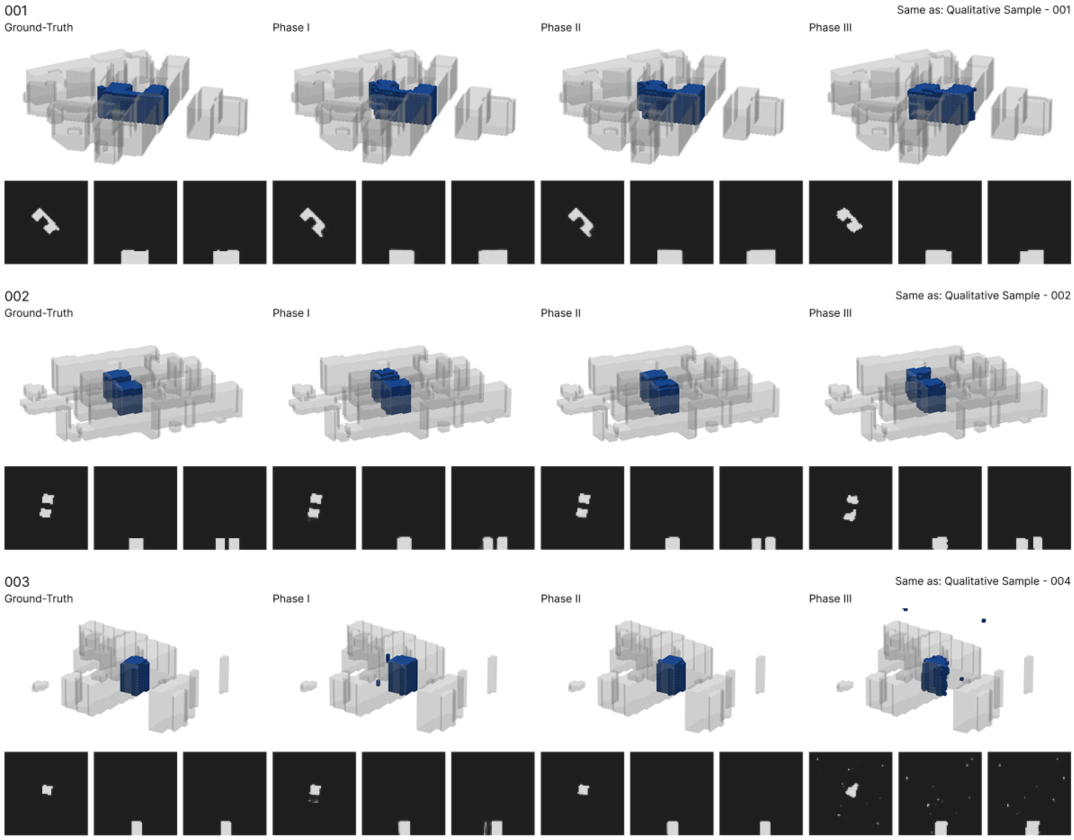

Figure 5.28: Final Predictions Across Phases

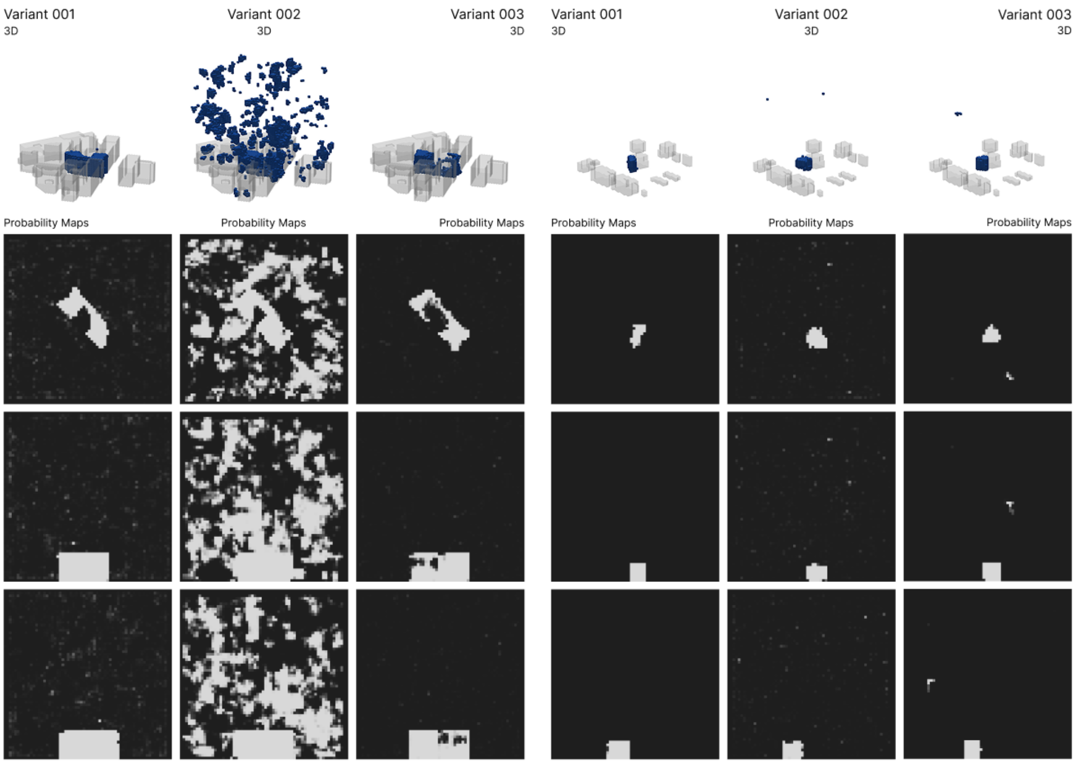

Figure 5.29: Phase III Variants Under Fixed Conditions

5.5.3 Synthesis

The cross-phase comparison reveals a trade-off between contextual plausibility, regulatory controllability, and stochastic variation, not a continuous improvement through increasing modelling complexity. Phase I shows that the chosen voxel representation and context-based modelling approach are already effective for parcel-scale massing, as the unconditional model approximates all three regulatory metrics with mean adjusted errors below 0.10. The strong implicit height agreement partly reflects the wide 2.0 m tolerance, while the small raw error still points to a useful vertical reference from the neighbour-height field.

Building on this context-derived prior, Phase II introduces regulatory conditioning and achieves the strongest balance between geometric fidelity, regulatory alignment, and controllability. Its gains appear most clearly in density-related alignment, regulatory target responsiveness, and variable-specific sensitivity under isolated conditioning changes. The no-dropout setting and the conditioning-channel experiments show that the conditioned U-Net benefits from full regulatory exposure and from a multi-dimensional target vector. At the same time, the weaker GRZ response indicates that footprint coverage remains more strongly shaped by the surrounding urban fabric, while floor-area intensity and vertical extent are more directly steerable.

Across the experiments, coordinate channels underperform in Phase I and Phase III and do not outperform the final no-dropout Phase II model, suggesting that the four spatial context channels already encode sufficient positional information for parcel-scale massing. Phases I and II also share a precision–recall asymmetry in which recall exceeds precision, reflecting mild over-prediction. The supplementary threshold analysis shows that stricter binarisation can reduce this effect, although overlap quality and regulatory closeness peak at different thresholds.

The IoU-versus-volume-error analyses in Figures 5.3 and 5.10 show that overlap quality and global mass consistency are only partially aligned. The loss experiments further indicate that boundary overlap and volumetric agreement remain distinct quality dimensions, while the narrow performance bands across the deterministic experiment series suggest a practical ceiling imposed by the dataset and representation. Within this regime, compact architectures already capture most of the learnable contextual form signal.

Phase III extends the modelling setup toward stochastic sampling, but the final pipeline does not turn this additional sampling capacity into a stable family of alternatives. The final VAE achieves near-perfect occupancy overlap and low voxel-space reconstruction errors on the weakly compressed 32^3 latent representation, supporting the autoencoding stage without establishing reliable conditional sampling in the full pipeline. The linear schedule improves denoising loss, yet decoded geometry and regulatory response are unstable. GRZ and GFZ shift opposite to the conditioning targets, and decoded occupancy frequently extends beyond the parcel boundary. Phase III is most informative as an exploratory boundary case, exposing unresolved issues around latent-space structure, sparsity handling, the coupling between Stage A and Stage B, and conditioning persistence in the reverse denoising process.

Overall, the experiments show that regulation- and context-aware generative modelling is most effective in the deterministic conditioning regime. Phase II provides the strongest balance between contextual plausibility, regulatory alignment, and controllability.

6 Discussion and Interpretation

The comparative evaluation frames the three phases as successive modelling assumptions, exposing a tension between contextual plausibility, regulatory controllability, and controlled variation. The shift from context-only prediction to regulatory conditioning shows that GRZ, GFZ, and height can steer a learned contextual prior. The two-stage latent diffusion pipeline, by contrast, shows that high-fidelity autoencoding does not automatically lead to reliable controlled sampling. Beyond the technical comparison, the sequence also exposes how much of planning knowledge a representation built on scalar targets and reduced spatial form can carry, and how much it leaves out.

6.1 From Contextual Prediction to Controlled Variation

The three phases can be read as successive tests of what forms of control can be introduced into a learned representation of urban context. Phase I tests whether context alone can predict plausible parcel-scale massing. Phase II adds explicit regulatory targets and shows that deterministic conditioning can stabilise target alignment. Phase III adds stochastic sampling, but the resulting outputs lose geometric and metric coherence under the tested setup. The results suggest that controlled variation depends not only on conditioning signals or sampling capacity, but also on how architecture, representation, loss design, and sampling procedure interact.

6.1.1 From Phase I to Phase II

Phase I achieves a test IoU of 0.7564 using only the spatial context representation, showing that the four context channels already provide substantial information about density, height, and parcel-scale massing. The regulatory evaluation supports this reading, as the unconditional model approximates GRZ, GFZ, and height with mean adjusted errors below 0.10 (Table 5.3).

With FiLM-based conditioning in Phase II, both precision and recall increase, while GRZ and GFZ errors decrease. This shows that the regulatory vector reduces ambiguity without sacrificing formal coherence. Height follows a different pattern, with mean adjusted error rising slightly from 0.0065 to 0.0187 while the median is zero. Vertical alignment already appears strongly constrained by the neighbour-height heatmap, so conditioning contributes most where the spatial prior is underdetermined, especially in footprint coverage and floor-area intensity.

Under joint $\pm 10\%$ scaling, all three predicted quantities move in the intended direction, with GFZ following the requested target range most closely at $+0.3768$ predicted change against $+0.3648$ requested change (Table 5.7). The isolated sensitivity analysis confirms variable-specific response patterns, with 65.4% span capture for GFZ, 61.2% for height, and 34.0% for GRZ (Table 5.8). The more conservative GRZ response suggests that footprint coverage remains more strongly anchored in the context-derived prior, while GFZ and height are more directly steerable within the coupled geometry of footprint, floor area, and vertical extent.

Both deterministic phases show a persistent precision–recall asymmetry, with recall exceeding precision and uncertain boundary voxels more often included than omitted. This produces slightly thicker or more extended volumes than the ground truth. Under the BCE–soft-Dice objective with positive class weighting, this pattern is consistent with a training setup that counteracts voxel sparsity and tends to penalise missed occupancy more strongly than excess mass. The supplementary threshold analysis shows that stricter binarisation can reduce this effect at evaluation time, although overlap quality and regulatory closeness peak at different thresholds.

The transition from Phase I to Phase II shows that regulatory conditioning can preserve the context-derived prior while resolving part of the ambiguity left by spatial context alone. Phase II is coherent with the surrounding urban fabric and responds predictably to GRZ, GFZ, and height, but its control is volumetrically coupled.

6.1.2 From Phase II to Phase III

Phase II produces one deterministic output for each spatial context and conditioning vector and cannot represent planning situations in which several geometrically distinct massing configurations satisfy the same GRZ, GFZ, and height targets. Phase III addresses this limit through a two-stage latent diffusion pipeline that separates representation learning from conditional sampling and thereby enables stochastic variation.

The final Stage A VAE reaches a validation loss of 8.03×10^{-7} on the 32^3 latent representation and reconstructs test-set occupancy volumes almost perfectly through single reparameterised forward-pass evaluation. The contrast between reconstruction and sampling becomes clear once latent codes are sampled under spatial and regulatory constraints. Performance drops, with the final decoded diffusion model reaching only 0.0691 test IoU and 0.0722 precision (Table 5.14), GRZ and GFZ moving opposite to the conditioning targets (Table 5.16), and qualitative outputs ranging from weakly localised to fragmented volumes (Figure 5.24).

The experimental design does not isolate individual causes, and a latent-standardisation check ruled out a simple latent-scale mismatch, so the observed failure pattern is interpreted through four linked factors.

Latent compression. The final 32^3 latent grid compresses the 64^3 input by only a factor of two per spatial axis, preserving the lowest validation $\mathcal{L}_{\text{VAE}}$ but leaving the diffusion model with $4 \times 32^3 = 131{,}072$ latent values. Stronger compression to 16^3 or 8^3 increases validation $\mathcal{L}_{\text{VAE}}$ by one to two orders of magnitude, suggesting a tension specific to sparse, sharply bounded occupancy, where the explored VAE family may admit no latent that is both learnable from the available samples and boundary-faithful. As only one family was examined, this is an open question, not a property of the task.

Sparsity and boundary sensitivity. The target volumes are highly sparse, so the latent space must preserve sharply bounded occupancy while avoiding diffuse activation. Small deviations in the sampled latent can produce large decoded errors, especially when activation spills beyond the parcel boundary. This appears in GRZ values above 1.0 and volume errors above $3{,}000$ voxels in Tables 5.14 and 5.15. The decoded outputs suggest that the pipeline has little tolerance for imprecision during denoising, as small latent errors can become large footprint, volume, and boundary errors in voxel space.

Conditioning persistence. In Phase II, FiLM modulation affects a single forward pass and propagates directly to the occupancy output. In Phase III, the same regulatory signal must persist across an iterative reverse process in latent space. The inverted GRZ and GFZ responses indicate that this signal is not reliably preserved through sampling, although the generally limited decoded prediction quality makes it difficult to separate failed conditioning from broader sampling instability.

Data scale. The training split of $9{,}215$ samples is sufficient for models that learn direct mappings from context and conditioning to occupancy, but remains limited for a latent diffusion model that must learn both the data distribution and its conditional structure. Berlin's urban-form diversity further spreads this signal across low-rise settlements, slab structures, perimeter blocks, and dense *Gründerzeit* conditions, reducing effective sample density across urban-form and regulatory subregions.

Together, these factors indicate that the tested conditional latent diffusion setup has not yet stabilised the transition from learned latent representation to sparse volumetric massing output. The final diffusion rerun supports this reading, as longer training only marginally improves validation loss while decoded geometry remains unstable. This suggests that Phase III's limited performance reflects a complex interaction among the size and heterogeneity of the training dataset, latent representation, denoising dynamics, conditioning, and sparse voxel geometry.

6.1.3 What the Sequence Reveals

Across both transitions, the U-Net models show that contextual fit and target steering can be brought into a productive relation. The move from Phase I to Phase II resolves residual ambiguities in the context-derived prior, reflected in higher precision and recall, stronger conditioning effects where neighbourhood cues remain underdetermined, and predictable responses to target perturbations. Regulatory conditioning does not replace the learned contextual prior, but narrows and redirects it within the coupled geometry of footprint, floor area, and height.

In Phase III, this alignment breaks down once the workflow shifts from deterministic mapping to iterative stochastic sampling. The factors discussed above indicate that this breakdown cannot be assigned to Stage B alone, but emerges from the coupling of latent representation, denoising, conditioning, data scale, and sparse voxel geometry. That this breakdown can be localised at all follows from the phased design itself, as keeping the dataset, voxel input-output representation, and evaluation protocol consistent across phases allows the loss of controllability to be read as a property of the shift toward stochastic sampling and not as an artefact of a changed setup. Read this way, Phase III contributes a delimited negative result instead of an inconclusive one.

The sequence thus clarifies both the capacity and the limitation of the tested approach. Phase II supports controlled target variation across GRZ, GFZ, and height while preserving contextual plausibility, but its architecture produces a single output for each context–target pair. Broader geometric diversity would require sampling mechanisms capable of producing multiple alternatives without losing parcel-boundary stability, volumetric coherence, and target alignment.

6.2 Knowledge, Normativity, and Responsibility

The results also raise questions about the forms of urban knowledge encoded by the dataset, the normative assumptions embedded in regulatory targets, and the redistribution of agency when the generation of urban form is mediated by statistical inference. This part of the discussion links the representational limits of the computational approach to questions of transparency, accountability, and responsibility.

6.2.1 Representation and Encoded Order

Both conventional planning practice and computational planning methods rely on abstraction, as the complexity of urban space must first be translated into an operational form before it can become a planning object (Scott, 1998; Kitchin, 2014). The models developed in this thesis operate within such an abstracted representation and infer patterns from data that have already been shaped by a long history of administrative, social, and technical decisions. Unlike parametric approaches, where assumptions are made explicit through rule-based relations, the assumptions in this work are distributed across the data, its representation, and the training process, shaping which spatial relations become inferable (Davis, 2013; Kitchin, 2014; Ananny & Crawford, 2018).

The experiments in Chapter 5 make these representational limits visible by testing prediction, conditioning, and sampling on the same reduced urban representation, without expanding the model beyond the information encoded in the source data, voxel grid, context channels, and regulatory targets. A first layer of abstraction lies in the source data, where CityGML LoD1 represents buildings as simplified volumetric envelopes and omits roof forms, material articulation, use distinctions, and many socially relevant differences. Voxelisation adds a further layer of abstraction by translating geometry into binary occupancy, while the context channels reduce the urban environment to cadastral structure, neighbouring building volumes, parcel edges, and streets.

Within this representation, GRZ, GFZ, and height function as conditioning targets derived from historically formed planning logics, making selected aspects of planning regulation computable while abstracting them from their institutional and interpretive context. The filtering process reinforces this by excluding outliers and privileging parcels aligned with their surroundings. This approach aligns with the aim of the thesis and with a simplified operationalisation of the contextual-analogy logic of § 34 BauGB, while also favouring formal continuity and orienting the models toward forms retained as comparable within the constructed dataset.

The representation turns selected planning concerns into numeric targets, differentiated environments into volumetric abstractions, and the heterogeneous city into a filtered distribution of comparable cases. The models do not learn urban form in a neutral sense, but a field of plausible forms already stabilised by planning history, available data, and study design. This makes dataset construction a central point of responsibility, since training conditions delimit the space of urban possibilities from which outputs can emerge (Kitchin, 2014; Ananny & Crawford, 2018; Pasquale, 2015). Within these bounds, such representations can support context-sensitive massing outputs under selected regulatory inputs, yet they remain dependent on the prior abstractions through which the city first becomes available as data.

6.2.2 Agency, Transparency, and Accountability

Once the generation of urban form is mediated by a learned representation, authorship no longer lies only in direct geometry production. It also lies in the construction of the conditions under which form becomes possible, comparable, and interpretable. Control is exercised before any output is produced, since choices about representation, conditioning, training data, and evaluation shape the space of possible results. Design agency shifts from direct form-making toward the orchestration and assessment of generative processes. Model outputs become intermediate artefacts for reflection and comparison, while decisions about relevance, desirability, and further development remain with human actors.

This redistribution of agency also reframes transparency, as it cannot depend only on the interpretability of learned weights but must include the chain of decisions through which urban reality is transformed into model input (Kitchin, 2014; Pasquale, 2015). Documented preprocessing, representational conventions, and evaluation procedures make the model's field of operation visible and define the conditions under which outputs can be trusted or questioned. Even where Phase II shows controllable behaviour and Phase III explores stochastic sampling, the system only reproduces and varies learned relations under specified conditions. It does not generate normative judgement.

Responsibility remains with those who define the problem, select the data, interpret the outputs, and embed the system within a planning process, since a modelled massing may be coherent within the dataset while remaining inadequate as an urban, social, or political proposition.

6.3 Implications for Urban Design Practice

The approach developed here is most relevant to small-scale, context-dependent infill planning, where formal limits are often not yet fixed and admissibility depends on contextual interpretation under the logic of § 34 BauGB. At this stage, parcel geometry, surrounding urban fabric, footprint, density, and height must be considered together before they become fixed design assumptions. Parametric workflows are particularly effective for larger or masterplanned projects, where rules, performance criteria, and dependencies can be established in advance and reused across many alternatives. For smaller infill projects, however, contextual constraints are more site-specific, and the effort required to encode a bespoke parametric workflow can be disproportionate to the project scope (Davis, 2013; Gerber & Lin, 2013; Turrin et al., 2011).

The Phase II model addresses this small-scale, context-dependent condition by learning relations between parcel geometry, surrounding urban fabric, and regulatory targets. Its controllability results show that GRZ, GFZ, and height can steer massing while preserving contextual plausibility, enabling comparison across different target settings without manually constructing each variant. As Phase III does not yet provide stable multi-sample alternatives, usable variation is tied to deliberate changes in the conditioning vector. The approach is strongest for small- and medium-scale infill parcels similar to the training distribution, while its outputs should be treated as preliminary study material requiring further design development, institutional review, and human judgement of relevance, plausibility, and desirability.

6.3.1 Conceptual Workflow for Early-Stage Planning Practice

Figure 6.1 presents a conceptual workflow for integrating the developed approach into an early-stage planning process. CAD and BIM support the development and coordination of defined geometry, while parametric methods enable systematic variation once spatial dependencies, parameters, and performance objectives have been specified (Eastman, 1999; Turrin et al., 2011). The selection of an appropriate method depends on the scale and maturity of the planning task. Projects with an established design direction and sufficiently defined constraints can proceed directly into conventional CAD or BIM development. Larger development sites may instead benefit from parametric exploration within a predefined design space.

The developed approach is positioned within small- and medium-scale infill situations where the spatial form has not yet been defined. This may include contexts governed by § 34 BauGB as well as planning settings in which quantitative targets are available without a fixed building envelope. Parcel geometry, surrounding massing, street context, and preliminary targets for GRZ, GFZ, and height are prepared as model inputs. The model then produces early massing studies that can be assessed in terms of contextual fit, footprint, height, volume, and target alignment. Implausible outputs require a return to the input data, target assumptions, or conventional site analysis. Selected directions can then be transferred into CAD, BIM, or parametric environments for further architectural development.

The workflow remains iterative. Findings from subsequent design development may lead to revised targets or a renewed comparison of early massing studies. This reflects a process-oriented understanding of design in which assumptions are tested, outcomes are evaluated, and conditions are revised through feedback (Pask, 1969). The model does not assess legal admissibility, and its outputs remain preliminary artefacts for comparison and reflection. Architectural judgement, legal interpretation, institutional review, and final planning decisions remain the responsibility of human actors.

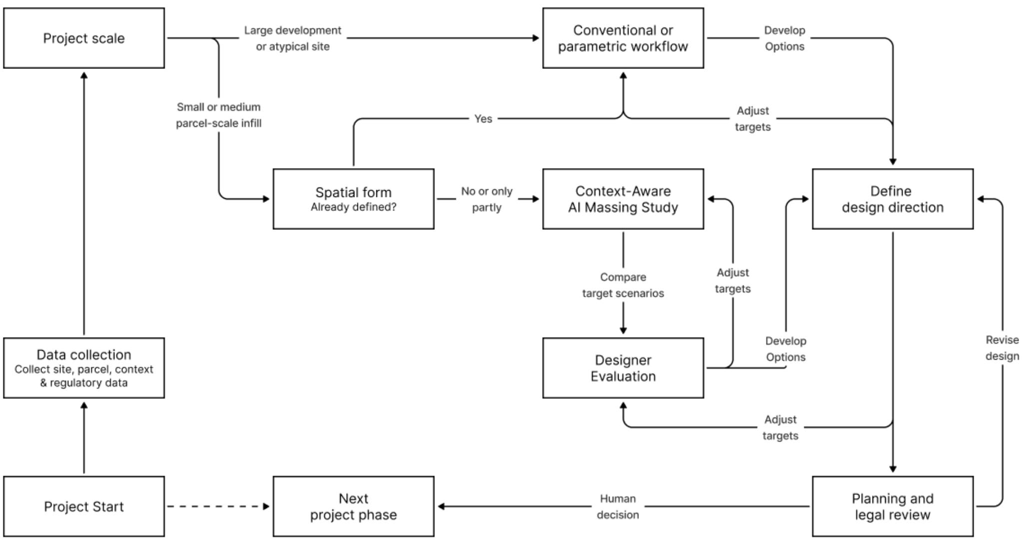

Figure 6.1: Conceptual Early-Stage Planning Workflow

7 Conclusion and Outlook

This thesis shows that regulation- and context-aware urban massing can be learned, produced, and evaluated within a shared voxel-based representation. The strongest results emerge when contextual information from the surrounding urban fabric is combined with explicit conditioning on GRZ, GFZ, and height. Stochastic multi-output sampling, by contrast, is unstable under the tested two-stage latent diffusion pipeline, exposing a gap between reliable deterministic steering and stable sampled variation. This final chapter summarises the key findings, defines their limits, outlines directions for future work, and reflects on the relation between computational abstraction and planning judgement.

7.1 Key Findings

The central research question asked whether a machine-learning-based generative model can produce context-sensitive urban massing that responds to regulatory targets, and under what conditions spatial variation can be introduced. Within the voxel-based representation used in this thesis, contextual fit and regulatory steering can be stabilised, while stochastic multi-output sampling is unresolved under the tested conditions.

The findings answer the first sub-question by showing that a shared voxel representation can translate urban context into a format that is learnable, interpretable, and evaluable. By reducing the city to parcel structure, simplified building volumes, streets, and voxelised occupancy, the representation allows urban form and regulatory metrics to be processed within a shared modelling and evaluation framework. The deterministic extraction of GRZ, GFZ, and height provides a consistent basis for comparing model outputs with parcel-specific targets. The resulting dataset of 13,164 samples captures only a limited share of Berlin's social and architectural reality, but provides a coherent basis for testing how urban form and selected regulatory metrics can enter a learned model together.

Phase I shows that the surrounding urban fabric contains substantial information about likely parcel-scale massing. Even without regulatory conditioning, the model recovers much of the density and height structure embedded in the data, as GRZ, GFZ, and height are partially reflected in recurring relationships between parcels, neighbours, streets, and existing building volumes, although the strong implicit height agreement partly reflects the wide $2.0\,$m tolerance. Its limitation lies in the model's inability to deliberately shift this context-derived prior toward alternative parcel-specific targets.

The findings answer the second sub-question by showing that GRZ, GFZ, and height can guide massing through explicit conditioning in Phase II. The final model improves both geometric fidelity and alignment with density-related regulatory targets relative to the unconditional model of Phase I, while off-target tests show monotonic responses when all three conditioning inputs are scaled jointly by $\pm 10\%$. A supplementary isolated sensitivity analysis shows variable-specific response patterns, with the clearest responses for GFZ and height and a more moderated response for GRZ. The regulatory vector functions as an effective steering mechanism within a context-sensitive deterministic model, although its behaviour remains coupled through regulatory metrics that jointly describe footprint, floor-area intensity, and vertical extent.

The third sub-question is only partly resolved. Phase II supports controlled changes across deliberately varied target settings, but Phase III does not achieve stable multi-output sampling under fixed conditions. Stage A reconstructs target volumes accurately under weak compression, but Stage B does not translate the final latent representation into reliable conditional sampling. The final latent diffusion model produces diffuse over-activation in many samples and fails the controllability test, with GRZ and GFZ shifting in the wrong direction under conditioning changes. Under fixed parcel surroundings and regulatory targets, the model does not yet produce a usable family of alternatives.

The fourth sub-question is addressed by positioning the deterministic Phase II approach as an exploratory instrument for comparing how parcel-specific GRZ, GFZ, and height targets interact with the surrounding urban fabric. Varying the conditioning vector supports comparisons across target settings, but not multiple alternatives for fixed targets. The outputs remain exploratory massing studies, are not planning-ready, and do not constitute an assessment of legal admissibility.

7.2 Limitations

As the source data are drawn from Berlin and the split is not spatially blocked, the results establish model behaviour only within this filtered urban case and do not determine transfer to other cities or planning systems. Comparable cities with similar parcel structures, building typologies, data availability, and regulatory conditions may support related model behaviour, but this was not examined. A spatial-overlap analysis found that 1,755 of the 1,974 test parcels (88.91%) had 128 m context windows overlapping with at least one training sample. On the remaining 219 non-overlapping cases, the final Phase II model achieved an IoU of 0.665, compared with 0.780 on the full test set, indicating weaker but still substantial performance on cases without training-window overlap. The dataset is also concentrated on small and medium-sized parcels, so performance is less certain for large or atypical configurations.

To make the problem computationally tractable, buildings are represented as LoD1 volumetric envelopes, and continuous geometry is translated into voxel occupancy. This makes the task learnable and enables deterministic evaluation of voxel-derived GRZ, GFZ, and height, while excluding architectural detail and many qualitative aspects of urban space. This abstraction is reinforced by the reduction to a 2.0 m training resolution, which preserves overall massing while smoothing finer-grained features such as setbacks and thin elements. Although the tolerance logic partly compensates for this discretisation, the outputs remain tied to the underlying grid and should be interpreted as massing-level approximations.

Regulation enters the model through GRZ, GFZ, and building height. These variables establish a link between planning logic, representation, and evaluation, but capture only a limited share of the factors relevant to admissibility, since planning decisions also depend on spatial arrangement, use, access, environmental effects, and institutional interpretation. The evaluation framework is similarly delimited. IoU, Dice, and related overlap metrics assess geometric similarity, while errors in GRZ, GFZ, and height assess regulatory target alignment. Design quality, planning legitimacy, and legal admissibility still require expert interpretation and cannot be inferred from metric agreement alone.

7.3 Outlook

The limitations point to several directions for extending the approach, while future work should distinguish between refining the comparatively stable deterministic Phase II model and treating stochastic variation as a separate technical problem requiring targeted responses to the failure modes observed in Phase III.

Higher-resolution and hierarchical generation offer the most direct path beyond the current 2.0 m training grid. A hierarchical approach could generate context- and metric-aware massing at a coarse level before refining it at higher resolution. This would allow the model to preserve parcel-scale reasoning, sharpen geometric detail, and reduce the burden on a single latent space to encode both global structure and local form.

Broader regulatory and semantic conditioning could extend the model beyond GRZ, GFZ, and height. Variables such as building arrangement, setback envelopes, land use, and access conditions could move the model from undifferentiated massing toward more structured spatial organisation while preserving interpretability.

Targeted development of stochastic sampling remains necessary if multi-output variation is to become usable. Progress depends on addressing the limitations identified in Phase III by testing latent representations better suited to sparse volumetric data, comparing latent-space and direct voxel-space sampling, improving denoising architectures for occupancy fields, and strengthening conditioning so that its influence persists throughout the reverse process. More stable training regimes and larger, better-balanced datasets may also be required before stochastic variation can be assessed as meaningful design diversity.

Multi-scale and block-level modelling could address the limits of the current parcel-centred representation and its fixed 128 m context window. Extending the approach to block or neighbourhood scale would allow the model to represent continuous street structures, shared open space, and coordinated density distributions. A multi-scale approach could first generate block-level organisation and then resolve individual parcels, while also improving coverage of larger sites.

Transfer and comparative urban datasets are necessary for assessing whether the learned relations extend beyond Berlin. An initial step would compare cases within the German planning system before extending the analysis to different regulatory contexts. Such studies could clarify whether regulation functions as a transferable conditioning signal or whether the representation and conditioning strategy must be adapted.

Design-facing evaluation and workflow integration require moving beyond geometric and regulatory metrics toward analysing how the model is used in planning and design practice. This includes examining how planners and designers use model-based variants to explore density, contextual fit, and early massing strategies, while treating outputs as intermediate artefacts for comparison and deliberation.

Future work should proceed as a differentiated research programme, not as a single enlarged model. The deterministic conditioned approach can be refined through higher resolution, richer conditioning, multi-scale context, transfer studies, and workflow integration. Stochastic sampling should be revisited separately through targeted responses to the limitations identified in Phase III.

7.4 Final Reflection

Architecture, urban planning, and urban design all depend on abstraction. Parcels, boundaries, ratios, and heights make spatial conflicts administratively manageable, yet their practical meaning emerges only in relation to context, precedent, judgement, and the expectations attached to a place. This thesis translates parts of Berlin's built form into a learnable representation and shows, under defined experimental conditions, that context-derived spatial relations and selected planning metrics can become computationally productive. Relations between the surrounding urban fabric and GRZ, GFZ, and height can be learned and turned into steerable model responses, so that selected regulatory targets enter not only as constraints to be evaluated, but also as signals through which built form becomes computationally explorable.

The same translation marks the limits of the approach. The representation captures only selected aspects of massing, context, and regulatory logic, and while this reduction is necessary for experimental control, it leaves out much of what gives urban space its meaning. A generated output may be geometrically coherent and aligned with selected regulatory targets, yet say little about use, ownership, conflict, atmosphere, memory, and public value. Its value lies in making relations between density, height, parcel geometry, and the surrounding urban fabric visible at an early stage, where it can support structured reflection and comparison. Questions of relevance, appropriateness, admissibility, and desirability remain matters of situated professional and political judgement.

The phased research-through-design process was central to this conclusion, as the shared representation and evaluation framework made both the improvement in deterministic controllability and the failure of stochastic sampling identifiable. The unsuccessful Phase III experiments show only that stable conditional multi-output sampling was not achieved with the tested latent representation, diffusion architecture, conditioning strategy, training scale, and sampling configuration. They do not establish that stochastic modelling is generally unsuitable for controlled urban massing.

The thesis brings selected regulatory metrics, spatial context, and generative computation into one shared representation at the scale of urban massing. Beyond its specific results, it points toward design tools that complement explicitly defined parametric rules by learning context-sensitive relations from existing urban form, while keeping measurable targets comparable and steerable. Realising this in practice will depend on the directions set out above, from higher-resolution and multi-scale generation to richer conditioning and stable stochastic variation. Within these bounds, generative artificial intelligence can serve as a mediating instrument between selected regulatory abstractions and design judgement, clarifying both the potential and the limitations of regulation-aware and context-sensitive urban massing.

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Appendix

Configuration Manifest

The following manifest maps the compact labels used in the thesis to their experiment group and source JSON file. The complete training configuration files are available in the accompanying code repository; the appendix retains only this manifest, one representative metadata JSON example, and one representative training-configuration JSON example.

Label	Experiment Group	JSON File
Metadata example	Dataset metadata	`DEBE00YY1HE0003y.json`
P1.01	Phase I — Baseline	`phase1_01_baseline.json`
P1.02	Phase I — Network capacity	`phase1_02_capacity.json`
P1.03	Phase I — Coordinate channels	`phase1_03_coords.json`
P1.04	Phase I — Surface weighting	`phase1_04_surface.json`
P1.05	Phase I — Capacity and coordinate channels	`phase1_05_capacity_coords.json`
P1.06	Phase I — Loss balance	`phase1_06_loss_balance.json`
P1.07	Phase I — Manual threshold	`phase1_07_manual_thresh.json`
P1-Final	Phase I — Compact U-Net rerun	`phase1_final.json`
P2.01	Phase II — Reference and coordinate sensitivity	`phase2_cond_01_reference_coords.json`
P2.02	Phase II — Conditioning-channel comparison	`phase2_cond_02_channel_ablation.json`
P2.03	Phase II — Conditioning dropout	`phase2_cond_03_cond_dropout.json`
P2.04	Phase II — Capacity transfer	`phase2_cond_04_capacity.json`
P2.05	Phase II — Loss balance	`phase2_cond_05_loss_balance.json`
P2.06	Phase II — Evaluation-time sensitivity	`phase2_cond_06_eval_sensitivity.json`
P2-Final	Phase II — Conditional U-Net rerun	`phase2_cond_final.json`
P3A.01	Phase III Stage A — VAE compression	`phase3_ldm_01_vae_compression.json`
P3A.02	Phase III Stage A — VAE capacity	`phase3_ldm_02_vae_capacity.json`
P3B.03	Phase III Stage B — Diffusion schedule	`phase3_ldm_03_diff_schedule.json`
P3B.04	Phase III Stage B — Diffusion capacity	`phase3_ldm_04_diff_capacity.json`
P3B.05	Phase III Stage B — Diffusion context and coordinates	`phase3_ldm_05_diff_context_coords.json`
P3B.06	Phase III Stage B — Diffusion conditioning dropout	`phase3_ldm_06_diff_cond_dropout.json`
P3A-Final	Phase III Stage A — Final VAE rerun	`phase3_final_vae.json`
P3B-Final	Phase III Stage B — Final diffusion rerun	`phase3_final_diff.json`

A Example Metadata JSON

The following record shows a representative metadata JSON file stored alongside a retained voxel sample. It documents parcel and grid identifiers, spatial bounds, derived regulatory metrics, geometric quantities, neighbourhood descriptors, and street-related measures used for filtering, analysis, and conditional training.

```
{
  "parcel_id": "DEBE00YY1HE0003y",
  "grid": {
    "grid_m": 128.0,
    "voxel_m": 0.5,
    "D": 256,
    "H": 256,
    "W": 256
  },
  "world_bbox_xy": [
    400013.1392216777,
    5819894.6572863525,
    400141.1392216777,
    5820022.6572863525
  ],
  "z_world_range": [
    53.049,
    181.049
  ],
  "max_height_m": 128.0,
  "target_z_grid": [
    1,
    17
  ],
  "channels": [
    "C0_build_mask",
    "C1_neighbor_height",
    "C2_parcel_edges",
    "C3_street_mask"
  ],
  "metrics": {
    "c1_mean": 0.0072288308292627335,
    "c1_max": 0.08314062654972076,
    "target_voxels": 6021,
    "neighbor_voxels": 127573,
    "num_neighbors": 26,
    "parcel_area_m2": 540.5104629920769,
    "target_footprint_area_m2": 99.36121200352106,
    "coverage_frac": 0.17945998577537597,
    "target_height_m": 8.0,
    "avg_neighbor_height_m": 5.530653846153846,
    "street_coverage_frac": 0.1270904541015625,
    "bgf_target_m2": 250.875,
    "grz_target": 0.17945998577537597,
    "gfz_target": 0.4641445766123448,
    "grz_neighbors": [
      0.17108352597261262,
      0.16659660425777342
    ],
    "gfz_neighbors": [
      0.4508050909378343,
      0.414159158184825
    ],
    "target_footprint_voxels_0p5m": 388,
    "target_voxels_0p5m": 6021,
    "target_height_m_0p5m": 8.0,
    "grz_target_0p5m": 0.17945998577537597,
    "gfz_target_0p5m": 0.4641445766123448,
    "bgf_target_m2_0p5m": 250.875,
    "target_footprint_voxels_2m": 24,
    "target_voxels_2m": 93,
    "target_height_m_2m": 8.0,
```

```
      "grz_target_2m": 0.17760988282923806,
      "gfz_target_2m": 0.4588255306421984,
      "bgf_target_m2_2m": 248.0,
      "voxel_metrics": {
        "0p5m": {
          "voxel_m": 0.5,
          "target_voxels": 6021,
          "target_footprint_voxels": 388,
          "target_height_m": 8.0,
          "grz_target": 0.17945998577537597,
          "gfz_target": 0.4641445766123448,
          "bgf_target_m2": 250.875
        },
        "2m": {
          "voxel_m": 2.0,
          "target_voxels": 93,
          "target_footprint_voxels": 24,
          "target_height_m": 8.0,
          "grz_target": 0.17760988282923806,
          "gfz_target": 0.4588255306421984,
          "bgf_target_m2": 248.0
        }
      }
    }
}
```

B Example Training Configuration JSON

The following record shows a representative training-configuration JSON file for the final Phase II conditional U-Net rerun. The full set of training configurations is available in the accompanying code repository.

```json
{
  "name": "phase2_cond_final",
  "description": "Final 50-epoch rerun for Phase 2 based on the strongest completed experiment under the shared Phase-II evaluation protocol.",
  "global": {
    "data_root": "00_Data/02_GeneratedDatasets/dataset_thesis",
    "split_manifest": "00_Data/02_GeneratedDatasets/dataset_thesis/report_split.json",
    "epochs": 50,
    "batch_size": 2,
    "accum_steps": 2,
    "lr": 0.0007,
    "optimizer": "adamw",
    "weight_decay": 0.01,
    "lr_schedule": "cosine",
    "seed": 42,
    "num_workers": 0,
    "downsample_stride": 4,
    "crop_size": 0,
    "add_coords": false,
    "auto_thresh": true,
    "thresh_grid": [
      0.40,
      0.45,
      0.50,
      0.55,
      0.60
    ],
    "tta": true,
    "tta_mode": "rot90",
    "early_stop_patience": 0,
    "log_every": 6,
    "save_every": 4,
    "vis_every": 5,
    "grad_clip": 1.0,
    "base_ch": 12,
    "depth": 4,
    "surface_weight": 1.0,
    "volume_reg_weight": 0.1,
    "cond_dim": 3,
    "cond_select": [
      0,
      1,
      2
    ],
    "cond_drop": 0.0
  },
  "overrides": [
    {
      "model": "cond_unet",
      "run_name": "phase2_final_drop_p00_tta_rot90",
      "description": "Final rerun of the best Phase-II validation-IoU experiment: compact depth-4 conditional UNet with full GRZ/GFZ/height conditioning, no coordinate channels, no conditioning dropout, and rot90-TTA for validation/inference."
    }
  ]
}
```